

Cell Injury, Cell Death, and Adaptations

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INTRODUCTION TO PATHOLOGY

Pathology is the study of the structural, biochemical, and functional changes in cells, tissues, and organs that underlie disease. By using morphologic, microbiologic, immunologic, and molecular techniques, pathology attempts to explain the whys and wherefores of the signs and symptoms manifested by patients while providing a rational basis for clinical care and therapy. It thus serves as the bridge between the basic sciences and clinical medicine and is the scientific foundation for all of medicine. In Chapter 1 we examined the cellular and molecular properties of healthy cells. In this chapter, we will build

on that knowledge to discuss the fundamental mechanisms underlying various forms of cell injury and death.

Traditionally, the study of pathology is divided into general pathology and systemic pathology. *General pathology* is concerned with the common reactions of cells and tissues to injurious stimuli. Such reactions are often not tissue specific; for example, acute inflammation in response to bacterial infections produces a very similar reaction in most tissues. *Systemic pathology* examines the alterations and underlying mechanisms in diseases of particular organ systems. In this book, we first cover the principles of general pathology before proceeding to specific disease processes as they affect different organs.

The four aspects of a disease process that form the core of pathology are causation (*etiology*), biochemical and molecular mechanisms (*pathogenesis*), the associated structural (*morphologic changes*) and functional alterations in cells and organs, and the resulting clinical consequences (*clinical manifestations*).

- *Etiology is the initiating cause of a disease.* Although there are myriad factors that cause disease, all can be grouped into two broad classes: genetic (e.g., inherited or acquired mutations, and disease-associated gene variants, or polymorphisms) and environmental (e.g., infectious, nutritional, chemical, physical). The idea that one etiologic agent is the cause of one disease arose from the study of infections and inherited disorders caused by single gene anomalies, but most diseases are not this simple. Most common afflictions are *multifactorial*, such as atherosclerosis and cancer, because they arise from the effects of diverse environmental insults on a genetically susceptible individual. The relative contribution of inherited susceptibility and environmental influences varies in different diseases, and it is challenging to precisely define their roles in most multifactorial diseases.
- *Pathogenesis refers to the sequence of molecular, biochemical, and cellular events that lead to the development of disease.* In other words, it refers to mechanism of disease. Thus, pathogenesis explains how the underlying causes produce the morphologic and clinical manifestations of the disease. The study of pathogenesis is a central focus of pathology. Even when the initial cause is known (e.g., infection or mutation), it is many steps removed from the expression of the disease. For example, to truly understand the disorder cystic fibrosis, it is essential to know not only the defective gene and gene product, but also the biochemical and morphologic events that lead to clinically significant disease in the lungs, pancreas, and other organs. New technological advances, particularly the use of so-called “omics” technologies (genomics, proteomics, metabolomics) to interrogate diseases, hold great promise for elucidating pathogenic mechanisms. Hopefully, the application of these methods and the analysis of mounds of “big data” so generated will lead not only to better understanding of pathogenesis but also to the identification of *biomarkers* that predict disease progression and therapeutic responses. This is the goal of *precision medicine*.
- *Morphologic changes refer to the structural alterations in cells or tissues that are characteristic of a disease and diagnostic of an etiologic process.* Traditionally, diagnostic pathology has employed morphology to determine the nature of disease and to follow its progression. Although morphology remains a cornerstone of diagnosis, it is now routinely supplemented by analysis of protein expression and molecular alterations. Nowhere is this more striking than in the study of neoplasms; breast cancers that are indistinguishable morphologically may be caused by different genetic abnormalities that result in widely different courses, therapeutic responses, and prognoses. Molecular analysis by techniques such as next-generation sequencing (see Chapter 7) has revealed genetic differences that predict the behavior of tumors as well as their response to therapies, an increasing number of which are now chosen based on the presence (or absence) of specific molecular alterations.
- *Clinical manifestations.* The end results of genetic, biochemical, and structural changes in cells and tissues are functional abnormalities that lead to the clinical manifestations (symptoms and signs) of disease, as well as its progression (clinical course and outcome). Hence, clinicopathologic correlations are very important in the study of disease.

Virtually all diseases start with molecular or structural alterations in cells. This concept of the cellular basis of disease was first put forth in the nineteenth century by Rudolf Virchow, known as the father of modern pathology. Virchow emphasized the idea that individuals are sick because their cells are sick. We therefore begin our consideration of pathology with the study of the causes, mechanisms, and morphologic

and biochemical correlates of *cell injury*. Injury to cells and to the extracellular matrix ultimately leads to *tissue and organ injury*, which determines the morphologic and clinical patterns of disease.

OVERVIEW OF CELLULAR RESPONSES TO STRESS AND NOXIOUS STIMULI

The normal cell is confined to a fairly narrow range of function and structure dictated by its state of metabolism, differentiation, and specialization; by constraints imposed by neighboring cells; and by the availability of metabolic substrates. It is nevertheless able to handle physiologic demands, maintaining a healthy steady state called *homeostasis*. *Adaptations* are reversible functional and structural responses to changes in physiologic states (e.g., pregnancy) and some pathologic stimuli, during which new but altered steady states are achieved, allowing the cell to survive and continue to function (Fig. 2.1). The adaptive response may consist of an increase in the size (hypertrophy) and functional activity of cells, an increase in cell number (hyperplasia), a decrease in the size and metabolic activity of cells (atrophy), or a change in the phenotype of cells (metaplasia). If the stress is eliminated, the cell can return to its original state without having suffered any harmful consequences.

If the limits of adaptive responses are exceeded or if cells are exposed to damaging insults, deprived of critical nutrients, or compromised by mutations that affect essential cellular functions, a sequence of events follows that is termed *cell injury* (see Fig. 2.1). Cell injury is *reversible* up to a point, but if the injurious stimulus is persistent or severe, the cell suffers *irreversible injury* and ultimately undergoes *cell death*. *Adaptation*, *reversible injury*, and *cell death* represent stages of progressive impairment following different types of insults. For instance, in response to increased hemodynamic loads, the heart muscle becomes enlarged, a form of adaptation, which because of increased metabolic demands is more susceptible to injury. If the blood supply to the myocardium is compromised or inadequate, the muscle first suffers reversible injury, manifested by certain cytoplasmic changes (described later). Unless the blood supply is quickly restored, the cells suffer irreversible injury and die (Fig. 2.2). It should be noted, however, that adaptation and reversible and irreversible injury do not always occur sequentially. For example, a sudden complete occlusion of a major coronary artery can lead to irreversible injury without preceding adaptation.

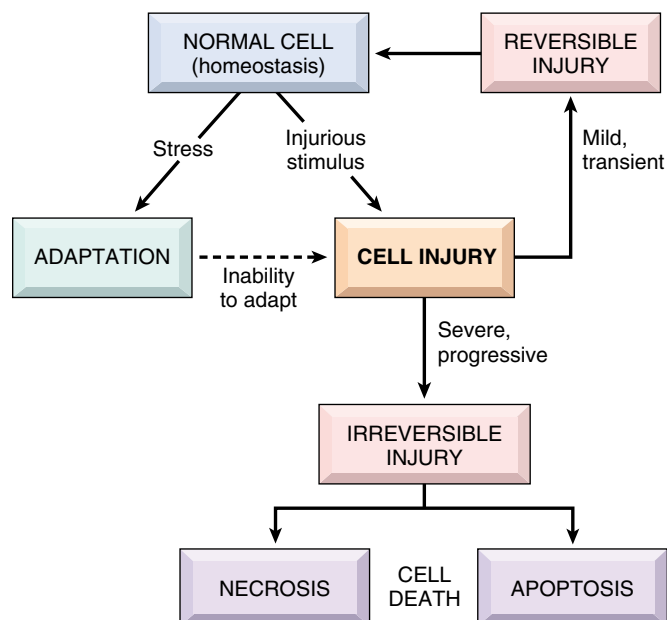


Fig. 2.1 Stages of the cellular response to stress and injurious stimuli.

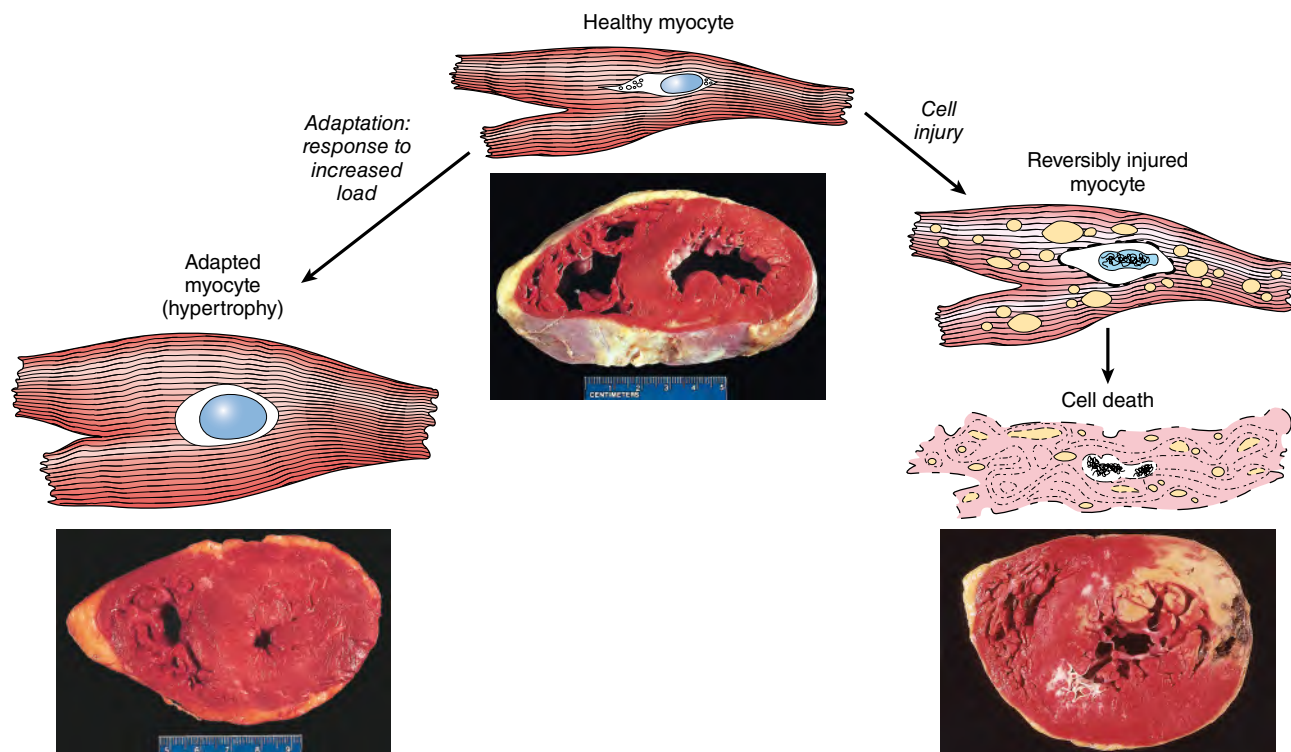


Fig. 2.2 The relationship among healthy, adapted, reversibly injured, and dead myocardial cells. All three transverse sections of the heart have been stained with triphenyltetrazolium chloride, an enzyme substrate that stains viable myocardium brick-red. The cellular adaptation shown here is myocardial hypertrophy (*lower left*), caused by increased blood pressure requiring greater mechanical effort by myocardial cells. This adaptation leads to thickening of the left ventricular wall (compare with the normal, healthy heart). In reversibly injured myocardium (illustrated schematically, *right*), there are functional alterations, usually without any gross changes but sometimes with microscopic changes such as cellular swelling and fat accumulation. In the specimen showing necrosis, a form of cell death (*lower right*), the unstained area in the posterolateral left ventricle represents an acute myocardial infarction caused by reduced blood flow (ischemia).

The removal of unneeded and aged cells through *cell death* is a normal and essential process in embryogenesis, the development of organs, and the maintenance of homeostasis into adulthood. Conversely, excessive cell death because of progressive injury is one of the most crucial events in the evolution of disease in any tissue or organ. It results from diverse causes, including ischemia (reduced blood flow), infection, and toxins. There are two principal pathways of cell death, *necrosis* and *apoptosis*, with recent evidence supporting the importance of additional pathways, such as *necroptosis*, *pyroptosis*, and *ferroptosis*. A detailed discussion of cell death pathways follows later in the chapter.

Stresses of different types may induce changes in cells and tissues other than typical adaptations, cell injury, and death. Metabolic derangements and chronic injury may be associated with *intracellular accumulations* of several substances, including proteins, lipids, and carbohydrates. Calcium may be deposited at sites of cell death, resulting in *pathologic calcification*. Finally, the normal process of *aging* is accompanied by characteristic morphologic and functional changes in cells.

This chapter discusses first the causes, mechanisms, and consequences of the various forms of cell damage, including reversible cell injury and cell death. We conclude with cellular adaptations to stress, and three other processes that affect cells and tissues: intracellular accumulations, pathologic calcification, and cell aging.

Causes of Cell Injury

The causes of cell injury range from the mechanical trauma of an automobile accident to subtle cellular abnormalities, such as a mutation causing

lack of a vital enzyme that impairs normal metabolic function. Most injurious stimuli can be grouped into the following broad categories.

Oxygen Deprivation

Hypoxia is a deficiency of oxygen, which causes cell injury by reducing aerobic oxidative respiration. Hypoxia is an extremely important and common cause of cell injury and cell death. Causes of hypoxia include reduced blood flow (*ischemia*); inadequate oxygenation of the blood due to cardiorespiratory failure; and decreased oxygen-carrying capacity of the blood, as in anemia or carbon monoxide poisoning and severe blood loss. Depending on the severity of the hypoxic state, cells may adapt, undergo injury, or die. For example, if an artery is narrowed, the tissue supplied by that vessel may initially shrink in size (*atrophy*), whereas more severe or sudden hypoxia induces cell injury and cell death.

Physical Agents

Physical agents capable of causing cell injury include mechanical trauma, extremes of temperature (burns and deep cold), sudden changes in atmospheric pressure, radiation, and electric shock (Chapter 9).

Chemical Agents and Drugs

The list of chemicals that may produce cell injury defies compilation. Simple chemicals such as glucose or salt in hypertonic concentrations may cause cell injury directly or by deranging electrolyte and fluid balance in cells. Even oxygen at high concentrations is toxic. Trace amounts of *poisons*, such as arsenic, cyanide, or mercury, may damage sufficient numbers of cells within minutes or hours to cause death.

Other potentially injurious substances are our daily companions: environmental pollutants; pesticides; industrial and occupational hazards, such as carbon monoxide and asbestos; legal and illegal drugs such as alcohol and cocaine; and the ever increasing variety of therapeutic drugs, many of which have toxic side effects. These are discussed further in Chapter 9.

Infectious Agents

These agents range from submicroscopic viruses to tapeworms several feet in length. In between are rickettsiae, bacteria, fungi, and parasites. The ways by which these biologic agents cause injury are diverse (Chapter 8).

Immunologic Reactions

The immune system serves an essential function in defense against infectious pathogens, but immune reactions may also cause cell injury. Injurious reactions to endogenous self-antigens are responsible for autoimmune diseases (Chapter 6). Immune reactions to many external agents, such as viruses and environmental substances, are also important causes of cell and tissue injury (Chapters 3 and 6).

Genetic Abnormalities

As described in Chapter 5, genetic aberrations as extreme as an extra chromosome, as in Down syndrome, or as subtle as a single base pair substitution leading to an amino acid substitution, as in sickle cell anemia, may produce highly characteristic clinical phenotypes ranging from congenital malformations to anemias. Genetic defects may cause cell injury because of deficient protein function, such as enzyme defects in inborn errors of metabolism, or accumulation of damaged DNA or misfolded proteins, both of which trigger cell death when they are beyond repair. DNA sequence variants that are common in human populations (polymorphisms) can also influence the susceptibility of cells to injury by chemicals and other environmental insults.

Nutritional Imbalances

Nutritional imbalances continue to be major causes of cell injury. Protein-calorie deficiencies cause an appalling number of deaths, chiefly among lower-resource populations. Deficiencies of specific vitamins are found throughout the world (Chapter 9). Nutritional shortages can be self-imposed, as in anorexia nervosa (a psychological disorder of inadequate food consumption) or stem from food shortages or poor diet. Ironically, nutritional excess is an even more important cause of cell injury. Obesity, rampant in the United States and increasing elsewhere in the world, is associated with several important diseases, including diabetes and cancer. In addition to the problems of undernutrition and overnutrition, the composition of the diet makes a significant contribution to several diseases. For example, diets high in certain lipids lead to elevated serum cholesterol and predispose to atherosclerosis, a leading risk factor for cardiovascular disease, the number one killer of adults in the United States.

PATHWAYS OF CELLULAR RESPONSE TO STRESS AND INJURY

On a cellular level, the response to injury commonly results in the activation of certain response pathways, including the *integrated stress response*, the *unfolded protein response* (ER stress), and *autophagy*. Generally, these pathways protect cells from the acute effects of injurious stimuli and promote adaptation over longer periods. However, when unable to mitigate untoward stresses in cells and tissues, all these pathways can also trigger cell death to eliminate damaged cells.

Integrated Stress Response

The integrated stress response (ISR) is an evolutionarily conserved network of intracellular signaling pathways that modulate gene expression and protein synthesis in order for cells to adapt to cell injury (Fig. 2.3). Increasing evidence implicates the ISR as an early response pathway to a broad array of cellular stressors, including hypoxia/ischemia, infection, nutrient starvation, membrane damage, and the accumulation of damaged, misfolded proteins. The ISR can be activated by one of four stress-sensing kinases: protein kinase R (PKR), PKR-like ER kinase (PERK), general control nonderepressible 2 kinase (GCN2), and heme-regulated eIF2 α kinase (HRI), which all phosphorylate a specific serine residue on the eukaryotic translation initiation factor eIF2 α , a key regulator of protein translation. These ISR stress-sensing kinases can act cooperatively in tuning the cellular response to a wide variety of stressors. The resulting phosphorylation of eIF2 α triggers wholesale changes in protein synthesis consisting of repression of translation of most mRNAs. By diminishing protein synthesis, the ISR provides a window of opportunity for an injured cell to conserve energy and to restore homeostasis. At the same time, the ISR promotes the translation of a select few mRNAs encoding transcription factors such as ATF4, which controls the expression of genes involved in antioxidant response and protein folding, hence facilitating the ability of the cell to repair itself. Remarkably, these kinases can also activate *autophagy* and, through activation of PERK, the *unfolded protein response*, illustrating the complex relationship between these arch-

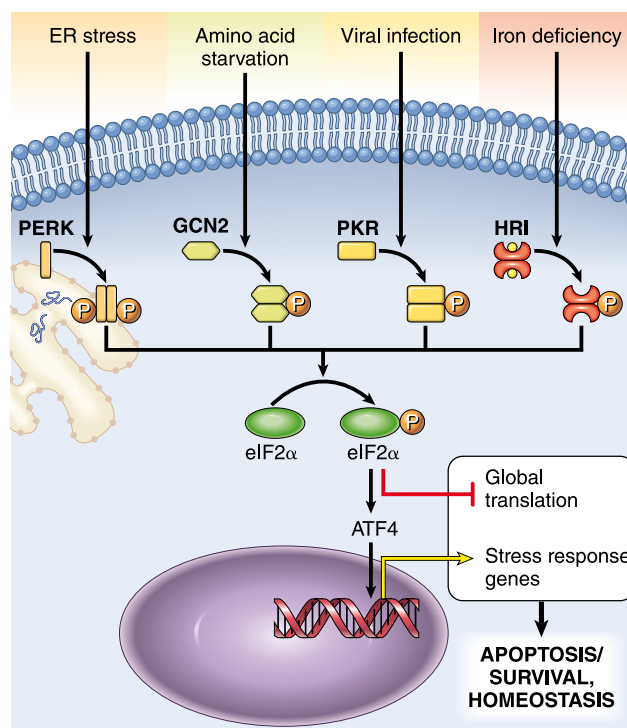


Fig. 2.3 The integrated stress response (ISR). Various types of cell injury result in the dimerization and activation of one of four upstream sensing kinases: PKR protein kinase R (*PKR*), PKR-like ER kinase (*PERK*), general control nonderepressible 2 (*GCN2*), and heme-regulated eIF2 α kinase (*HRI*). These kinases all phosphorylate a key mediator of protein synthesis, the eukaryotic translation initiation factor eIF2 α , on serine 51. The phosphorylation of eIF2 α is considered the core signaling event in the ISR and results in the global repression of protein translation in a cell. At the same time, the ISR promotes the translation of select transcription factors, such as ATF4, a master regulator of genes critical for cell survival and adaptation. Together, these pathways promote cell survival, recovery, and the restoration of physiological homeostasis. ER, Endoplasmic reticulum.

typical cellular responses to injury. Of further importance, a dysfunctional ISR has been linked to the pathogenesis of various diseases, including cancer, autoimmune disease, type 2 diabetes, and age-related neurodegenerative diseases.

Unfolded Protein Response and Endoplasmic Reticulum Stress

The accumulation of misfolded proteins in the ER activates stress adaptive mechanisms and, if unchecked, can trigger cell death via apoptosis. Chaperones in the ER control the proper folding of newly synthesized proteins, and misfolded polypeptides are shuttled into the cytoplasm where they are ubiquitinated and targeted for proteolysis in proteasomes (Chapter 1). If, however, unfolded or misfolded proteins accumulate in the ER, they trigger a number of alterations that are collectively called the *unfolded protein response* (UPR). The unfolded protein response activates signaling pathways that increase the production of chaperones, enhance proteasomal degradation of abnormal proteins, and slow protein translation, thus reducing the load of misfolded proteins in the cell (Fig. 2.4). However, if this cytoprotective response is unable to cope with the accumulation of misfolded proteins, the cell activates caspases, which trigger cell death through apoptosis. This process is called *ER stress*. Intracellular accumulation of misfolded proteins may be caused by an increased rate of misfolding or a reduction in the cell's ability to repair or eliminate them. Increased misfolding may be a consequence of deleterious mutations or decreased capacity to correct misfolded proteins, as occurs during aging, or increased or altered protein synthesis, as occurs in plasma cell tumors (Chapter 13). Protein misfolding may also be increased in viral infections when proteins encoded by the viral genome are synthesized in such large quantities that they overwhelm the quality control system that normally ensures proper protein folding. Increased demand for secretory proteins such as insulin in insulin-resistant states and changes in intracellular pH and redox state are other stressors

KEY CONCEPTS

Integrated Stress Response (ISR)

- The ISR consists of a network of intracellular signaling pathways through which cells respond to diverse stressors, including hypoxia/ischemia, infection, nutrient starvation, and the accumulation of damaged, misfolded proteins.
- The ISR is triggered by the activation of stress-sensing protein kinases, which all phosphorylate a specific serine residue on the eukaryotic translation initiation factor eIF2 α .
- eIF2 α phosphorylation mediates two key effector responses: (1) global repression of protein synthesis, which allows for an injured cell to conserve energy; and (2) activation of transcription factors, such as ATF4, which govern gene programs that promote cell repair and survival.
- Dysfunctions in both the ISR and UPR (see below) are implicated in the pathogenesis of various diseases, including type 2 diabetes, cancer, and age-related degenerative diseases.

Unfolded Protein Response (UPR) and Endoplasmic Reticulum (ER) Stress

- ER stress is a cell state in which protein folding demands exceed the protein folding capacity of the ER.
- The excessive intracellular accumulation of unfolded or misfolded proteins during ER stress activates the UPR.
- Initially, the UPR is adaptive and includes the production of chaperones, enhanced proteasomal degradation of abnormal proteins, and diminished protein translation, which altogether reduce the load of misfolded proteins in the cell.
- If unable to cope with the accumulation of misfolded proteins, the UPR in a cell triggers apoptosis.

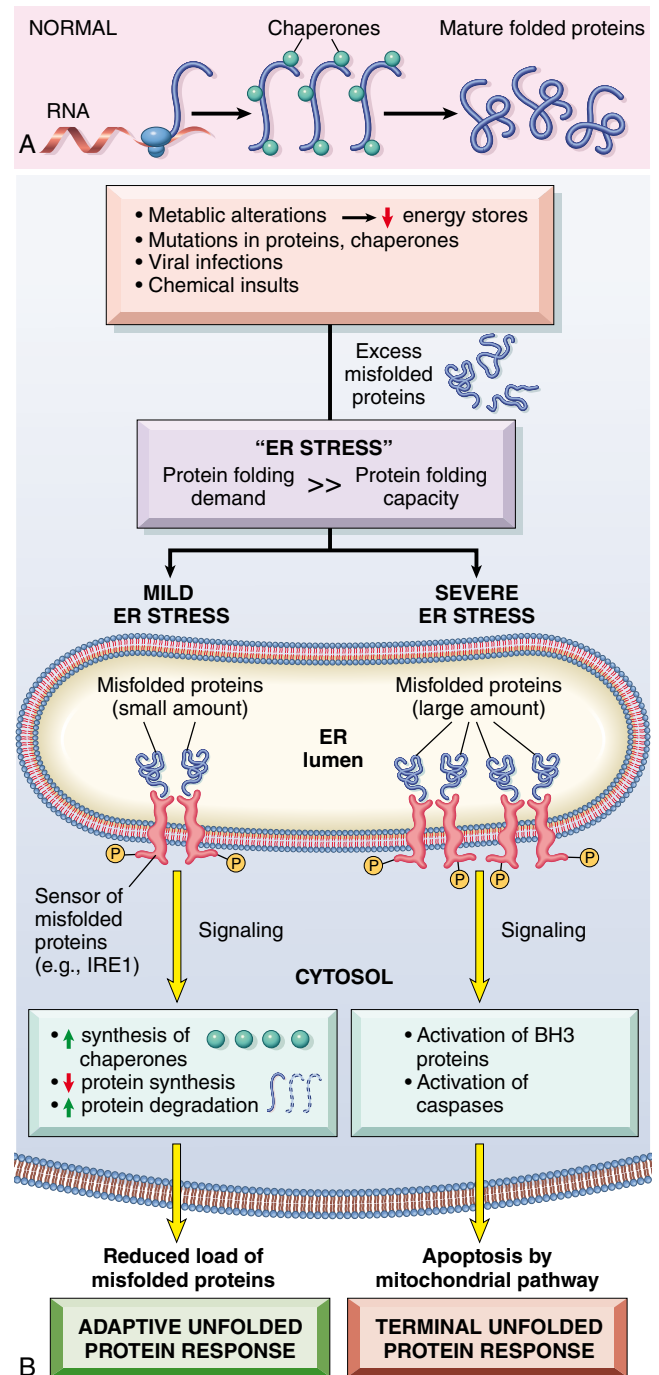


Fig. 2.4 The unfolded protein response and endoplasmic reticulum (ER) stress. (A) In healthy cells, newly synthesized proteins are folded in the ER with the help of chaperones and are then incorporated into the cell or secreted. (B) Various external stresses or mutations induce a state called *ER stress*, in which the cell is unable to efficiently handle the load of misfolded proteins. Accumulation of these proteins in the ER triggers the *unfolded protein response* (UPR) to restore protein homeostasis; if this response is inadequate, the unfolded protein response actively induces apoptosis.

that result in misfolded protein accumulation. Protein misfolding is thought to be the causative cellular abnormality in several neurodegenerative diseases (Chapter 28). Since many “foldases” require adenosine triphosphate (ATP) to function, deprivation of glucose and oxygen, as in ischemia and hypoxia, may also increase the burden of misfolded proteins. Diseases caused by misfolded proteins are listed in Table 2.1.

TABLE 2.1 Selected Examples of Diseases Caused by Misfolding of Proteins

Disease	Affected Protein	Pathogenesis
Cystic fibrosis	CFTR	Loss of CFTR leads to defects in chloride transport
Familial hypercholesterolemia	LDL receptor	Loss of LDL receptor leads to hypercholesterolemia
Tay-Sachs disease	Hexosaminidase β subunit	Lack of the lysosomal enzyme leads to storage of GM ₂ gangliosides in neurons
α_1 -Antitrypsin deficiency	α_1 -Antitrypsin	Storage of nonfunctional protein in hepatocytes causes apoptosis; absence of enzymatic activity in lungs causes destruction of elastic tissue giving rise to emphysema
Creutzfeldt-Jacob disease	Prions	Abnormal folding of PrP ^{Sc} causes neuronal cell death
Alzheimer disease	A β peptide	Abnormal folding of A β peptides causes aggregation within neurons and apoptosis

CFTR, Cystic fibrosis transmembrane conductance regulator; LDL, low-density lipoprotein; PrP^{Sc}, scrapie isoform of the prion protein.

Autophagy

Autophagy is a process in which a cell eats its own contents (Greek: *auto*, self; *phagy*, eating) by delivering cytoplasmic materials to the lysosome for degradation. In addition to serving as an evolutionarily conserved survival mechanism during nutrient deprivation, where starved cells live by cannibalizing themselves and recycling the digested contents, autophagy is a fundamental response to stress in physiologic states (e.g., aging and exercise) and in pathologic processes (e.g., hypoxia, oxidative stress, organelle, and membrane damage). Autophagy proceeds through several steps (Fig. 2.5):

- Nucleation and formation of an isolation membrane, also called a *phagophore*; the isolation membrane is believed to be derived from the ER, though other sources such as the plasma membrane and mitochondria may contribute
- Formation of a vesicle, called the *autophagosome*, from the isolation membrane, inside which intracellular organelles and cytosolic structures are sequestered
- Maturation of the autophagosome by fusion with lysosomes, to deliver digestive enzymes that degrade the contents of the autophagosome

In recent years, over 30 “autophagy-related genes” called *Atgs* have been identified whose products are required for the creation of the autophagosome. Environmental cues like nutrient deprivation or depletion of growth factors activate an initiation complex that promotes the hierarchical recruitment of *Atgs* to nucleate the initiation membrane. The initiation membrane elongates further, surrounds and captures its cytosolic cargo, and closes to form the autophagosome. The elongation and closure of the initiation membrane require the coordinated action of two ubiquitin-like conjugation systems that result in the covalent linkage of the lipid phosphatidylethanolamine (PE) to microtubule-associated protein light chain 3 (LC3). PE-lipidated LC3 is increased during autophagy and is therefore a useful marker for identifying cells in which autophagy is occurring. The newly formed autophagosome fuses with lysosomes to form an autophagolysosome. In the terminal step, the inner membrane and enclosed cytosolic cargoes are degraded by lysosomal enzymes. Increasing evidence indicates that autophagy is not a random process that engulfs cytosolic contents indiscriminately. Rather, the loading of cargo into the

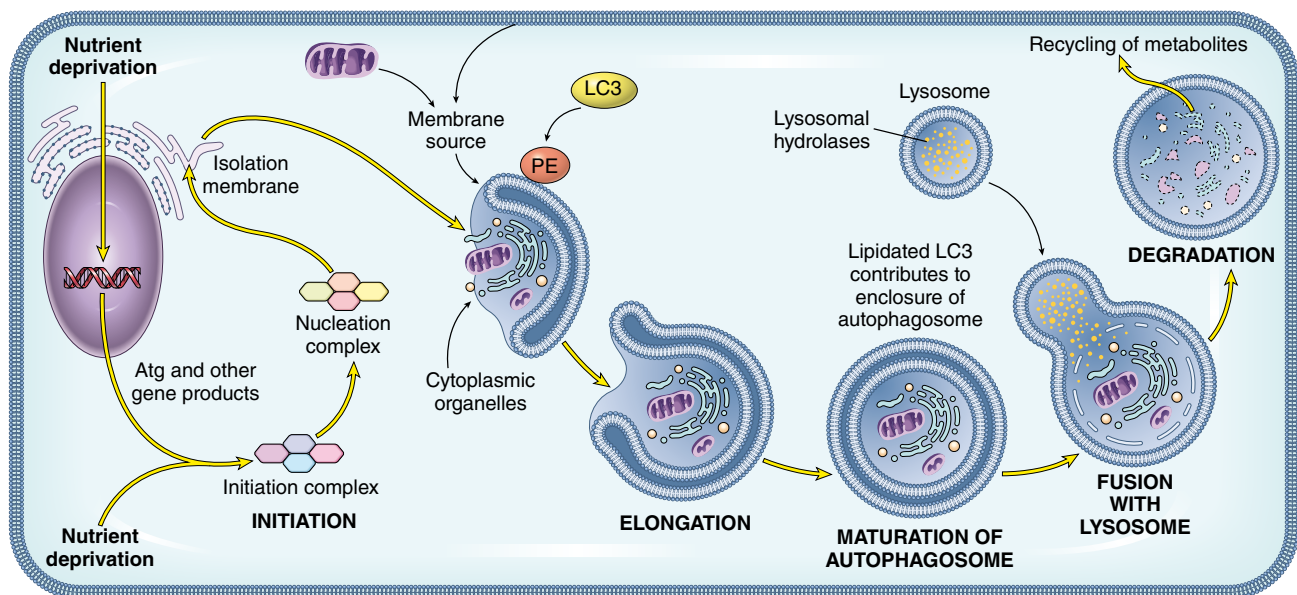


Fig. 2.5 Autophagy. Cellular stresses such as nutrient deprivation activate the autophagy pathway via nuclear and cytoplasmic signaling pathways. The early steps of autophagy proceed through several phases (initiation, nucleation, and elongation of isolation membrane) mediated by Atg proteins, creating double-membrane-bound vacuoles (autophagosome) in which cytoplasmic materials, including organelles, are sequestered and then degraded after fusion of the vesicles with lysosomes. In the final stage, the digested materials are released for recycling of metabolites. See text for details. LC3, Light chain 3; PE, phosphatidylethanolamine.

autophagosome is selective, and one of the functions of the lipidated LC3 is to target protein aggregates and effete organelles.

Autophagy frequently functions as a survival mechanism under various stress conditions, maintaining the integrity of cells by recycling essential metabolites and clearing intracellular debris. It is therefore prominent in atrophic cells exposed to severe nutrient deprivation. Autophagy is also involved in the turnover of organelles like the ER, mitochondria, and lysosomes and the clearance of intracellular aggregates that accumulate during aging, stress, and various disease states. Like the unfolded protein response, autophagy can trigger cell death if it is unable to cope with the stressor. Although the precise mechanisms remain a subject of intense scrutiny, multiple cell death pathways, including apoptosis, necroptosis, and ferroptosis, can be activated in the setting of excessive autophagy. However, it is not clear whether cell death is caused by autophagy itself or by the stress that triggered autophagy.

There is accumulating evidence that autophagy plays a role in human diseases, including:

- **Cancer:** Autophagy can both promote cancer growth and act as a defense against cancers. This is an area of active investigation, as discussed in Chapter 7.
- **Neurodegenerative disorders:** Many neurodegenerative disorders are associated with dysregulation of autophagy, which exacerbates the defects in proteostasis and organelle dysfunction observed in these diseases. Alzheimer disease is characterized by impaired autophagosome maturation, and in mouse models of the disease, genetic defects in autophagy accelerate neurodegeneration. In Huntington disease, mutant huntingtin impairs autophagy.
- **Infectious diseases:** Many pathogens are degraded by autophagy; these include mycobacteria, *Shigella* spp., and HSV-1. This is one way by which microbial proteins are digested and delivered to antigen presentation pathways. Macrophage-specific deletion of Atg5 increases susceptibility to tuberculosis.
- **Inflammatory bowel diseases:** Genome-wide association studies have linked Crohn disease and ulcerative colitis to single-nucleotide polymorphisms (SNPs) in the autophagy-related gene *ATG16L1*. How these polymorphisms promote intestinal inflammation is not known.

KEY CONCEPTS

Autophagy

- Autophagy involves sequestration of cellular organelles into cytoplasmic autophagic vacuoles (autophagosomes) that fuse with lysosomes, where the enclosed material is digested.
- Autophagy is an adaptive response during nutrient deprivation that allows the cell to cannibalize itself to survive.
- Autophagosome formation is regulated by more than a dozen proteins that act in a coordinated and sequential manner.
- Dysregulation of autophagy occurs in many disease states, including cancer, inflammatory bowel diseases, and neurodegenerative disorders. Autophagy plays a role in host defense against certain microbes.

PROGRESSION OF CELL INJURY AND DEATH

It is useful to describe the basic alterations that occur in damaged cells before discussing the mechanisms that bring about these changes. All stresses and noxious influences exert their effects first at the molecular or biochemical level. There is a time lag between the stress and the morphologic changes of cell injury or death; understandably, the early changes are subtle and are only detected with highly sensitive methods of examination (Fig. 2.6). With histochemical, ultrastructural, or biochemical techniques, changes may be seen in minutes to hours after injury, whereas changes visible by light microscopy or the naked eye may take considerably longer (hours to days) to appear. As would be

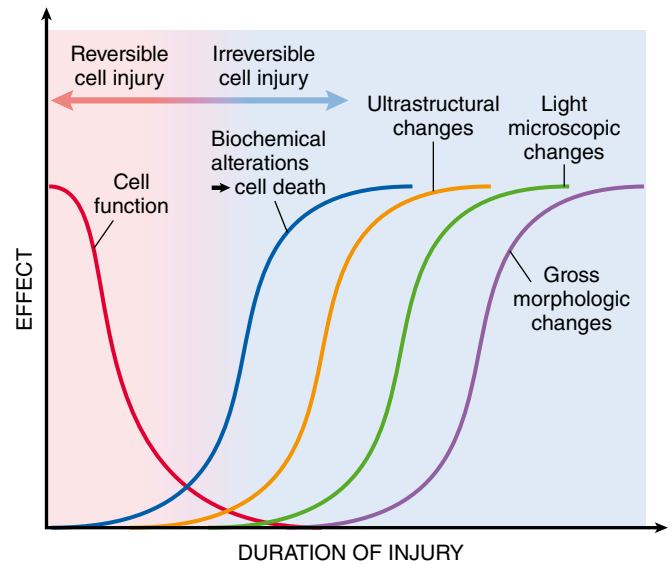


Fig. 2.6 Sequential development of biochemical and morphologic changes in cell injury. Cells may become rapidly nonfunctional after the onset of injury, although they may still be viable, with potentially reversible damage; a longer duration of injury may lead to irreversible damage and cell death. Note that ultrastructural, light microscopic, and grossly visible morphologic changes lag behind the biochemical alterations that lead to cell death.

expected, the morphologic manifestations of necrosis take more time to develop than those of reversible damage. For example, in ischemia of the myocardium, cell swelling is a reversible morphologic change that may occur in a matter of minutes, and is an indicator of ongoing cellular damage that may progress to irreversibility within 1 or 2 hours. Unmistakable light microscopic evidence of cell death, however, may not be seen until 4 to 12 hours after onset of ischemia.

The sequential structural changes in cell injury progressing to cell death are illustrated in Fig. 2.7 and described later. Within limits, the cell can repair the alterations seen in reversible injury and, if the injurious stimulus abates, may return to normalcy. Persistent or excessive injury, however, causes cells to pass the rather nebulous “point of no return” into irreversible injury and cell death. Different injurious stimuli induce death mainly by necrosis and/or apoptosis (see Fig. 2.7 and Table 2.2).

REVERSIBLE CELL INJURY

Reversible cell injury is characterized by functional and structural alterations in early stages or mild forms of injury, which are correctable if the damaging stimulus is removed. Two features are consistently seen in reversibly injured cells.

- Early alterations in reversible injury include *generalized swelling of the cell* and its organelles, blebbing of the plasma membrane, detachment of ribosomes from the ER, and clumping of nuclear chromatin. Swelling of cells results from influx of water. This is usually caused by failure of the ATP-dependent $\text{Na}^+ - \text{K}^+$ plasma membrane pump due to depletion of ATP resulting from oxygen deficiency, which interferes with mitochondrial oxidative phosphorylation, or mitochondrial damage by radiation or toxins (discussed later).
- *Fatty change* occurs in organs that are actively involved in lipid metabolism (e.g., liver). It results when toxic injury disrupts metabolic pathways and leads to rapid accumulation of triglyceride-filled lipid vacuoles. This is discussed in Chapter 18.
- Other alterations are described in the following sections.

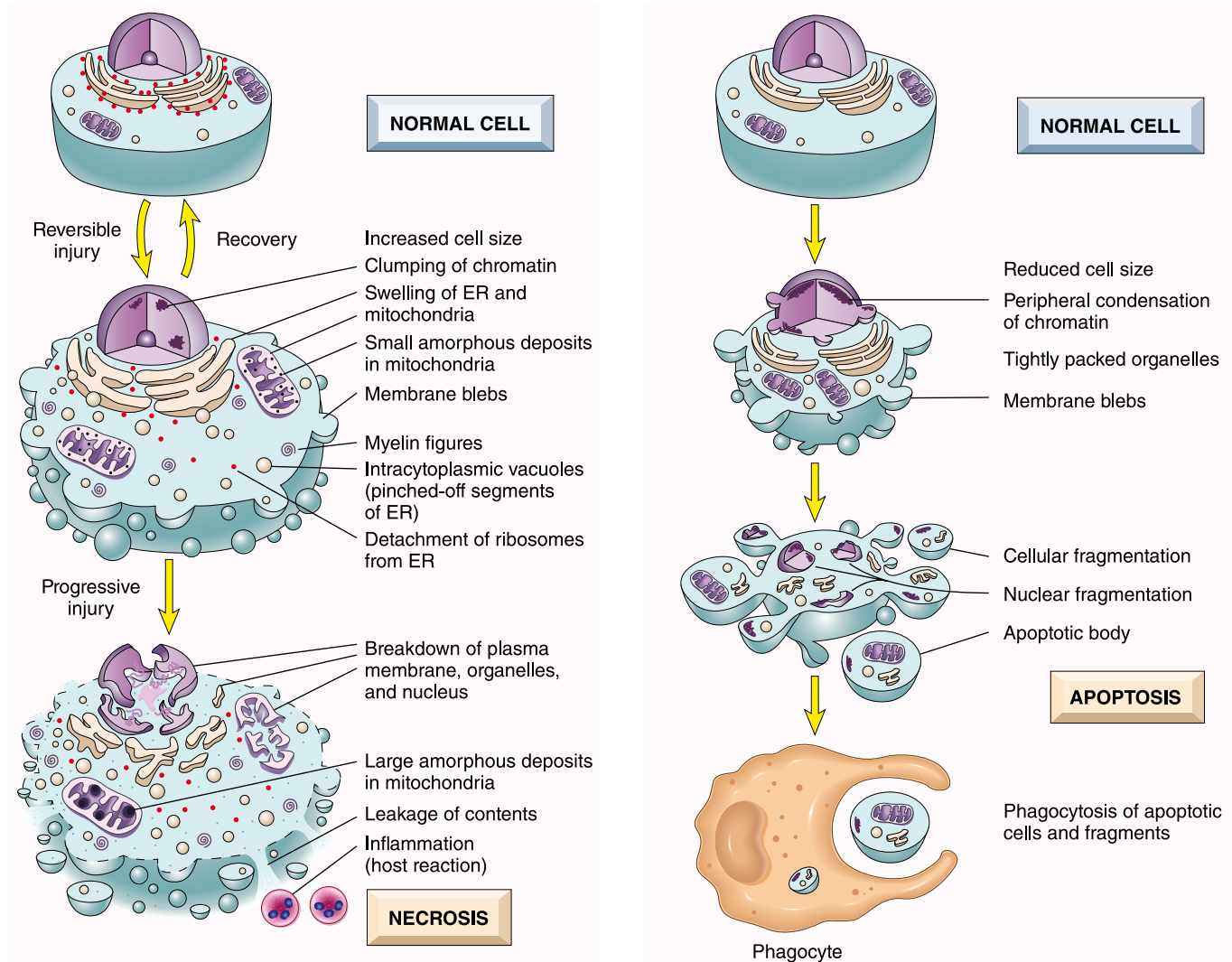


Fig. 2.7 Morphologic changes in cell injury culminating in necrosis or apoptosis.

MORPHOLOGY

Cellular swelling is the earliest manifestation of almost all forms of injury to cells (Fig. 2.8B). When it affects many cells, it causes pallor, increased turgor, and increased weight of the affected organ. On microscopic examination, small clear vacuoles may be seen within the cytoplasm; these represent distended and pinched-off segments of the ER. This pattern of nonlethal injury is sometimes called **hydropic change** or **vacuolar degeneration**. The cytoplasm of injured cells appears red (eosinophilic) when stained with hematoxylin and eosin (H&E) due to loss of RNA, which binds the blue hematoxylin dye. The eosinophilia becomes more pronounced with progression toward necrosis.

The ultrastructural changes of reversible cell injury, visible by electron microscopy (Fig. 2.9B), include:

1. Plasma membrane alterations, such as blebbing, blunting, and loss of microvilli
2. Mitochondrial changes, including swelling and the appearance of small amorphous densities
3. Accumulation of "myelin figures" in the cytosol composed of phospholipids derived from damaged cellular membranes
4. Dilation of the ER, with detachment of polysomes
5. Nuclear alterations, with disaggregation of granular and fibrillar elements

CELL DEATH

There are two principal types of cell death, **necrosis** and **apoptosis**, which differ in their mechanisms, morphology, and roles in physiology and disease (see Table 2.2). Severe mitochondrial damage with depletion of ATP and rupture of lysosomal and plasma membranes are typically associated with necrosis. Necrosis occurs in many commonly encountered injuries, such as those following ischemia, exposure to toxins, various infections, and trauma. Apoptosis has many unique features (see later).

Necrosis has historically been thought of as "accidental" cell death, reflecting severe injury that irreparably damages so many cellular components that the cell simply "falls apart." If early (reversible) injury progresses because the injurious stimulus persists, the end result is death by necrosis. When cells die by necrosis, there is a local inflammatory response that clears the "scene of the accident." In contrast, apoptosis is "regulated" cell death, mediated by defined molecular pathways that are activated under specific circumstances and kill cells with surgical precision, without inflammation or the associated collateral damage. The separation of necrosis and apoptosis is not always so clear, however, and some forms of necrosis are genetically controlled through a defined

TABLE 2.2 Features of Necrosis and Apoptosis

Feature	Necrosis	Apoptosis
Cell size	Enlarged (swelling)	Reduced (shrinkage)
Nucleus	Pyknosis, karyorrhexis, karyolysis	Fragmentation into nucleosome-size fragments
Plasma membrane	Disrupted	Intact; altered structure, especially orientation of lipids
Cellular contents	Enzymatic digestion; may leak out of cell	Intact; may be released in apoptotic bodies
Adjacent inflammation	Frequent	No
Physiologic or pathologic role	Usually pathologic (culmination of irreversible cell injury)	Often physiologic, means of eliminating unwanted cells; may be pathologic after some forms of cell injury, especially DNA damage

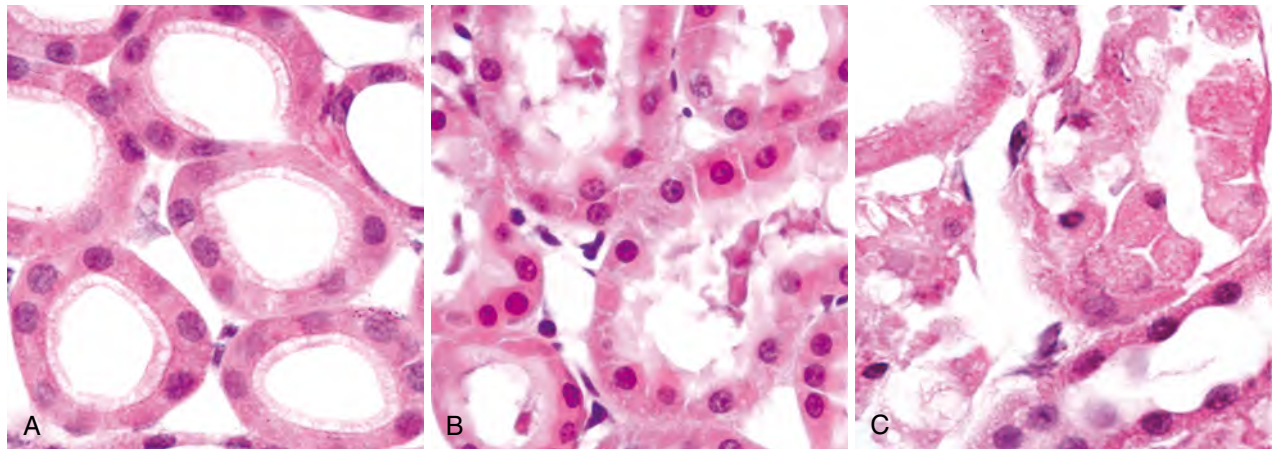


Fig. 2.8 Morphologic changes in reversible cell injury and necrosis. (A) Normal, healthy kidney tubules with viable epithelial cells. (B) Early (reversible) ischemic injury showing surface blebs, increased eosinophilia of cytoplasm, and swelling of occasional cells. (C) Necrosis (irreversible injury) of epithelial cells, with loss of nuclei, fragmentation of cells, and leakage of contents. The ultrastructural features of these stages of cell injury are shown in Fig. 2.9. (Courtesy Drs. Neal Pinckard and MA Venkatachalam, University of Texas Health Sciences Center, San Antonio.)

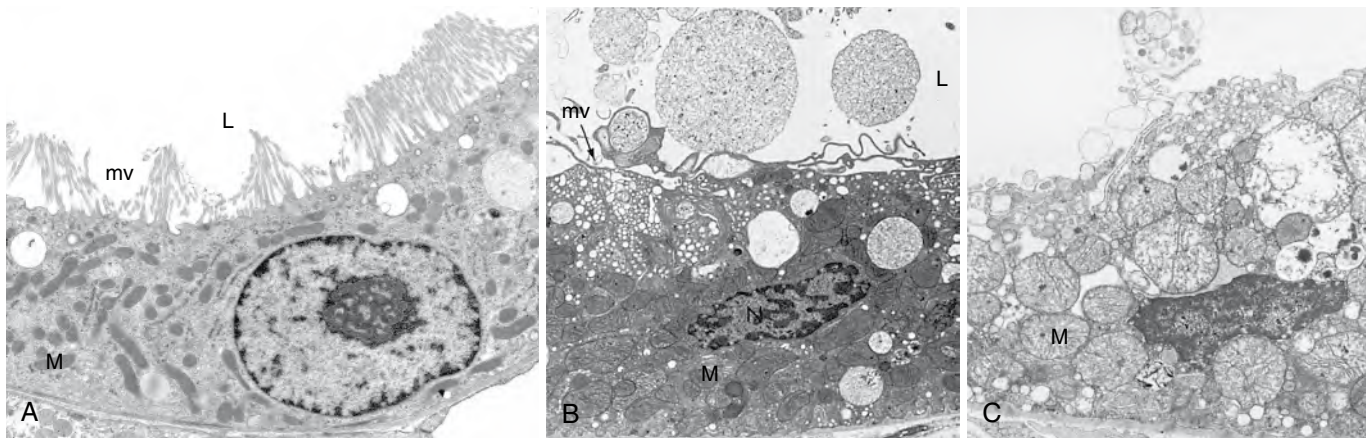


Fig. 2.9 Ultrastructural features of reversible and irreversible cell injury (necrosis) in a rabbit kidney. (A) Electron micrograph of a normal epithelial cell of the proximal kidney tubule. Note abundant microvilli (*mv*) lining the luminal surface (*L*). (B) Epithelial cell of the proximal tubule showing early cell injury resulting from reperfusion following ischemia. The microvilli are lost and have been incorporated in apical cytoplasm; blebs have formed and are extruded in the lumen. Mitochondria (*M*) would have been swollen during ischemia; with reperfusion, they rapidly undergo condensation and become electron dense. (C) Proximal tubular cell showing late injury, expected to be irreversible. Note the markedly swollen mitochondria containing electron-dense deposits, expected to contain precipitated calcium and proteins. Higher magnification micrographs of the cell would show disrupted plasma membrane and swelling and fragmentation of organelles. (A, Courtesy Dr. Brigitte Kaisslin, Institute of Anatomy, University of Zurich, Switzerland. B–C, Courtesy Dr. M.A. Venkatachalam, University of Texas Health Sciences Center, San Antonio.)

molecular pathway, called *necroptosis* (discussed later). Moreover, in some situations, cell death may show morphologic features of both apoptosis and necrosis, and apoptosis may progress to necrosis, so the distinctions may not be absolute. Nevertheless, it is useful to consider these as largely nonoverlapping pathways of cell death because their principal mechanisms, morphological features, and functional consequences are usually different.

Necrosis

Necrosis is a pathologic process that is the consequence of severe injury. The main causes of necrosis include loss of oxygen supply (ischemia), exposure to microbial toxins, burns and other forms of chemical and physical injury, and unusual situations in which active proteases leak out of cells and damage surrounding tissues (as in pancreatitis). All these initiating triggers lead to irreparable damage of numerous cellular components.

Necrosis is characterized by denaturation of cellular proteins, leakage of cellular contents through damaged membranes, local inflammation, and enzymatic digestion of the lethally injured cell. When damage to membranes is severe, lysosomal enzymes enter the cytoplasm and digest the cell. Cellular contents also leak through the damaged plasma membrane into the extracellular space, where they elicit a host reaction (inflammation). Some specific substances released from injured cells have been called *damage-associated molecular patterns* (DAMPs). These include ATP (released from damaged mitochondria), uric acid (a breakdown product of DNA), and numerous other molecules that are normally confined within healthy cells and whose release is an indicator of severe cell injury. These molecules are recognized by receptors present in macrophages and most other cell types and trigger phagocytosis of the debris as well as the production of cytokines that induce inflammation (Chapter 3). Inflammatory cells produce more proteolytic enzymes, and the combination of phagocytosis and enzymatic digestion usually leads to clearance of the necrotic cells.

Necrosis-associated leakage of intracellular proteins through damaged plasma membranes and ultimately into the circulation is the basis for blood tests that detect tissue-specific cellular injury. Cardiac muscle cells, for example, express cardiac-specific variants of the contractile protein troponin, while bile duct epithelium expresses a specific isoform of the enzyme alkaline phosphatase, and hepatocytes express transaminases. Upon necrosis of these cell types, the associated loss of membrane integrity is reflected in increased serum levels of these proteins, which serve as biomarkers used clinically to assess and quantify tissue damage. Cardiac-specific troponins can be detected in the blood as early as 2 hours after myocardial cell necrosis, well before histologic evidence of myocardial infarction becomes apparent. Because of their sensitivity and specificity, serial measurement of serum cardiac troponins has a central role in the diagnosis and management of patients with myocardial infarction (Chapter 12).

MORPHOLOGY

Necrotic cells show **eosinophilia** in H&E stains, attributable in part to the loss of cytoplasmic RNA and in part to accumulation of denatured cytoplasmic proteins (which bind the red dye eosin). The necrotic cell may have a glassy homogeneous appearance relative to normal cells, mainly as a result of the loss of glycogen particles (see Fig. 2.8C). When enzymes have digested the cell's organelles, the cytoplasm becomes vacuolated and appears moth-eaten. Dead cells may be replaced by large whorled phospholipid precipitates called **myelin figures**, which are either phagocytosed by other cells or further degraded into fatty acids; calcification of such fatty acid residues results in

deposition of calcium-rich precipitates. Using electron microscopy, necrotic cells are characterized by discontinuities in plasma and organelle membranes, marked dilation of mitochondria with the appearance of large amorphous densities, intracytoplasmic myelin figures, amorphous debris, and aggregates of fluffy material representing denatured protein (see Fig. 2.9C).

Nuclear changes appear in one of three patterns, all due to the breakdown of DNA. The basophilia of the chromatin may fade (**karyolysis**), a change that presumably reflects loss of DNA because of enzymatic degradation by endonucleases. A second pattern (also seen in apoptotic cell death) is **pyknosis**, characterized by nuclear shrinkage and increased basophilia. Here the chromatin condenses into a dense, shrunken basophilic mass. In the third pattern, known as **karyorrhexis**, the pyknotic nucleus undergoes fragmentation. With the passage of time (1 or 2 days), the nucleus in the necrotic cell totally disappears.

It is useful to consider the possible events that determine when reversible injury becomes irreversible and progresses to necrosis. The clinical relevance of this question is obvious—if we can answer it, we may be able to devise strategies for preventing cell injury from having permanent deleterious consequences. Although the “point of no return,” at which the damage becomes irreversible, is still largely undefined, *two phenomena consistently characterize irreversibility—the inability to reverse mitochondrial dysfunction* (lack of oxidative phosphorylation and ATP generation) even after resolution of the original injury and *profound disturbances in membrane function*. As mentioned earlier, injury to lysosomal membranes results in the enzymatic dissolution of the injured cell that is characteristic of necrosis.

Patterns of Tissue Necrosis

The discussion of necrosis has focused so far on changes in individual cells. When large numbers of cells die, the tissue or organ is said to be necrotic; thus, a myocardial infarct is necrosis of a portion of the heart caused by death of many myocardial cells. Necrosis of tissues has several morphologically distinct patterns, which are important to recognize because they provide clues about the underlying cause. Although the terms describing these patterns are somewhat dated, they are often used and their implications are understood by pathologists and clinicians.

MORPHOLOGY

Coagulative necrosis is a form of necrosis in which the architecture of dead tissue is preserved for a span of at least some days (Fig. 2.10). The affected tissue has a firm texture. Presumably, the injury denatures not only structural proteins but also enzymes and so blocks the proteolysis of the dead cells; as a result, intensely eosinophilic cells with indistinct or reddish nuclei may persist for days or weeks. Ultimately, the necrotic cells are broken down by the action of lysosomal enzymes derived from infiltrating leukocytes, which also remove the debris of the dead cells by phagocytosis. Ischemia caused by obstruction in a vessel may lead to coagulative necrosis of the supplied tissue in all organs except the brain (see next paragraph for explanation). A localized area of coagulative necrosis is called an **infarct**.

Liquefactive necrosis, in contrast to coagulative necrosis, is characterized by digestion of the dead cells, resulting in transformation of the tissue into a viscous liquid. It is seen in focal bacterial or, occasionally, fungal infections, because microbes stimulate the accumulation of leukocytes and the liberation of enzymes from these cells. The necrotic material is frequently creamy yellow because of the presence of leukocytes and is called **pus**. For unknown reasons, hypoxic death of cells within the central nervous system often manifests as liquefactive necrosis (Fig. 2.11).

Gangrenous necrosis is not a specific pattern of cell death, but the term is commonly used in clinical practice. It is usually applied to a limb, generally

the lower leg, that has lost its blood supply and has undergone necrosis (typically coagulative necrosis) involving multiple tissue planes. When bacterial infection is superimposed, there is increased liquefactive necrosis because of the actions of degradative enzymes in the bacteria and the attracted leukocytes (giving rise to so-called **wet gangrene**).

Caseous necrosis is encountered most often in foci of tuberculous infection (Chapter 8). The term *caseous* (cheeselike) is derived from the friable white appearance of the area of necrosis (Fig. 2.12). On microscopic examination, the necrotic area appears as a structureless collection of fragmented or lysed cells and amorphous granular debris enclosed within a distinctive collection of macrophages called epithelioid cells; this appearance is characteristic of a focus of inflammation known as a **granuloma** (Chapter 3).

Fat necrosis refers to focal areas of fat destruction, typically resulting from release of activated pancreatic lipases into the substance of the pancreas and the peritoneal cavity. This occurs in the calamitous abdominal emergency known as acute pancreatitis (Chapter 19). In this disorder, pancreatic enzymes leak out of damaged acinar cells and liquefy the membranes of fat cells in the peritoneum, releasing triglyceride esters that are split by pancreatic lipases. Fatty acids are generated that combine with calcium to produce grossly visible chalky-white areas (**fat saponification**), which enable the surgeon and the pathologist to identify the underlying disorder (Fig. 2.13). On histologic examination, the necrotic areas contain the shadowy outlines of necrotic fat cells, basophilic calcium deposits, and an inflammatory reaction.

Fibrinoid necrosis is a special form of vascular damage usually seen in immune reactions involving blood vessels (Chapter 11). It typically occurs when complexes of antigens and antibodies are deposited in the walls of arteries. Deposits of these immune complexes, together with plasma proteins that have leaked out of vessels, result in a bright pink, amorphous appearance in H&E stains called “fibrinoid” (fibrin-like) by pathologists (Fig. 2.14).

Ultimately, in the living patient most necrotic cells and their contents disappear due to enzymatic digestion and phagocytosis of the debris by leukocytes. If necrotic cells and cellular debris are not promptly destroyed and reabsorbed, they provide a nidus for the deposition of calcium salts and other minerals and thus tend to become calcified. This phenomenon, called *dystrophic calcification*, is considered later in the chapter.

KEY CONCEPTS

Cell Injury and Necrosis

- Exposure of cells to stress or noxious agents causes cell injury that is reversible to a point but may progress to death of the cells, principally by necrosis.
- **Reversible cell injury:** Characterized by cellular swelling, fatty change, plasma membrane blebbing and loss of microvilli, mitochondrial swelling, dilation of the ER, and eosinophilia (due to decreased cytoplasmic RNA).
- **Necrosis:** A pathologic process in which cellular membranes are destroyed, enzymes and other constituents leak out, and local inflammation is induced to clear the damaged cells. Morphologic features are eosinophilia; nuclear shrinkage, fragmentation, and dissolution; breakdown of plasma membrane and organellar membranes; abundant myelin figures; and leakage and enzymatic digestion of cellular contents.
- **Patterns of tissue necrosis:** Under different conditions, necrosis in tissues may assume specific patterns: coagulative, liquefactive, gangrenous, caseous, fat, and fibrinoid.

Apoptosis

Apoptosis is a type of cell death that is induced by a tightly regulated suicide program in which cells destined to die activate intrinsic enzymes that degrade cellular DNA and nuclear and cytoplasmic proteins. Apoptotic cells break up into plasma membrane-bound fragments, called *apoptotic bodies*, which contain portions of the cytoplasm and nucleus. While the plasma membrane remains intact, its surface components are altered so as to produce “find me” and “eat me” signals for phagocytes, discussed later. As a result, the dead cell and its fragments are rapidly devoured, before the contents leak out, and therefore apoptosis does not elicit an inflammatory reaction. Apoptosis was first recognized in 1972 by the distinctive morphologic appearance of membrane-bound fragments derived from cells and named after the Greek designation for “falling off.” It was subsequently discovered in model organisms such as worms that certain cells undergo apoptosis at precise times during development. This phenomenon, termed *programmed cell death*, is controlled by the action of a small number of genes and is required for normal embryogenesis. Thus, apoptosis is a unique mechanism of cell death, distinct from necrosis in many respects (see Fig. 2.7 and Table 2.2).

Causes of Apoptosis

Apoptosis occurs in two broad contexts: as part of normal physiologic processes, and as a pathophysiologic mechanism of cell loss in many different diseases.

Apoptosis in Physiologic Situations. Death by apoptosis is a normal phenomenon that serves to eliminate cells that are no longer needed, or as a mechanism to maintain a constant number of various cell populations in tissues. It is estimated that humans turn over almost 1 million cells per second! Central to this process is death of cells by apoptosis and their removal by phagocytes. Apoptosis is important in the following physiologic situations:

- *The removal of supernumerary cells (in excess of the required number) during development.* Cell death is critical for involution of primordial structures and remodeling of maturing tissues. Apoptosis is a generic term for this pattern of cell death, regardless of the context, while programmed cell death refers only to apoptosis during development.
- *Involution of hormone-dependent tissues on hormone withdrawal,* such as endometrial cell breakdown during the menstrual cycle, ovarian follicular atresia in menopause, and regression of the lactating breast after weaning.
- *Cell turnover in proliferating cells populations* such as epithelial cells in intestinal crypts to maintain constant cell numbers (*homeostasis*).
- *Elimination of lymphocytes that fail to produce functional antigens receptors,* such as immature lymphocytes in the bone marrow and thymus and germinal center B cells.
- *Elimination of potentially harmful self-reactive lymphocytes* to prevent immune reactions against one’s own tissues (Chapter 6).
- Death of host cells that have served their useful purpose, such as neutrophils in an *acute inflammatory response*, and lymphocytes at the end of an *immune response*.

In all of these situations, cells undergo apoptosis because they are deprived of necessary survival signals, such as growth factors and interactions with the extracellular matrix, or they receive pro-apoptotic signals from other cells or the surrounding environment.

Apoptosis in Pathologic Conditions. Apoptosis eliminates cells that are injured beyond repair without eliciting a host reaction, thus limiting collateral tissue damage. Death by apoptosis is responsible for loss of cells in a variety of pathologic states:

- **DNA damage.** Radiation and cytotoxic anticancer drugs can damage DNA, either directly or via production of free radicals.

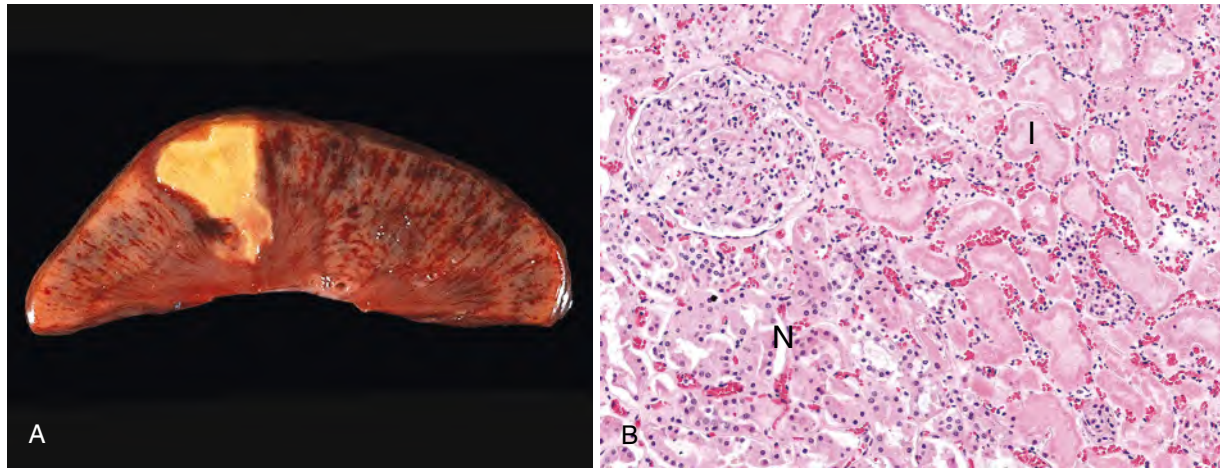


Fig. 2.10 Coagulative necrosis. (A) A wedge-shaped kidney infarct (yellow). (B) Microscopic view of the edge of the infarct, with healthy kidney (N) and necrotic cells in the infarct (I) showing preserved cellular outlines with loss of nuclei and an inflammatory infiltrate (seen as nuclei of inflammatory cells in between necrotic tubules).

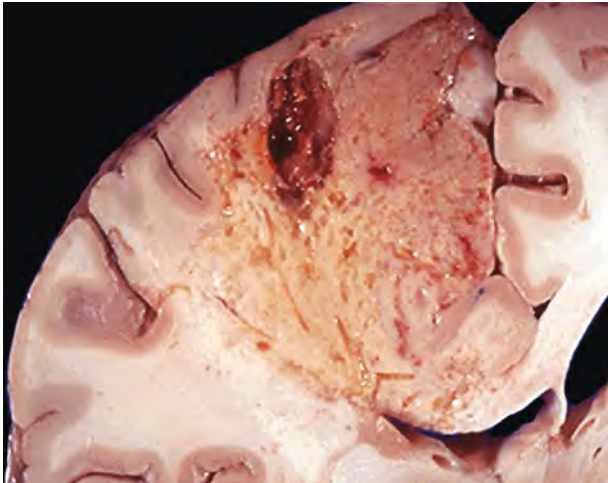


Fig. 2.11 Liquefactive necrosis. An infarct in the brain showing dissolution of the tissue.

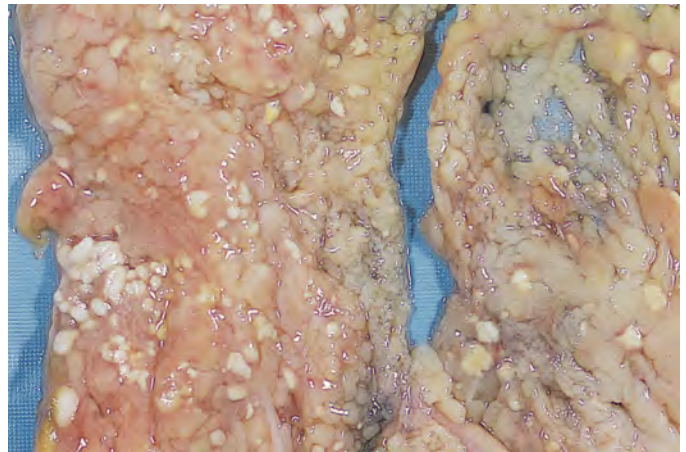


Fig. 2.13 Fat necrosis. The areas of white chalky deposits represent foci of fat necrosis with calcium soap formation (saponification) at sites of lipid breakdown in the mesentery.

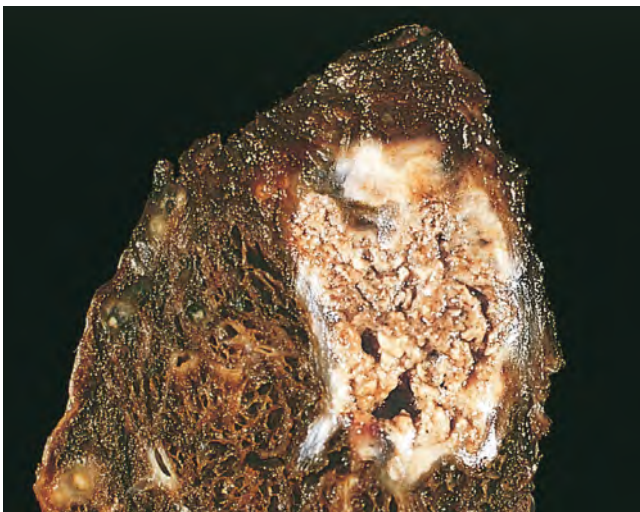


Fig. 2.12 Caseous necrosis. Tuberculosis of the lung, with a large area of caseous necrosis containing yellow-white and "cheesy" appearing debris.

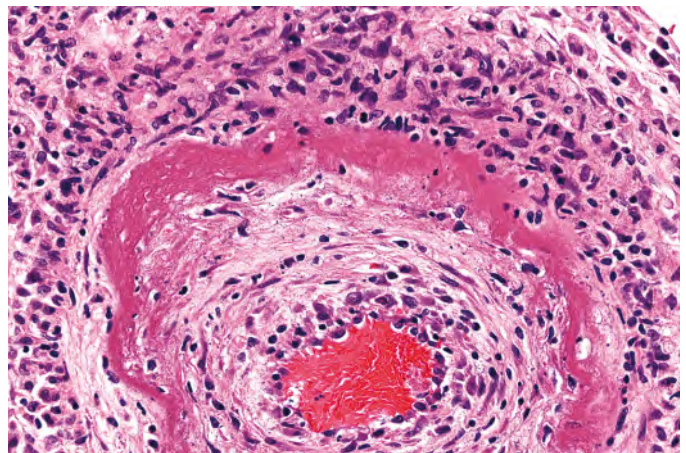


Fig. 2.14 Fibrinoid necrosis in an artery. The wall of the artery shows a circumferential bright pink area of necrosis with inflammation (neutrophils with dark nuclei).

If repair mechanisms cannot correct the damage, the cell triggers intrinsic mechanisms that induce apoptosis. In these situations, apoptosis has a protective effect by preventing the survival of cells with DNA mutations that can lead to malignant transformation.

- **Accumulation of misfolded proteins.** Cell death triggered by improperly folded intracellular proteins and the subsequent ER stress response as discussed earlier.
- Apoptosis can be induced during certain *infections*, particularly viral infections, because of the virus itself (as in adenovirus and HIV infections) or the host immune response (as in viral hepatitis). An important host response to viruses consists of cytotoxic T lymphocytes (CTLs) specific for viral proteins, which induce apoptosis of infected cells in an attempt to eliminate reservoirs of infection. During this process, there can be significant tissue damage. The same CTL-mediated mechanism is responsible for killing tumor cells, cellular rejection of transplants, and tissue damage in graft-versus-host disease.
- Apoptosis may also contribute to *pathologic atrophy in parenchymal organs after duct obstruction*, such as occurs in the pancreas, parotid gland, and kidney.

Morphologic and Biochemical Changes in Apoptosis

Before discussing underlying mechanisms, the morphologic and biochemical characteristics of apoptosis are described.

MORPHOLOGY

The following morphologic features, some best seen with the electron microscope, characterize cells undergoing apoptosis (Fig. 2.15; also see Fig. 2.7).

- **Cell shrinkage.** Cell size is reduced, the cytoplasm is dense and eosinophilic (see Fig. 2.15A), and the organelles, although relatively normal, are more tightly packed. This contrasts with necrosis, in which an early feature is cell swelling, not shrinkage.
- **Chromatin condensation.** This is the most characteristic feature of apoptosis. The chromatin aggregates peripherally, under the nuclear membrane, into dense masses of various shapes and sizes (see Fig. 2.15B). The nucleus itself may break up into two or more fragments.
- **Formation of cytoplasmic blebs and apoptotic bodies.** The apoptotic cell first shows extensive surface membrane blebbing, which is followed by fragmentation of the dead cells into membrane-bound apoptotic bodies composed of cytoplasm and tightly packed organelles, with or without nuclear fragments (see Fig. 2.15C).
- **Phagocytosis of apoptotic cells or cell bodies, usually by macrophages.** The apoptotic bodies are rapidly ingested by phagocytes and degraded by the phagocyte's lysosomal enzymes.

In H&E-stained tissue, the apoptotic cell appears as a round or oval mass of intensely eosinophilic cytoplasm with fragments of dense nuclear chromatin (see Fig. 2.15A). Because the cell shrinkage and formation of apoptotic bodies are rapid and the pieces are quickly cleared by phagocytes, considerable apoptosis may occur in tissues before it is apparent in histologic sections. The absence of an inflammatory response can also make it difficult to detect apoptosis by light microscopy.

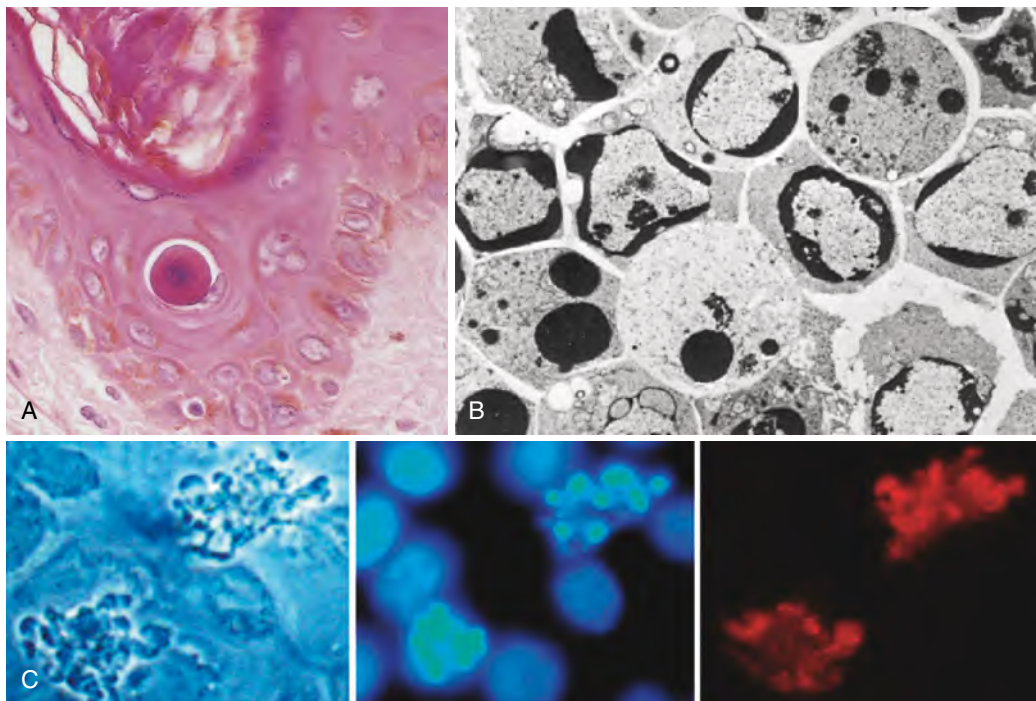


Fig. 2.15 Morphologic features of apoptosis. (A) Apoptosis of an epidermal cell. The cell is reduced in size and contains brightly eosinophilic cytoplasm and a condensed nucleus. (B) Electron micrograph of cultured cells undergoing apoptosis showing some nuclei with peripheral crescents of compacted chromatin, and others that are uniformly dense or fragmented. (C) Images of cultured cells undergoing apoptosis show blebbing and formation of apoptotic bodies (*left*, phase contrast micrograph), a stain for DNA showing nuclear fragmentation (*middle*), and activation of caspase-3 (*right*) (immunofluorescence stain with an antibody specific for the active form of caspase-3, revealed as red color). (B, From Kerr JFR, Harmon BV. Definition and incidence of apoptosis: a historical perspective. In Tomei LD, Cope FO, eds. *Apoptosis: The Molecular Basis of Cell Death*. Cold Spring Harbor Laboratory Press; 1991. C, Courtesy Dr. Zheng Dong, Medical College of Georgia.)

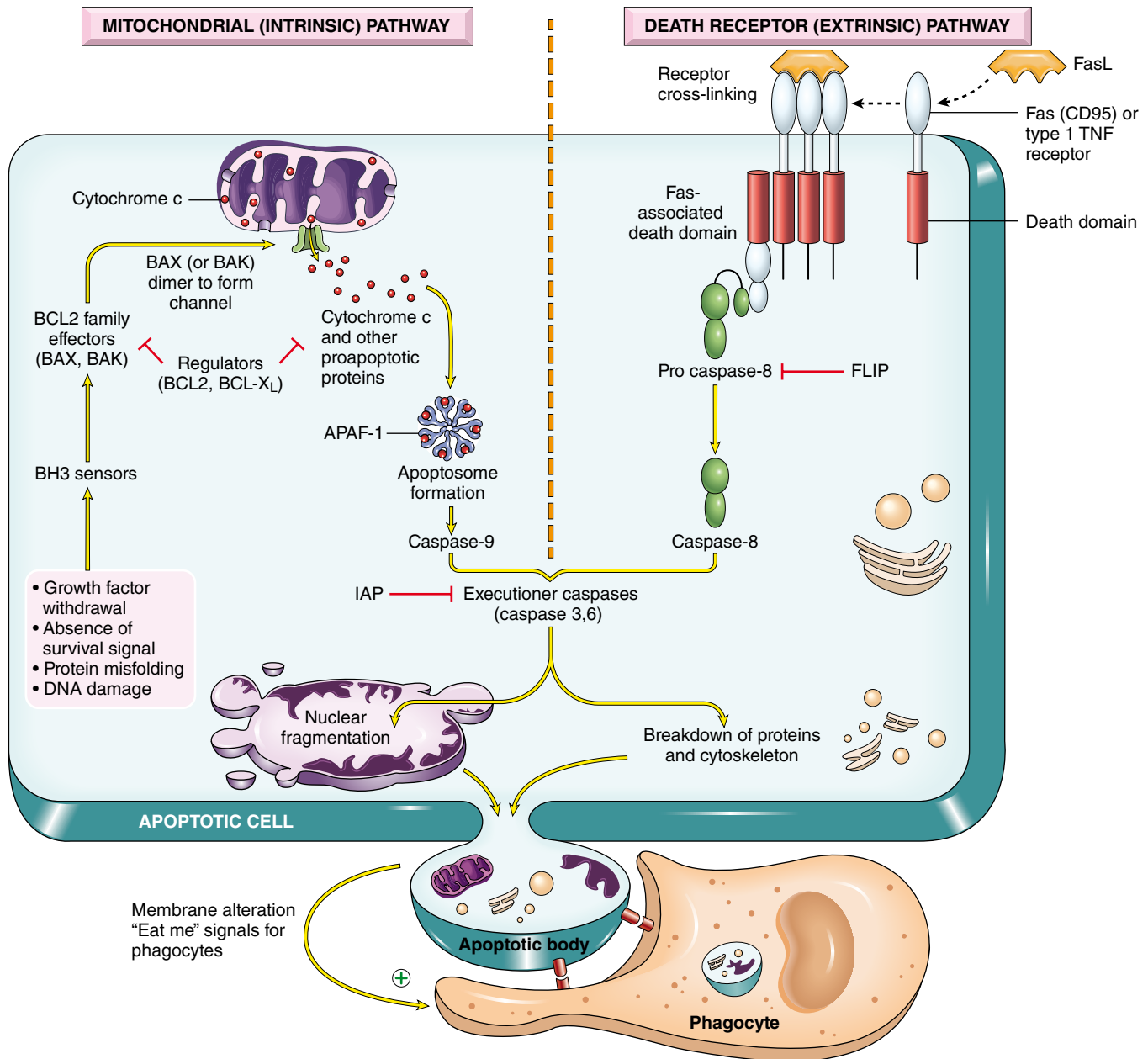


Fig. 2.16 Mechanisms of apoptosis. While the mitochondrial and death receptor pathways of apoptosis differ in their induction and regulation, they both culminate in the activation of caspases. In the mitochondrial or intrinsic pathway, anti-apoptotic BCL2 proteins maintain the integrity of mitochondrial membranes. Loss of survival signals, DNA damage, and other insults activate sensors (BH3-only family proteins) that antagonize anti-apoptotic BCL2. This results in the activation of the pro-apoptotic proteins BAX and BAK, which form channels in the mitochondrial membrane and elicit the leakage of mitochondria proteins including cytochrome c, which leads to apoptosome formation and executioner caspase activation. In the death receptor or extrinsic pathway, ligand engagement of plasma membrane death receptors of the TNF receptor superfamily, such as Fas, lead to the assembly of adaptor proteins into a "death-inducing signaling complex," which activates caspase-8 and the downstream executioner caspase-3 and -6. *FADD*, Fas-associated death domain; *FasL*, Fas ligand; *TNF*, tumor necrosis factor.

Mechanisms of Apoptosis

Apoptosis results from the activation of enzymes called **caspases** (so named because they are proteases containing a cysteine in their active site that cleave proteins after aspartic residues). Like many proteases, caspases exist as inactive proenzymes and must undergo enzymatic cleavage to become active. The presence of active caspases is therefore a marker for cells undergoing apoptosis (see Fig. 2.15C). The process of apoptosis may be divided into an *initiation phase*, during which

some caspases become catalytically active and unleash a cascade of other caspases, and an *execution phase*, during which the terminal caspases trigger cellular fragmentation and demise. Regulation of these enzymes depends on a finely tuned balance between the abundance and activity of pro-apoptotic and anti-apoptotic proteins.

Two distinct pathways converge on caspase activation: the mitochondrial pathway and the death receptor pathway (Fig. 2.16). Although these pathways intersect, they are generally induced under

different conditions, involve different initiating molecules, and serve distinct roles in physiology and disease.

Mitochondrial (Intrinsic) Pathway of Apoptosis. The mitochondrial pathway is responsible for apoptosis in most physiologic and pathologic situations. It results from increased permeability of the mitochondrial outer membrane with consequent release of death-inducing (pro-apoptotic) molecules from the mitochondrial intermembrane space into the cytoplasm (see Fig. 2.16). Mitochondria are organelles that contain remarkable proteins such as cytochrome c, a double-edged sword that is essential for producing the energy (e.g., ATP) that sustains cell viability, but that when released into the cytoplasm (an indication that the cell is not healthy) initiates the suicide program of apoptosis. The release of pro-apoptotic proteins such as cytochrome c is determined by the integrity of the outer mitochondrial membrane, which is tightly controlled by the BCL2 family of proteins. This family is named after *BCL2*, a gene that is frequently overexpressed due to chromosomal translocations and other aberrations in certain B cell lymphomas (Chapter 13). There are more than 20 members of the BCL family, which can be divided into three groups based on their pro-apoptotic or anti-apoptotic functions and the BCL2 homology (BH) domains they possess.

- **Anti-apoptotic.** BCL2, BCL-X_L, and MCL1 are the principal members of this group; they possess four BH domains (called BH1–4). These proteins reside in the outer mitochondrial membrane as well as in the cytosol and ER membranes. By keeping the mitochondrial outer membrane impermeable, they prevent leakage of cytochrome c and other death-inducing proteins into the cytosol (see Fig. 2.16).
- **Pro-apoptotic.** BAX and BAK are the two prototypic members of this group; they contain the first three BH domains (BH1–3). On activation, BAX and/or BAK oligomerize within the outer mitochondrial membrane and enhance its permeability. The precise mechanism by which BAX-BAK oligomers permeabilize membranes is not settled. According to one model illustrated in Fig. 2.16, they form a channel in the outer mitochondrial membrane that allows cytochrome c leakage from the intermembranous space.
- **Regulated apoptosis initiators.** Members of this group, including BAD, BIM, BID, Puma, and Noxa, contain only one BH domain, the third of the four BH domains, and hence, they are called *BH3-only proteins*. The activity of BH3-only proteins is modulated by sensors of cellular stress and damage; when upregulated and activated, they can initiate apoptosis.

Growth factors and other survival signals stimulate the production of anti-apoptotic proteins such as BCL2, thus protecting cells from apoptosis. When cells are deprived of survival signals, suffer DNA damage, or develop ER stress due to the accumulation of misfolded proteins, BH3-only proteins are upregulated through increased transcription and/or posttranslational modifications (e.g., phosphorylation). These BH3-only proteins in turn directly activate the two critical pro-apoptotic family members, BAX and BAK, which form oligomers that insert into the mitochondrial membrane and allow proteins from the inner mitochondrial membrane to leak out into the cytoplasm. BH3-only proteins may also bind and block the function of BCL2 and BCL-X_L. At the same time, synthesis of BCL2 and BCL-X_L may decline because their transcription relies on survival signals. The net result of BAX-BAK activation coupled with loss of the protective functions of the anti-apoptotic BCL2 family members is the release of mitochondrial proteins such as cytochrome c into the cytoplasm that can activate the caspase cascade (see Fig. 2.16).

Once released into the cytosol, cytochrome c binds to a protein called APAF-1 (apoptosis-activating factor-1), forming a multimeric

structure called the *apoptosome*. This complex binds to caspase-9, the critical initiator caspase of the mitochondrial pathway, and promotes its autocatalytic cleavage, generating catalytically active forms of the enzyme. Active caspase-9 then triggers a cascade of caspase activation by cleaving and thereby activating other pro-caspases (such as caspase-3), which mediate the execution phase of apoptosis (discussed later). Other mitochondrial proteins with arcane names like Smac/DIABLO enter the cytoplasm, where they bind to and neutralize cytoplasmic proteins that function as physiologic *inhibitors of apoptosis* (IAPs). The normal function of the IAPs is to block the inappropriate activation of caspases, including executioners like caspase-3, and keep cells alive. Thus, IAP inhibition permits initiation of the caspase cascade.

Extrinsic (Death Receptor–Initiated) Pathway of Apoptosis.

This pathway is initiated by engagement of plasma membrane death receptors. Death receptors are members of the tumor necrosis factor (TNF) receptor family that contain a cytoplasmic domain involved in protein-protein interactions. This *death domain* is essential for delivering apoptotic signals. (Some TNF receptor family members do not contain cytoplasmic death domains; their function is to activate inflammatory cascades [Chapter 3], and their role in triggering apoptosis is much less established.) The best-known death receptors are the type 1 TNF receptor (TNFR1) and a related protein called Fas (CD95), but several others have been described. The mechanism of apoptosis induced by these death receptors is well illustrated by Fas, a death receptor expressed on many cell types (Fig. 2.16). The ligand for Fas is called Fas ligand (FasL). FasL is expressed on T cells that recognize self-antigens (and functions to eliminate self-reactive lymphocytes that also express the receptor Fas upon recognition of self-antigens) and on some CTLs that kill virus-infected and tumor cells. When FasL binds to Fas, three or more molecules of Fas are brought together, and their cytoplasmic death domains form a binding site for an adaptor protein called FADD (*Fas-associated death domain*). Once attached to this complex, FADD binds inactive caspase-8 (or caspase-10), bringing together multiple caspase molecules and leading to autocatalytic cleavage and generation of active caspase-8. In turn, active caspase-8 initiates the same executioner caspase sequence as in the mitochondrial pathway. This extrinsic apoptosis pathway can be inhibited by a protein called FLIP, which binds to pro-caspase-8, thereby blocking FADD binding. Some viruses and normal cells produce FLIP as a mechanism to protect themselves from Fas-mediated apoptosis.

The extrinsic and intrinsic pathways of apoptosis are initiated in fundamentally different ways by distinct molecules, but there may be interconnections between them. For instance, in hepatocytes and pancreatic β cells, caspase-8 produced by Fas signaling cleaves and activates the BH3-only protein BID, which then feeds into the mitochondrial pathway. The combined activation of both pathways delivers a fatal blow to the cells.

Execution Phase of Apoptosis. The intrinsic and extrinsic pathways converge to activate a caspase cascade that mediates the final phase of apoptosis. The intrinsic mitochondrial pathway activates the initiator caspase-9, whereas the extrinsic death receptor pathway activates caspase-8 and caspase-10. The active forms of these caspases trigger the rapid and sequential activation of the executioner caspases, such as caspase-3 and caspase-6, which then act on many cellular components (Fig. 2.16). For instance, once activated these caspases cleave an inhibitor of a DNase, making the DNase enzymatically active and allowing DNA degradation to commence. Caspases also proteolyze structural components of the nuclear matrix and thus promote fragmentation of nuclei. Other steps in apoptosis are less well-defined. For instance, we do not know how membrane blebs and apoptotic bodies are formed.

Removal of Dead Cells. The formation of apoptotic bodies breaks cells up into “bite-sized” fragments that are edible for phagocytes. Apoptotic cells and their fragments also undergo several changes in their membranes that actively promote their phagocytosis so they are most often cleared before they lose membrane integrity and release their cellular contents (Fig. 2.16). In healthy cells, phosphatidylserine is present on the inner leaflet of the plasma membrane, but in apoptotic cells this phospholipid “flips” out and is expressed on the outer layer of the membrane, where it is recognized by several macrophage receptors. Cells that are dying by apoptosis also secrete soluble factors that recruit phagocytes, and macrophages themselves may produce proteins that bind to apoptotic cells (but not live cells), leading to their engulfment. Apoptotic bodies may also become coated with natural antibodies and proteins of the complement system, notably C1q, which are recognized by phagocytes. Thus, numerous ligands induced on apoptotic cells serve as “eat me” signals and are recognized by receptors on phagocytes that bind and engulf these cells. This process of apoptotic cell phagocytosis is called *efferyocytosis*; it is so efficient that dead cells disappear, often within minutes, without leaving a trace. In addition, production of proinflammatory cytokines is reduced in macrophages that have ingested apoptotic cells. Together with rapid clearance, this limits inflammatory reactions, even in the face of extensive apoptosis.

KEY CONCEPTS

Apoptosis

- Regulated mechanism of cell death that serves to eliminate unwanted and irreparably damaged cells, with the least possible host reaction.
- Characterized by enzymatic degradation of proteins and DNA, initiated by caspases, and by recognition and removal of dead cells by phagocytes.
- Initiated by two major pathways:
 - The mitochondrial (intrinsic) pathway is triggered by loss of survival signals, DNA damage, and accumulation of misfolded proteins (ER stress), all of which lead to leakage of pro-apoptotic proteins from the mitochondrial membrane into the cytoplasm and subsequent caspase activation; inhibited by anti-apoptotic members of the BCL2 family, which are induced by survival signals including growth factors.
 - The death receptor (extrinsic) pathway eliminates self-reactive lymphocytes and is a mechanism of cell killing by cytotoxic T lymphocytes. It is initiated by engagement of death receptors (members of the TNF receptor family). The responsible ligands can be soluble or expressed on the surface of adjacent cells.

Other Mechanisms of Cell Death

Although necrosis and apoptosis are the best-defined mechanisms of cell death, several other ways by which cells die have been described. Their importance in human diseases remains a topic of investigation, but students should be aware of these processes and their unique features.

Necroptosis

As the name indicates, this form of cell death is a hybrid that shares aspects of necrosis and apoptosis. Morphologically, and to some extent biochemically, it resembles necrosis, as both are characterized by loss of ATP, swelling of the cell and organelles, generation of reactive oxygen species (ROS), release of lysosomal enzymes, and ultimately rupture of the plasma membrane. Mechanistically, it is triggered by signal transduction pathways that culminate in cell death, a feature similar to apoptosis. Because of these overlapping features, necroptosis is sometimes called *programmed necrosis* to distinguish it from forms

of necrosis driven passively by toxic or ischemic injury to the cell. In sharp contrast to apoptosis, the signals leading to necroptosis do not result in caspase activation, and hence it is also sometimes referred to as “caspase-independent” programmed cell death. The process of necroptosis starts in a manner similar to that of the extrinsic form of apoptosis, that is, by ligation of a receptor by its ligand. Ligation of TNFR1 is the most widely studied model of necroptosis, but many other signals, including ligation of Fas and yet to be identified sensors of viral DNA and RNA, can also trigger necroptosis. Since TNF can cause both apoptosis and necroptosis, the mechanisms underlying these effects of TNF are especially illustrative (Fig. 2.17).

Although the entire set of signaling molecules and their interactions are not known, necroptosis involves two kinases called *receptor-interacting protein kinase 1* and 3 (RIPK1 and RIPK3). As indicated in Fig. 2.17, ligation of TNFR1 recruits these kinases into a multiprotein complex, and RIPK3 phosphorylates a cytoplasmic protein called MLKL. In response to its phosphorylation, MLKL monomers assemble into oligomers, translocate from the cytosol to the plasma membrane, and cause the plasma membrane disruption that is characteristic of necrosis. This explains the morphologic similarity of necroptosis with necrosis initiated by other injuries.

Necroptosis is postulated to be an important death pathway in both physiologic and pathologic conditions. For example, physiologic necroptosis occurs during the formation of the mammalian bone growth plate. In pathologic states, it is associated with cell death in

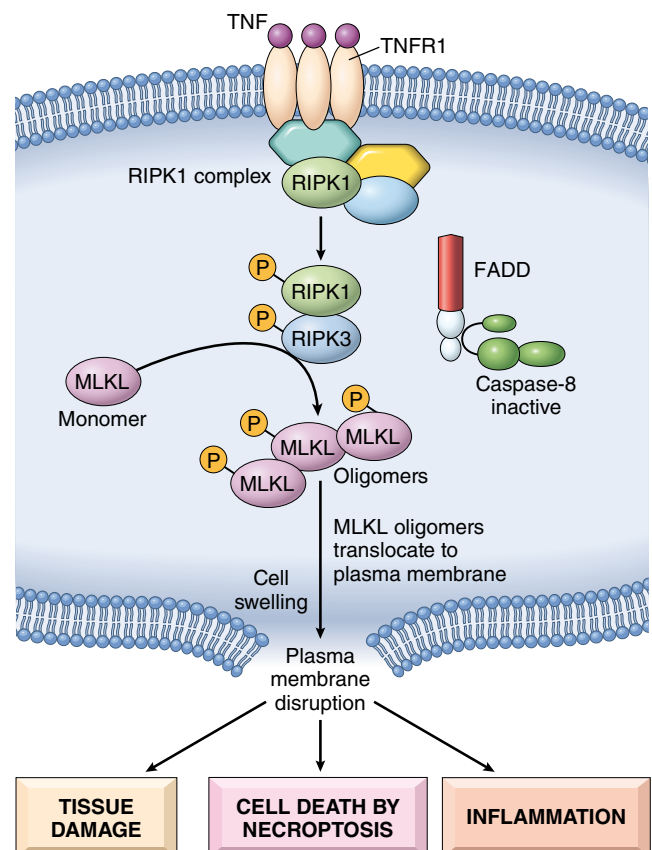


Fig. 2.17 Molecular mechanism of TNF-mediated necroptosis. When caspase-8 is inactive in a cell, the cross-linking of TNFR1 by TNF initiates the illustrated series of downstream events, which ultimately lead to plasma membrane disruption, cell death, and inflammation. See text for details. (Modified from Galluzzi L, Vanden Berghe T, Vanlangenakker N, et al. Programmed necrosis from molecules to health and disease. *Int Rev Cell Molec Biol*. 2011;289:1–35.)

steatohepatitis, acute pancreatitis, ischemia-reperfusion injury, and neurodegenerative diseases such as Parkinson disease. Necroptosis also acts as a backup mechanism in host defense against certain viruses that encode caspase inhibitors (e.g., cytomegalovirus).

Pyroptosis

Pyroptosis (“fiery death”) is a lytic form of cell death accompanied by the release of the inflammatory mediators, including the fever-inducing cytokine IL-1 (*pyro* refers to fever) (Fig. 2.18). Pyroptosis is thought to be the mechanism by which some microbes cause the death of infected cells while triggering local inflammation. Excessive pyroptosis may contribute to various inflammatory disorders, including sepsis and autoimmune disease.

Pyroptosis is activated by canonical and noncanonical pathways, both of which converge on the activation of pore-forming proteins called *gasdermins*, the best characterized of which is gasdermin D (GSDMD). In the canonical pathway, microbial products that enter

infected cells are recognized by cytoplasmic innate immune receptors that can activate the multiprotein complex called the *inflammasome* (Chapter 6). The function of the inflammasome is to activate caspase-1 (also known as interleukin-1 β -converting enzyme), which cleaves a precursor form of interleukin-1 β (IL-1 β) and releases its biologically active form. IL-1 β is a mediator of many aspects of inflammation, including leukocyte recruitment and fever (Chapter 3). Noncanonical pyroptosis is activated by intracellular lipopolysaccharide (LPS) produced from gram-negative bacteria, which results in the activation of caspases-4/5. Both caspase-1 and caspases-4/5 cleave GSDMD, which oligomerizes to form pores in the plasma membrane. These pores not only mediate the release of IL-1 β and other proinflammatory mediators, but profoundly disrupt local osmotic potential, resulting in the formation of pyroptotic bodies and eventually plasma membrane rupture and lytic cell death.

Ferroptosis

Ferroptosis is a distinct form of cell death triggered when excessive intracellular levels of iron or ROS overwhelm the glutathione-dependent antioxidant defenses (discussed later) to cause unchecked membrane lipid peroxidation. The widespread peroxidation of lipids disrupts many aspects of membrane function, including fluidity, lipid-protein interactions, ion and nutrient transport, and signaling pathways. The overall effect is the loss of plasma membrane permeability, which ultimately leads to cell death resembling necrosis. The process is, however, regulated by specific signals (unlike necrosis) and can be prevented by reducing iron levels (hence its name). Ultrastructurally, the most prominent features are the loss of mitochondrial cristae and ruptured outer mitochondrial membrane. While its role in normal development and physiology remains controversial, ferroptosis has been linked to cell death in a variety of human pathologies, including cancer, neurodegenerative diseases, and stroke.

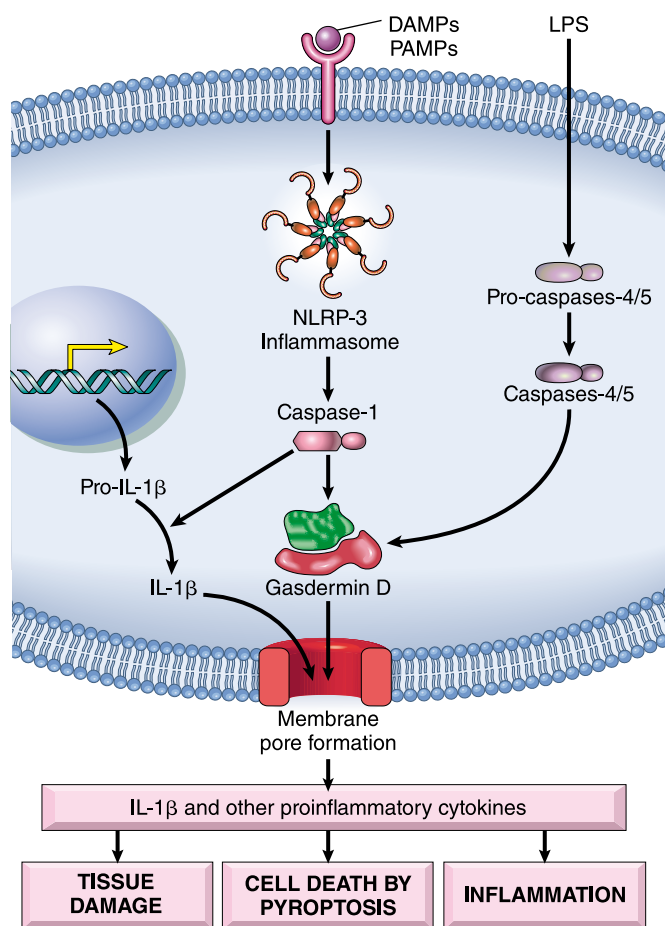


Fig. 2.18 Canonical and noncanonical pyroptosis. Canonical pyroptosis is mediated by the activation of the inflammasome by innate immune receptors, which results in caspase-1-mediated cleavage of the precursor form of IL-1 β (pro-IL-1 β) to generate mature cytokine. During noncanonical pyroptosis, lipopolysaccharide (LPS) produced by gram-negative bacteria activates caspases-4/5. Either caspase-1 or caspases-4/5 can cleave gasdermin D, which then oligomerizes to form pores in the plasma membrane. These pores mediate the release of IL-1 β and other proinflammatory mediators, as well as disrupt local osmotic potential, resulting in plasma membrane rupture and lytic cell death. DAMPs, Damage-associated molecular patterns; IL-1 β , interleukin-1 beta; PAMPs, pathogen-associated molecular patterns.

KEY CONCEPTS

Necroptosis, Pyroptosis and Ferroptosis

- **Necroptosis** resembles necrosis morphologically, but like apoptosis is a genetically controlled form of cell death.
 - Necroptosis is triggered by ligation of TNFR1 and by proteins found in RNA and DNA viruses.
 - Necroptosis is caspase independent and depends on the RIPK1–RIPK3 complex. RIPK1–RIPK3 signaling leads to the phosphorylation of MLKL, which then forms pores in the plasma membrane.
 - Release of cellular contents evokes an inflammatory reaction as in necrosis.
- **Pyroptosis** occurs in cells infected by microbes. It involves activation of caspases, which cleave and activate the pore-forming function of GSDMD, resulting in lytic death of the infected cell and the release of inflammatory mediators such as IL-1 β .
- **Ferroptosis** is an iron-dependent pathway of cell death induced by lipid peroxidation.

MECHANISMS OF CELL INJURY

The discussion of the pathways of cell death sets the stage for a consideration of the underlying biochemical mechanisms of cell injury. The molecular alterations that lead to cell injury are complex, but several principles are relevant to most forms of cell injury:

- **The cellular response to injurious stimuli depends on the nature of the injury, its duration, and its severity.** Small doses of a chemical toxin or brief periods of ischemia may induce reversible injury, whereas large doses of the same toxin or more prolonged ischemia

might result in rapid cell death or in slowly progressive injury that with time becomes irreversible and leads to cell death.

- **The consequences of cell injury depend on the type, state, and adaptability of the injured cell.** The cell's nutritional and hormonal status, metabolic demands, and functions dictate the response to injury. How vulnerable is a cell, for example, to loss of blood supply and hypoxia? When a skeletal muscle cell in the leg is deprived of its blood supply, it can be rested and preserved; in contrast, cardiac muscle cells have no such option. Exposure of two individuals to identical concentrations of a toxin, such as carbon tetrachloride (CCl_4), may produce no effect in one and cell death in the other. This may be due to polymorphisms in genes encoding hepatic enzymes that metabolize carbon tetrachloride to toxic by-products (Chapter 9). With the complete mapping of the human genome, there is great interest in identifying genetic polymorphisms that affect the responses of different individuals to various injurious agents.
- **Any injurious stimulus may simultaneously trigger multiple interconnected mechanisms that damage cells.** This is one reason why it can be difficult to ascribe cell injury in a particular situation to a single or even dominant biochemical derangement.

Cell injury results from abnormalities in one or more essential cellular components (Fig. 2.19). Below, we discuss general mechanisms involved in cell injury as well as the principal targets of injurious stimuli, which include mitochondria, cell membranes, the machinery of protein synthesis and secretion, and DNA. The consequences of injury of each of these cellular components are distinct but overlapping.

Mitochondrial Damage

Mitochondria are critical players in all pathways leading to cell injury and death. This should be expected because mitochondria supply life-sustaining energy by producing ATP and are also targets of many injurious stimuli. Thus, in many ways, they are the arbiters of life and death of cells. Mitochondria can be damaged by increases of cytosolic

Ca^{2+} , ROS (discussed later), and oxygen deprivation, which makes them sensitive to virtually all types of injurious stimuli, including hypoxia and toxins. In addition, mutations in mitochondrial genes are the cause of some inherited diseases (Chapter 5).

There are several major consequences of mitochondrial damage:

- **ATP depletion.** Decreased ATP synthesis and ATP depletion are frequently associated with hypoxic and chemical (toxic) injury (Fig. 2.20). ATP is produced in two ways. The major pathway in mammalian cells, particularly those that are nondividing (e.g., brain and liver), is *oxidative phosphorylation* of adenosine diphosphate in a reaction that results in reduction of oxygen by the mitochondrial electron transport system. The second is the *glycolytic pathway*, which can generate ATP, albeit in far smaller amounts, in the absence of oxygen using glucose derived either from body fluids or from the hydrolysis of glycogen. In addition to mitochondrial damage, the major causes of ATP depletion are reduced supply of oxygen and nutrients (because of ischemia and hypoxia), and the actions of some toxins (e.g., cyanide). Mitochondrial damage often results in the formation of a high-conductance channel in the mitochondrial membrane, called the *mitochondrial permeability transition pore* (see Fig. 2.20). The opening of this conductance channel leads to the loss of mitochondrial membrane potential, resulting in failure of oxidative phosphorylation and progressive ATP depletion that culminates in necrosis.

High-energy phosphate in the form of ATP is required for virtually all synthetic and degradative processes within the cell. These include membrane transport, protein synthesis, lipogenesis, and the deacylation-reacylation reactions necessary for phospholipid turnover. Hence, depletion of ATP to 5% to 10% of normal levels has widespread effects on many critical cellular systems.

- The activity of the *plasma membrane energy-dependent sodium pump* ($\text{Na}^+\text{K}^+\text{-ATPase}$) is reduced (Chapter 1). Failure of this active transport system causes sodium to enter and accumulate inside cells and potassium concentrations to fall. The net solute gain results in osmotically driven water accumulation that leads to *cell swelling* and ER dilation.

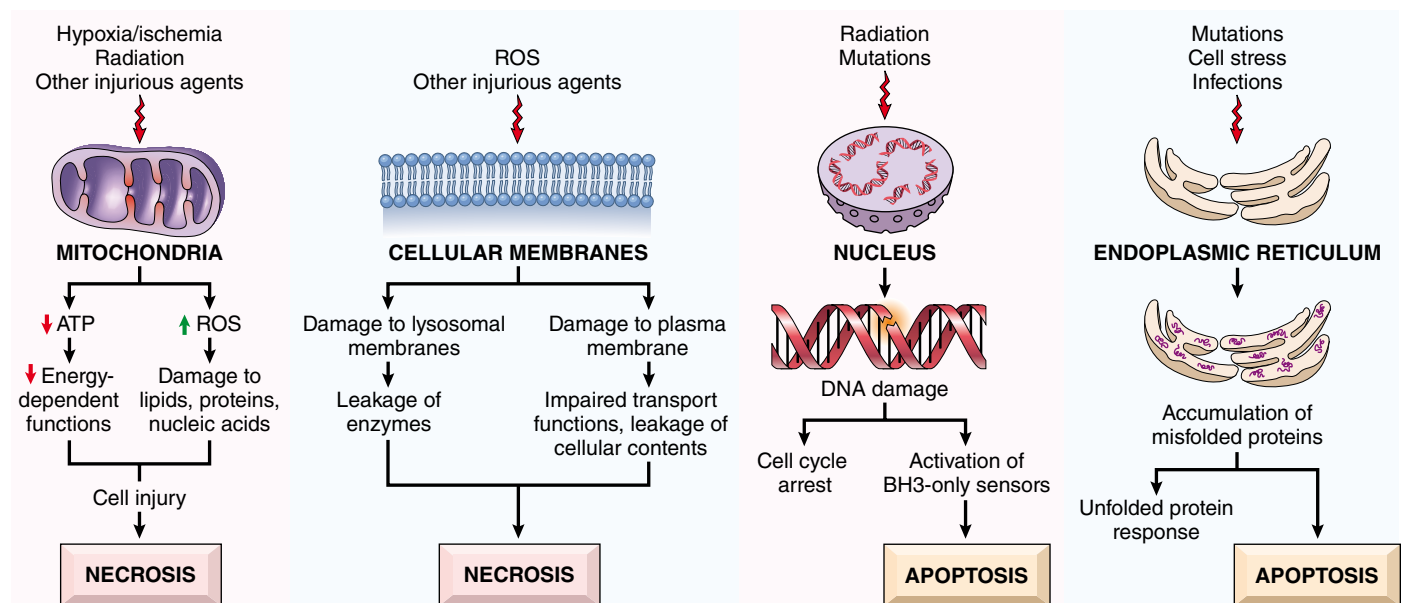


Fig. 2.19 Principal forms and sites of damage in cell injury. *ATP*, Adenosine triphosphate; *ROS*, reactive oxygen species.

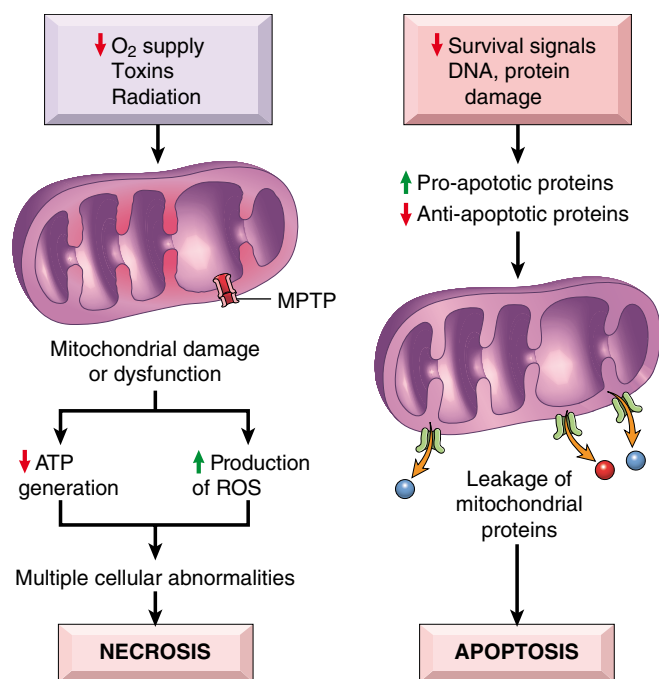


Fig. 2.20 Role of mitochondria in cell injury and death. Mitochondria are affected by a variety of injurious stimuli, and their abnormalities lead to necrosis or apoptosis. *ATP*, Adenosine triphosphate; *MPTP*, mitochondrial permeability transition pore; *ROS*, reactive oxygen species.

- **Cellular energy metabolism is altered.** If the supply of oxygen to cells is reduced, as in ischemia, oxidative phosphorylation ceases, resulting in a decrease in cellular ATP and associated increase in adenosine monophosphate. These changes stimulate phosphorylase and phosphofructokinase activities, leading to increased rates of glycogenolysis and glycolysis, respectively, in an effort to maintain energy supplies by generating ATP through metabolism of glucose derived from glycogen. Consequently, *glycogen stores are rapidly depleted*. Glycolysis under anaerobic conditions results in the accumulation of *lactic acid* and inorganic phosphates from the hydrolysis of phosphate esters. This reduces the intracellular pH, resulting in decreased activity of many cytosolic enzymes.
- With prolonged or worsening depletion of ATP, structural disruption of the protein synthetic apparatus occurs, manifest as detachment of ribosomes from the rough ER and dissociation of polysomes, with a consequent *reduction in protein synthesis*. There may also be increased protein misfolding, with injurious effects that are discussed later.
- Ultimately, there is irreversible damage to mitochondrial and lysosomal membranes, and the cell undergoes *necrosis*.
- Incomplete oxidative phosphorylation also leads to the formation of ROS, which have many deleterious effects, described later.
- As discussed earlier, leakage of mitochondrial proteins due to channel formation by pro-apoptotic BAX and BAK is the initial step in apoptosis by the intrinsic pathway. This action of BAX and BAK is specific to mitochondrial membranes and leads to damage of other organelles indirectly.

Membrane Damage

Early loss of selective membrane permeability, leading ultimately to overt membrane damage, is a consistent feature of most forms of cell

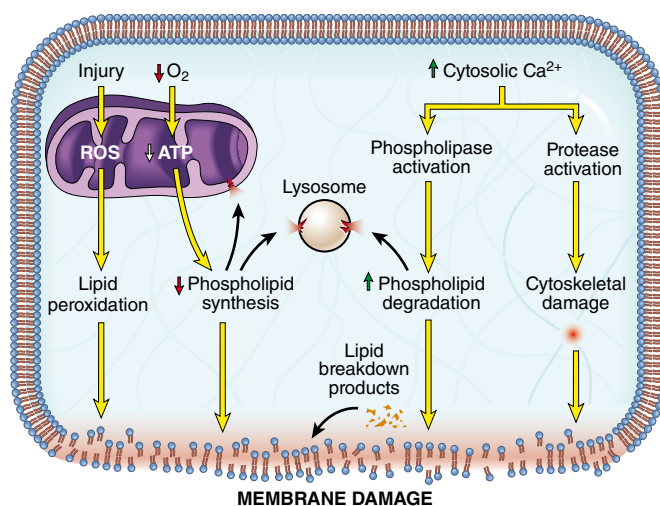


Fig. 2.21 Mechanisms of membrane damage in cell injury. Decreased O_2 and increased cytosolic Ca^{2+} are typically seen in ischemia, but may accompany other forms of cell injury. Reactive oxygen species, which are often produced on reperfusion of ischemic tissues, also cause membrane damage (not shown).

injury (except apoptosis). Membrane damage may affect the integrity and functions of all cellular membranes. In ischemic cells, membrane defects may be the result of ATP depletion and calcium-mediated activation of phospholipases. The plasma membrane can also be damaged directly by bacterial toxins, viral proteins, lytic complement components, and a variety of physical and chemical agents. Several biochemical mechanisms may contribute to membrane damage (Fig. 2.21):

- **ROS.** Oxygen free radicals cause injury to cell membranes by lipid peroxidation, discussed later.
- **Decreased phospholipid synthesis.** The production of phospholipids in cells may be reduced because of defective mitochondrial function or hypoxia, both of which decrease ATP production and thus affect energy-dependent biosynthetic pathways. Decreased phospholipid synthesis may affect all cellular membranes, including those of mitochondria.
- **Increased phospholipid breakdown.** Severe cell injury is associated with increased degradation of membrane phospholipids, probably due to activation of calcium-dependent phospholipases by increased cytosolic and mitochondrial Ca^{2+} . Phospholipid breakdown leads to the accumulation of *lipid breakdown products*, including unesterified free fatty acids, acyl carnitine, and lysophospholipids, which have a detergent effect on membranes. They may also insert into the lipid bilayer of the membrane or exchange with membrane phospholipids, potentially causing changes in permeability and electrophysiologic alterations.
- **Cytoskeletal abnormalities.** Cytoskeletal filaments serve as anchors connecting the plasma membrane to the cell interior, and proteases activated by cytosolic Ca^{2+} may damage these tethers. When exacerbated by cell swelling, particularly in myocardial cells, this damage leads to detachment of the cell membrane from the cytoskeleton, rendering it susceptible to stretching and rupture.

Damage to DNA

Damage to nuclear DNA activates sensors that trigger p53-dependent pathways (Chapter 7). DNA damage may be caused by exposure to radiation, chemotherapeutic (anticancer) drugs, and ROS, or may occur

spontaneously as a part of aging, due largely to deamination of cytosine residues to uracil residues. DNA damage activates p53, which arrests cells in the G1 phase of the cell cycle and activates DNA repair mechanisms. If these mechanisms fail to correct the DNA damage, p53 triggers apoptosis by the mitochondrial pathway. Thus, the cell chooses to die rather than survive with abnormal DNA that has the potential to induce malignant transformation. Predictably, mutations in p53 that interfere with its ability to arrest cell cycling or to induce apoptosis are associated with numerous cancers (Chapter 7).

In addition to damage to various organelles, some biochemical alterations occur in many situations that lead to cell injury. Two of these general pathways are discussed next.

Oxidative Stress: Accumulation of Oxygen-Derived Free Radicals

Cell injury induced by free radicals, particularly reactive oxygen species (ROS), is an important mechanism of cell damage in many pathologic conditions, such as chemical and radiation injury, ischemia-reperfusion injury (induced by restoration of blood flow in ischemic tissue), cellular aging, and microbial killing by phagocytes. *Free radicals* are chemical species that have a single unpaired electron in an outer orbit. Unpaired electrons are highly reactive and “attack” and modify adjacent molecules, such as organic chemicals—proteins, lipids, carbohydrates, nucleic acids—that are key components of cell membranes and nuclei. Some of these reactions are autocatalytic, whereby molecules that react with free radicals are themselves converted into free radicals, thus propagating the chain of damage.

ROS are a type of oxygen-derived free radical whose role in cell injury is well established. ROS are produced normally in cells during mitochondrial respiration and energy generation, but they are degraded and removed by intracellular ROS scavengers. These defense systems allow cells to maintain a steady state in which free radicals may be present at low concentrations but do not cause damage. Increased production or decreased scavenging of ROS may lead to an excess of free radicals, a condition called *oxidative stress*. Oxidative stress has been implicated in a wide variety of pathologic processes, including cell injury, cancer, aging, and some degenerative diseases, such as Alzheimer disease. ROS are also produced in large amounts by activated leukocytes, particularly neutrophils and macrophages, during inflammatory reactions aimed at destroying microbes and cleaning up dead cells and other unwanted substances (Chapter 3).

The following section discusses the generation and removal of ROS, and how they contribute to cell injury. The properties of some of the most important free radicals are summarized in Table 2.3.

Generation of Free Radicals

Free radicals may be generated within cells in several ways (Fig. 2.22):

- *The reduction-oxidation reactions that occur during normal metabolic processes.* As a part of normal respiration, molecular O_2 is reduced by the transfer of four electrons to H_2 to generate two water molecules. This conversion is catalyzed by oxidative enzymes in the ER, cytosol, mitochondria, peroxisomes, and lysosomes. During this process, small amounts of partially reduced intermediates are produced in which different numbers of electrons have been transferred from O_2 ; these include superoxide anion (O_2^- , one electron), hydrogen peroxide (H_2O_2 , two electrons), and hydroxyl radicals ($\cdot OH$, three electrons).
- *Absorption of radiant energy* (e.g., ultraviolet light, x-rays). For example, ionizing radiation can hydrolyze water into $\cdot OH$ and hydrogen (H) free radicals.
- *Rapid bursts of ROS are produced in activated leukocytes during inflammation.* This occurs in a precisely controlled reaction carried out by a plasma membrane multiprotein complex that uses NADPH oxidase for the redox reaction (Chapter 3). In addition, some intracellular oxidases (e.g., xanthine oxidase) generate O_2^- . Defects in leukocytic superoxide production lead to chronic granulomatous disease (Chapter 6).
- *Enzymatic metabolism of exogenous chemicals or drugs* can generate free radicals that are not ROS but have similar effects (e.g., CCl_4 can generate $\cdot CCl_3$, described later in the chapter).
- *Transition metals* such as iron and copper donate or accept free electrons during intracellular reactions and catalyze free radical formation, as in the Fenton reaction ($H_2O_2 + Fe^{2+} \rightarrow Fe^{3+} + \cdot OH + OH^-$). Because most of the intracellular free iron is in the ferric (Fe^{3+}) state, it must be reduced to the ferrous (Fe^{2+}) form to participate in the Fenton reaction. This reduction can be enhanced by O_2^- , and thus sources of iron and O_2^- may cooperate in oxidative cell damage.
- *Nitric oxide (NO)*, an important chemical mediator generated by endothelial cells, macrophages, neurons, and other cell types (Chapter 3), can act as a free radical and may also be converted to highly reactive peroxynitrite anion ($ONOO^-$) as well as NO_2 and NO_3^- .

TABLE 2.3 Properties of the Principal Free Radicals Involved in Cell Injury.

Properties	O_2^-	H_2O_2	$\cdot OH$	$ONOO^-$
Mechanisms of production	Incomplete reduction of O_2 during oxidative phosphorylation; by phagocyte oxidase in leukocytes	Generated by SOD from O_2^- and by oxidases in peroxisomes	Generated from H_2O by hydrolysis (e.g., by radiation); from H_2O_2 by Fenton reaction; from O_2^-	Produced by interaction of O_2^- and NO generated by NO synthase in many cell types (endothelial cells, leukocytes, neurons, others) No enzymatic means of inactivation
Enzymatic mechanisms of inactivation	Conversion to H_2O_2 and O_2 by SOD	Conversion to H_2O and O_2 by catalase (peroxisomes), glutathione peroxidase (cytosol, mitochondria)	None	Conversion to HNO_2 by peroxiredoxins (cytosol, mitochondria)
Pathologic effects	Stimulates production of degradative enzymes in leukocytes and other cells; may directly damage lipids, proteins, DNA; acts close to site of production	Can be converted to OH^- and OCl^- , which destroy microbes and cells; can act distant from site of production	Most reactive oxygen-derived free radical; principal ROS responsible for damaging lipids, proteins, and DNA	Damages lipids, proteins, DNA

HNO₂, Nitrite; *H₂O₂*, hydrogen peroxide; *NO*, nitric oxide; O_2^- , superoxide anion; OCl^- , hypochlorite; *OH*, hydroxyl radical; *ONOO⁻*, peroxynitrite; *ROS*, reactive oxygen species; *SOD*, superoxide dismutase.

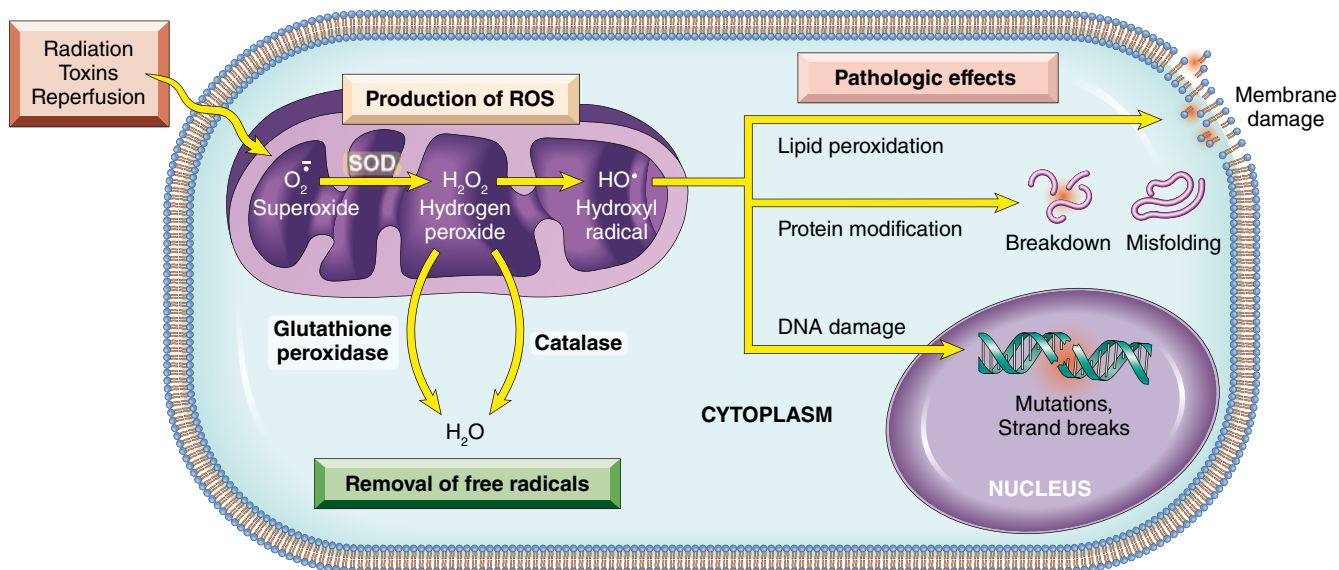


Fig. 2.22 The generation, removal, and role of reactive oxygen species (ROS) in cell injury. The production of ROS is increased by many injurious stimuli. These free radicals are removed by spontaneous decay and by specialized enzymatic systems. Excessive production or inadequate removal leads to accumulation of free radicals in cells, which may damage lipids (by peroxidation), proteins, and deoxyribonucleic acid (DNA), resulting in cell injury. SOD, Superoxide dismutase.

Removal of Free Radicals

Free radicals are inherently unstable and generally decay spontaneously. O_2^- , for example, is unstable and decays (dismutates) spontaneously to O_2 and H_2O_2 in the presence of water. In addition, cells have developed multiple nonenzymatic and enzymatic mechanisms to remove free radicals and thereby minimize injury (see Fig. 2.22). These include:

- **Antioxidants** either block free radical formation or inactivate (e.g., scavenge) free radicals. Examples are the lipid-soluble vitamins E and A as well as ascorbic acid and glutathione in the cytosol.
- As we have seen, free *iron* and *copper* can catalyze the formation of ROS. Under normal circumstances, the reactivity of these metals is minimized by their binding to storage and transport proteins (e.g., transferrin, ferritin, lactoferrin, and ceruloplasmin), which prevents these metals from participating in reactions that generate ROS.
- Several *enzymes* act as free radical-scavenging systems and break down H_2O_2 and O_2^- . These enzymes are located near the sites of generation of the oxidants and include:
 1. *Catalase*, present in peroxisomes, decomposes H_2O_2 ($2H_2O_2 \rightarrow O_2 + 2H_2O$).
 2. *Superoxide dismutases* (SODs) are found in many cell types and convert O_2^- to H_2O_2 ($2O_2^- + 2H^+ \rightarrow H_2O_2 + O_2$). This group of enzymes includes both manganese-SOD, which is localized in mitochondria, and copper-zinc-SOD, which is found in the cytoplasm.
 3. *Glutathione peroxidase* also protects against injury by catalyzing free radical breakdown ($H_2O_2 + 2GSH \rightarrow GSSG$ [oxidized glutathione] + $2H_2O$). The intracellular ratio of oxidized glutathione (GSSG) to reduced glutathione (GSH) reflects the oxidative state of the cell and is an important indicator of the cell's ability to detoxify ROS.

Pathologic Effects of Free Radicals

The effects of ROS and other free radicals are wide-ranging, but three reactions are particularly relevant to cell injury (see Fig. 2.22):

- **Lipid peroxidation in membranes.** In the presence of O_2 , free radicals may cause peroxidation of lipids within plasma and organellar

membranes. Oxidative damage is initiated when the double bonds in unsaturated fatty acids of membrane lipids are attacked by O_2 -derived free radicals, particularly by O_2^- . The lipid-free radical interactions yield peroxides, which are themselves unstable and reactive, and an autocatalytic chain reaction ensues (called *propagation*) that can result in extensive membrane damage.

- **Oxidative modification of proteins.** Free radicals promote oxidation of amino acid side chains, formation of covalent protein-protein cross-links (e.g., disulfide bonds), and oxidation of the protein backbone. Oxidative modifications may also damage the active sites of enzymes, disrupt the conformation of structural proteins, and enhance proteasomal degradation of unfolded or misfolded proteins, thereby raising havoc throughout the cell.
- **Lesions in DNA.** Free radicals can cause single- and double-strand breaks in DNA, cross-linking DNA strands, and forming adducts. Oxidative DNA damage has been implicated in cell aging (discussed later in the chapter) and in malignant transformation of cells (Chapter 7). The traditional thinking about free radicals was that they cause cell injury and death by necrosis. However, it is now clear that free radicals can also trigger apoptosis. It is also possible that these potentially deadly molecules, when produced under controlled conditions in the "right" dose, serve important physiologic functions in signaling by cellular receptors and other pathways.

Disturbance in Calcium Homeostasis

Calcium ions normally serve as second messengers in several signaling pathways, but if released into the cytoplasm of cells in excessive amounts, are also an important source of cell injury. In keeping with this, calcium depletion protects cultured cells from injury induced by a variety of harmful stimuli. Cytosolic free Ca^{2+} is normally maintained at very low concentrations ($\sim 0.1 \mu\text{mol}$) compared with extracellular levels of 1.3 mmol , and most intracellular Ca^{2+} is sequestered in mitochondria and the ER. Ischemia and certain toxins cause an excessive increase in cytosolic Ca^{2+} , initially because of release from intracellular stores, and later due to

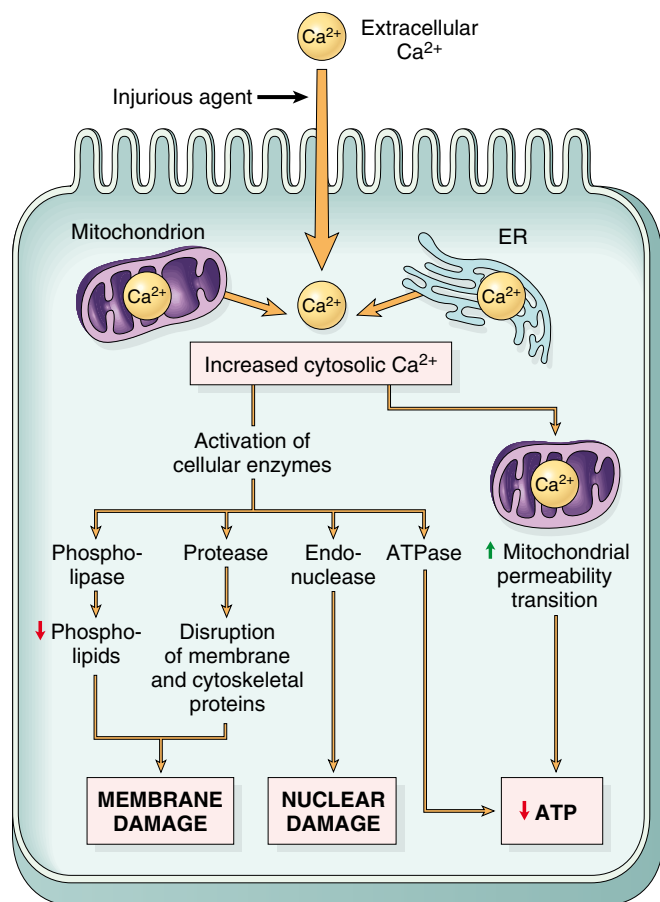


Fig. 2.23 Role of increased cytosolic calcium in cell injury. ER, Endoplasmic reticulum.

increased influx across the plasma membrane (Fig. 2.23). Excessive intracellular Ca^{2+} may cause cell injury by several mechanisms, although the significance of these mechanisms in cell injury in vivo is not established.

- The accumulation of Ca^{2+} in mitochondria results in opening of the mitochondrial permeability transition pore and, as described earlier, failure of ATP generation.
- Increased cytosolic Ca^{2+} activates several enzymes with potentially deleterious effects on cells. These enzymes include *phospholipases* (which cause membrane damage), *proteases* (which break down both membrane and cytoskeletal proteins), *endonucleases* (which are responsible for DNA and chromatin fragmentation), and *ATPases* (thereby hastening ATP depletion).

KEY CONCEPTS

Mechanisms of Cell Injury

- ATP depletion: failure of energy-dependent functions → reversible injury → necrosis
- Mitochondrial damage: ATP depletion → failure of energy-dependent cellular functions → ultimately, necrosis; under some conditions, leakage of mitochondrial proteins that cause apoptosis
- Increased permeability of cellular membranes: may affect plasma membrane, lysosomal membranes, mitochondrial membranes; typically culminates in necrosis

- Accumulation of damaged DNA and misfolded proteins: triggers apoptosis
- Accumulation of ROS: covalent modification of cellular proteins, lipids, nucleic acids
- Influx of calcium: activation of enzymes that damage cellular components and may also trigger apoptosis

CLINICOPATHOLOGIC CORRELATIONS: SELECTED EXAMPLES OF CELL INJURY AND DEATH

Having briefly reviewed the causes, morphology, response to and general mechanisms of cell injury and death, we now describe some common and clinically significant forms of cell injury. These examples illustrate many of the mechanisms and sequence of events in cell injury described earlier.

Hypoxia and Ischemia

Ischemia, a common cause of cell injury in clinical medicine, results from hypoxia induced by reduced blood flow, most often due to a mechanical arterial obstruction. It can also occur due to reduced venous drainage. In contrast to hypoxia, in which blood flow is maintained and during which energy production by anaerobic glycolysis can continue, ischemia compromises the delivery of substrates for glycolysis. Thus, in ischemic tissues, not only does aerobic metabolism cease but anaerobic energy generation also fails after glycolytic substrates are exhausted or glycolysis is inhibited by the accumulation of metabolites, which otherwise would be washed out by flowing blood. For this reason, *ischemia causes more rapid and severe cell and tissue injury than hypoxia.*

Mechanisms of Ischemic Cell Injury

The sequence of events following hypoxia or ischemia reflects many of the biochemical alterations in cell injury, summarized in Fig. 2.24 and shown earlier in Figs. 2.8 and 2.9. As intracellular oxygen tension falls, oxidative phosphorylation fails, and ATP generation decreases. The consequences of ATP depletion were described earlier in the Mitochondrial Damage section. In brief, loss of ATP results initially in reversible cell injury (cell and organelle swelling) and later in cell death by necrosis.

Mammalian cells have developed protective responses to deal with hypoxic stress. The best defined of these is induction of a transcription factor called *hypoxia-inducible factor-1* (HIF-1), which promotes new blood vessel formation, stimulates cell survival pathways, and enhances glycolysis. Several promising investigational compounds that promote HIF-1 signaling are being developed. Nevertheless, there are still no reliable therapeutic approaches for reducing the injurious consequences of ischemia in clinical situations. The strategy that is perhaps the most useful in ischemic (and traumatic) brain and spinal cord injury is the transient induction of hypothermia (lowering the core body temperature to 92°F). This treatment reduces the metabolic demands of the stressed cells, decreases cell swelling, suppresses the formation of free radicals, and inhibits the host inflammatory response. All of these may contribute to decreased cell and tissue injury.

Ischemia-Reperfusion Injury

Restoration of blood flow to ischemic tissues can promote recovery of cells if they are reversibly injured but can also paradoxically exacerbate cell injury and cause cell death. Consequently, reperfused tissues

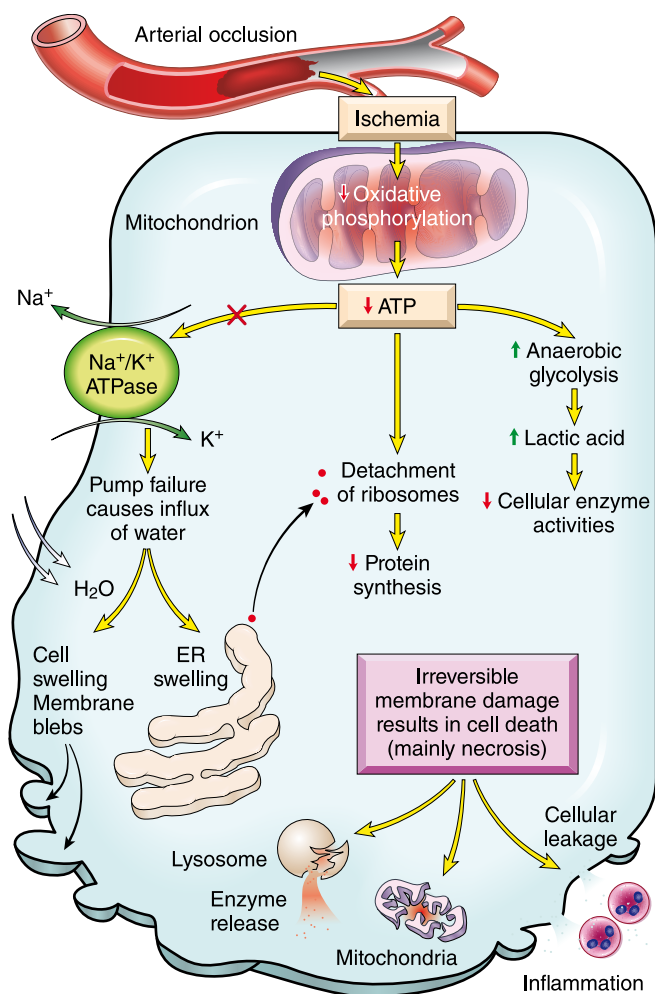


Fig. 2.24 Functional and morphologic consequences of decreased intracellular adenosine triphosphate (ATP) in ischemic cell injury. The morphologic changes shown here are indicative of reversible cell injury. Further depletion of ATP results in cell death, typically by necrosis. ER, Endoplasmic reticulum.

may sustain loss of viable cells in addition to those that are irreversibly damaged by the ischemia. This process, called *ischemia-reperfusion injury*, is clinically important because it contributes to tissue damage during myocardial and cerebral infarction following therapies that restore blood flow (Chapters 12 and 28).

How does reperfusion injury occur? The likely answer is that new damaging processes are set in motion during reperfusion, causing the death of cells that might have recovered otherwise. Several mechanisms have been proposed:

- **Oxidative stress.** New damage may be initiated during reoxygenation by increased generation of *reactive oxygen and nitrogen species*. These free radicals may be produced in reperfused tissue because of incomplete reduction of oxygen in leukocytes, and in damaged endothelial cells and parenchymal cells. Compromise of cellular antioxidant defense mechanisms during ischemia may sensitize cells to free radical damage.
- **Intracellular calcium overload.** As mentioned earlier, intracellular and mitochondrial calcium overload begins during acute ischemia; it is exacerbated during reperfusion due to an influx of calcium that

results from cell membrane damage and ROS-mediated injury to the sarcoplasmic reticulum. Calcium overload favors opening of the mitochondrial permeability transition pore with resultant ATP depletion. This, in turn, causes further cell injury.

- **Inflammation.** Ischemic injury is associated with inflammation because of “danger signals” released from dead cells, cytokines secreted by resident immune cells such as macrophages, and increased expression of adhesion molecules by hypoxic parenchymal and endothelial cells, all of which act to recruit circulating neutrophils to reperfused tissue. The inflammation causes additional tissue injury (Chapter 3). The importance of neutrophil influx in reperfusion injury has been demonstrated experimentally by the salutary effects of treatment with antibodies that block cytokines or adhesion molecules and thereby reduce neutrophil extravasation.
- Activation of the *complement system* may contribute to ischemia-reperfusion injury. For unknown reasons, some IgM antibodies have a propensity to deposit in ischemic tissues. When blood flow is resumed, complement proteins bind to the deposited antibodies, are activated, and exacerbate cell injury and inflammation.

Chemical (Toxic) Injury

Chemical injury remains a frequent problem in clinical medicine and is an important limitation to drug therapy. Because many drugs are metabolized in the liver, this organ is a major target of drug toxicity. In fact, toxic liver injury is often the reason for terminating the therapeutic use or development of a drug. The mechanisms by which chemicals, certain drugs, and toxins produce injury are described in greater detail in Chapter 9 in the discussion of environmental diseases. Here the major pathways of chemically induced injury with selected examples are described.

Chemicals induce cell injury by one of two general mechanisms:

- **Direct toxicity.** Some chemicals injure cells directly by combining with critical molecular components. For example, in mercuric chloride poisoning, *mercury* binds to the sulfhydryl groups of cell membrane proteins, causing increased membrane permeability and inhibition of ion transport. In such instances, the greatest damage is usually to the cells that use, absorb, excrete, or concentrate the chemicals—in the case of mercuric chloride, the cells of the gastrointestinal tract and kidney (Chapter 9). *Cyanide* poisons mitochondrial cytochrome oxidase and thus inhibits oxidative phosphorylation. Many antineoplastic chemotherapeutic agents and antibiotics also induce cell damage by direct cytotoxic effects.
- **Conversion to toxic metabolites.** Most toxic chemicals are not biologically active in their native form but must be converted to reactive toxic metabolites, which then act on target molecules. This modification is usually accomplished by the cytochrome P-450 mixed-function oxidases in the smooth ER of the liver and other organs. The toxic metabolites cause membrane damage and cell injury mainly by formation of free radicals and subsequent lipid peroxidation; direct covalent binding to membrane proteins and lipids may also contribute. For instance, CCl_4 , which was once widely used in the dry cleaning industry, is converted by cytochrome P-450 to the highly reactive free radical $\cdot\text{CCl}_3$, which causes lipid peroxidation and damages many cellular structures. The analgesic drug acetaminophen is also converted to a toxic product during metabolism in the liver, leading to cell injury. These and other examples of chemical injury are described in Chapter 9.

KEY CONCEPTS

Examples of Cell Injury

- Mild ischemia: Reduced oxidative phosphorylation → ATP depletion → failure of Na pump → influx of sodium and water → organelle and cellular swelling (reversible).
- Severe/prolonged ischemia: Severe swelling of mitochondria, calcium influx into mitochondria and into the cell with rupture of lysosomes and plasma membrane. Death by necrosis and apoptosis due to release of cytochrome c from mitochondria.
- Reperfusion injury follows restoration of blood flow to ischemic tissues and is caused by oxidative stress due to release of free radicals from leukocytes and endothelial cells. Blood brings calcium that overloads reversibly injured cells with consequent mitochondrial injury, as well as oxygen and leukocytes, which generate free radicals and cytokines. Complement may be activated locally by IgM antibodies deposited in ischemic tissues.
- Chemicals may cause injury directly or by conversion into toxic metabolites. The organs chiefly affected are those involved in absorption or excretion of chemicals or others such as the liver, where the chemicals are converted to toxic metabolites. Direct injury to critical organelles such as mitochondria or indirect injury from free radicals generated from the chemicals/toxins is involved.

ADAPTATIONS OF CELLULAR GROWTH AND DIFFERENTIATION

Adaptations are reversible changes in the size, number, phenotype, metabolic activity, or functions of cells in response to changes in their environment. Such adaptations may take several distinct forms.

Hypertrophy

Hypertrophy is an increase in the size of cells that results in an increase in the size of the affected organ. The hypertrophied organ has no new cells, just larger cells. The increased size of the cells is due to the synthesis and assembly of additional intracellular structural components. Cells capable of division may respond to stress by undergoing

both hyperplasia (described later) and hypertrophy, whereas nondividing cells (e.g., myocardial fibers) increase tissue mass due to hypertrophy. In many sites, hypertrophy and hyperplasia may coexist, with both contributing to increased organ size.

Hypertrophy can be *physiologic* or *pathologic*; the former is caused by increased functional demand or stimulation by hormones and growth factors.

- **Pathologic hypertrophy.** The striated muscle cells in the heart and skeletal muscles have only a limited capacity for division and respond to increased metabolic demands mainly by undergoing hypertrophy. *The most common stimulus for hypertrophy of skeletal and cardiac muscle is increased workload.* In both tissue types, muscle cells respond by synthesizing more protein and increasing the number of myofilaments per cell. This in turn increases the amount of force each myocyte can generate and thus the strength and work capacity of the muscle as a whole. A classic example of pathologic hypertrophy is enlargement of the heart in response to pressure overload, usually resulting from hypertension or valvular disease (see Fig. 2.2). Initially, cardiac hypertrophy improves function, but over time this adaptation often fails, setting the stage for heart failure and other significant forms of heart disease (Chapter 12).
- **Physiologic hypertrophy.** The massive physiologic growth of the uterus during pregnancy is a good example of hormone-induced enlargement of an organ that results mainly from hypertrophy of smooth muscle fibers (Fig. 2.25). Uterine hypertrophy during pregnancy is stimulated by estrogenic hormone signaling through estrogen receptors that eventually results in increased synthesis of smooth muscle proteins and increased cell size. The bulging muscles of bodybuilders engaged in “pumping iron” result from enlargement of individual skeletal muscle fibers in response to increased demand.

Mechanisms of Hypertrophy

Hypertrophy is a result of increased cellular protein production. Much of our understanding of hypertrophy is based on studies of the heart. There is great interest in defining the molecular basis of myocardial hypertrophy because beyond a certain point it becomes maladaptive. Hypertrophy results from the action of growth factors and direct

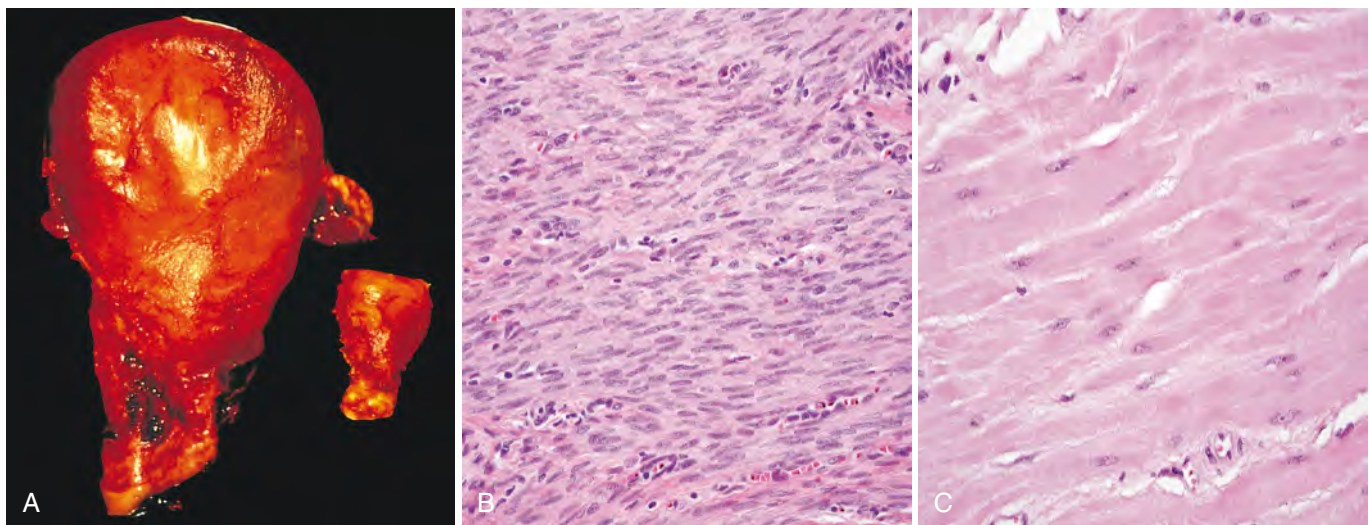


Fig. 2.25 Physiologic hypertrophy of the uterus during pregnancy. (A) Gross appearance of a normal uterus (right) and a gravid uterus (removed for postpartum bleeding) (left). (B) Small spindle-shaped uterine smooth muscle cells from a normal uterus, compared with (C) large plump cells from the gravid uterus, at the same magnification.

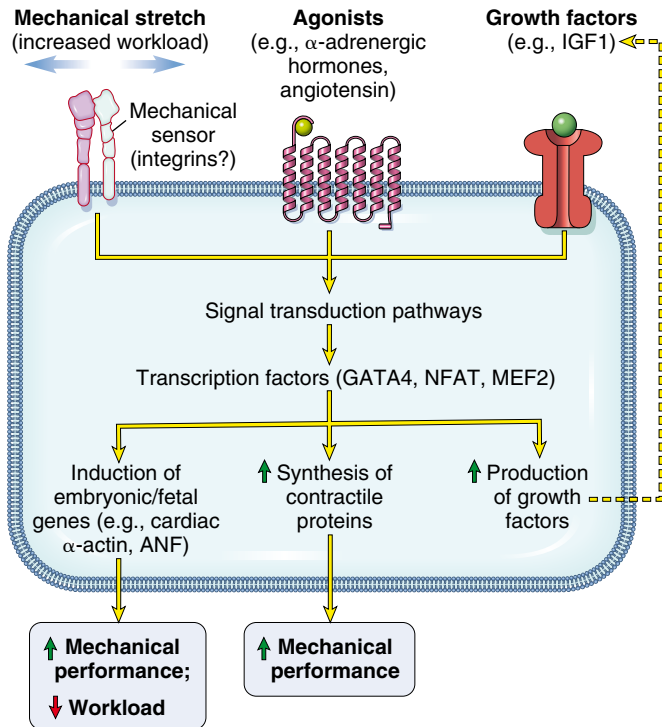


Fig. 2.26 Biochemical mechanisms of myocardial hypertrophy. The major known signaling pathways and their functional effects are shown. Mechanical sensors appear to be the major triggers for physiologic hypertrophy, and agonists and growth factors may be more important in pathologic states. *ANF*, Atrial natriuretic factor; *GATA4*, transcription factor that binds to DNA sequence GATA; *IGF1*, insulin-like growth factor; *NFAT*, nuclear factor-activated T cells; *MEF2*, myocyte enhancing factor 2.

effects of mechanical force on pathways that stimulate protein synthesis (Fig. 2.26):

- Mechanical sensors in the cell detect the increased load.
- These sensors activate a complex downstream web of signaling pathways, including the phosphoinositide 3-kinase (PI3K)/AKT pathway (postulated to be most important in physiologic, e.g., exercise-induced, hypertrophy) and G-protein-coupled receptor-initiated pathways (activated by many growth factors and vasoactive agents, and thought to be more important in pathologic hypertrophy).
- Some of the signaling pathways stimulate increased production of growth factors (e.g. TGF- β , insulin-like growth factor-1 [IGF-1], fibroblast growth factor) and vasoactive agents (e.g., α -adrenergic agonists, endothelin-1, and angiotensin II).
- These and other pathways activate transcription factors, including GATA4, nuclear factor of activated T cells (NFAT), and myocyte enhancer factor 2 (MEF2), which increase the expression of genes that encode contractile proteins.

Cardiac hypertrophy is also associated with a switch in gene expression from genes that encode adult-type contractile proteins to genes that encode functionally distinct fetal isoforms. For example, the α isoform of myosin heavy chain is replaced by the β isoform, which produces slower, more energetically economical contractions. Other proteins that are altered in hypertrophic myocardial cells are the products of genes that participate in the cellular response to stress. For example, cardiac hypertrophy is associated with increased atrial natriuretic factor gene expression. Atrial natriuretic factor is a peptide

hormone that causes salt secretion by the kidney, decreases blood volume and pressure, and therefore serves to reduce hemodynamic load.

Whatever the exact cause and mechanism of cardiac hypertrophy, it eventually reaches a limit beyond which enlargement of muscle mass is no longer able to cope with the increased burden. At this stage, several regressive changes occur in the myocardial fibers, of which the most important are degradation and loss of myofibrillar contractile elements. In extreme cases, myocyte death can occur. The net result of these changes is cardiac failure, a sequence of events that illustrates how an adaptation to stress can progress to significant cell injury if the stress is not relieved.

Hyperplasia

Hyperplasia is an increase in the number of cells in an organ or tissue in response to a stimulus. Although hyperplasia and hypertrophy are distinct processes, they frequently occur together, and may be triggered by the same external stimuli. Hyperplasia can only take place if the tissue contains cells capable of dividing. It can be physiologic or pathologic.

- **Physiologic hyperplasia.** Physiologic hyperplasia due to the action of hormones or growth factors occurs when there is a need to increase functional capacity of hormone-sensitive organs, or when there is need for compensatory increase after damage or resection. Hormonal hyperplasia is well illustrated by the proliferation of the glandular epithelium of the female breast at puberty and during pregnancy, usually accompanied by enlargement (hypertrophy) of the glandular epithelial cells. The classic illustration of compensatory hyperplasia comes from the study of liver regeneration. In individuals who donate one lobe of the liver for transplantation, the remaining cells proliferate so that the organ soon grows back to its original size. Experimental models of partial hepatectomy have been very useful for defining the mechanisms that stimulate regeneration of the liver (Chapter 3). The bone marrow is also remarkable in its capacity to undergo rapid hyperplasia in response to a deficiency of mature blood cells. For example, in the setting of an acute bleeding or premature breakdown of red cells (hemolysis), feedback loops involving the growth factor erythropoietin are activated that stimulate the growth of red cell progenitors, allowing red cell production to increase as much as eightfold. The regulation of hematopoiesis is discussed further in Chapter 13.
- **Pathologic hyperplasia.** Most forms of pathologic hyperplasia are caused by excessive or inappropriate actions of hormones or growth factors acting on target cells. Endometrial hyperplasia is an example of abnormal hormone-induced hyperplasia. Normally, after a menstrual period there is a rapid burst of proliferative activity in the endometrium that is stimulated by pituitary hormones and ovarian estrogen. It is brought to a halt by the rising levels of progesterone, usually about 10 to 14 days before the end of the menstrual period. In some instances, however, the balance between estrogen and progesterone is disturbed, resulting in absolute or relative increases in the amount of estrogen, with consequent hyperplasia of the endometrial glands. This form of pathologic hyperplasia is a common cause of abnormal uterine bleeding. Benign prostatic hyperplasia is another common example of pathologic hyperplasia, in this case as a response to hormonal stimulation by androgens. Although these forms of pathologic hyperplasia are abnormal, the process remains controlled, and the hyperplasia can either regress or stabilize if the hormonal stimulation is eliminated. As discussed in Chapter 7, the increased cell division associated with hyperplasia elevates the risk of acquiring genetic aberrations that can drive unrestrained proliferation and give rise

to cancer. Thus, *although hyperplasia is distinct from cancer, pathologic hyperplasia constitutes a fertile soil in which cancerous proliferations may eventually arise*. For instance, patients with hyperplasia of the endometrium are at increased risk for developing endometrial cancer (Chapter 22).

Hyperplasia is a characteristic response to certain *viral infections*, such as papillomaviruses, which cause skin warts and several mucosal lesions composed of masses of hyperplastic epithelium. Here, the viruses make factors that interfere with host proteins that regulate cell proliferation. Like other forms of hyperplasia, some of these virally induced proliferations are also precursors to cancer (Chapter 7).

Mechanisms of Hyperplasia

Hyperplasia is the result of growth factor–driven proliferation of mature cells and, in some cases, by increased output of new cells from tissue stem cells. For instance, after partial hepatectomy, growth factors are produced in the liver that engage receptors on the surviving liver cells and activate signaling pathways that stimulate cell proliferation. But if the proliferative capacity of the liver cells is compromised, as in some forms of hepatitis causing cell injury, hepatocytes can instead regenerate from intrahepatic stem cells. The roles of growth factors and stem cells in cellular replication and tissue regeneration are discussed in more detail in Chapter 3.

Atrophy

Atrophy is a reduction in the size of an organ or tissue due to a decrease in cell size and number. Atrophy can be physiologic or pathologic. *Physiologic atrophy* is common during normal development. Some embryonic structures, such as the notochord and thyroglossal duct, undergo atrophy during fetal development.

Pathologic atrophy has several causes, which can be local or generalized. Common causes of atrophy include:

- *Decreased workload (disuse atrophy)*. When a fractured bone is immobilized in a plaster cast or when a patient is restricted to complete bed rest, skeletal muscle atrophy rapidly ensues. The initial decrease in cell size is reversible once activity is resumed. With more prolonged disuse, skeletal muscle fibers decrease in number (due to apoptosis) as well as in size; muscle atrophy can be accompanied by increased bone resorption, leading to disuse osteoporosis.
- *Loss of innervation (denervation atrophy)*. The normal metabolism and function of skeletal muscle depend on its nerve supply. Damage to the nerves leads to atrophy of the muscle fibers supplied by those nerves (Chapter 27).
- *Diminished blood supply*. A gradual decrease in blood supply (chronic ischemia) to a tissue because of slowly developing arterial occlusive disease results in tissue atrophy. In late adult life, the brain may undergo progressive atrophy, mainly because of reduced blood supply as a result of atherosclerosis (Fig. 2.27). This is called *senile atrophy*.
- *Inadequate nutrition*. Profound protein-calorie malnutrition (marasmus) is associated with the utilization of skeletal muscle proteins as a source of energy after other reserves such as adipose stores have been depleted. This results in marked muscle wasting (*cachexia*; Chapter 9). Cachexia is also seen in patients with chronic inflammatory diseases and cancer. In some cachectic states, chronic overproduction of the inflammatory cytokine TNF is thought to be responsible for appetite suppression and lipid depletion, culminating in muscle atrophy.
- *Loss of endocrine stimulation*. Many hormone-responsive tissues, such as the breast and reproductive organs, are depend on endocrine stimulation for normal metabolism and function. The loss of estrogen stimulation after menopause results in atrophy of the endometrium, vaginal epithelium, and breast. Similarly, the prostrate atrophies following chemical or surgical castration (e.g., for treatment of prostate cancer).
- *Pressure*. Tissue compression for any length of time can cause atrophy. An enlarging benign tumor can cause atrophy in the surrounding uninvolved tissues. Atrophy in this setting is probably the result of ischemic changes caused by compromise of the blood supply from the pressure exerted by the expanding mass.

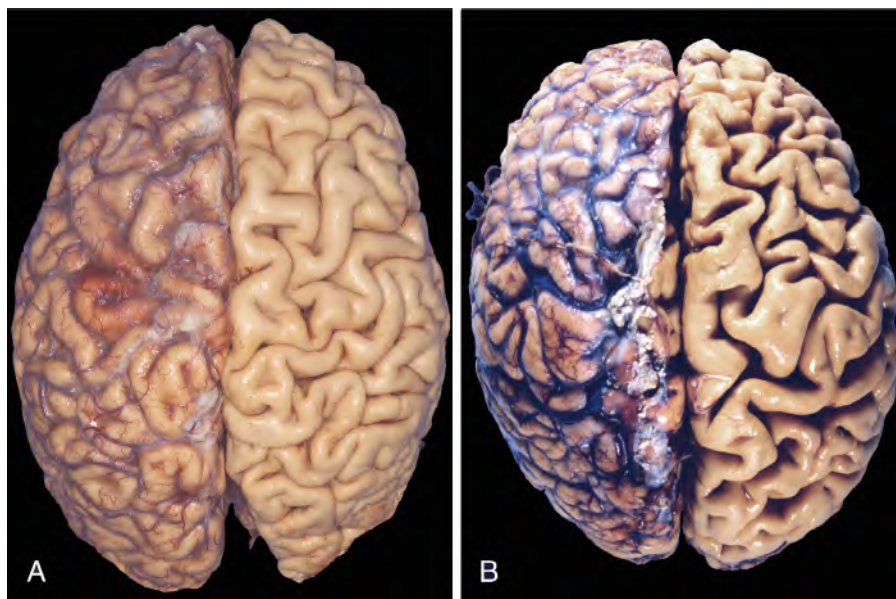


Fig. 2.27 Atrophy. (A) Normal brain of a young adult. (B) Atrophy of the brain in an 82-year-old male with atherosclerotic cerebrovascular disease resulting in reduced blood supply. Note that loss of brain substance narrows the gyri and widens the sulci. The meninges have been stripped from the right half of each specimen to reveal the surface of the brain.

The fundamental cellular changes associated with atrophy are similar in all these settings. The initial response is a decrease in cell size and organelles, which may reduce the metabolic needs of the cell sufficiently to permit its survival. In atrophic muscle, the cells contain fewer mitochondria and myofilaments and a reduced amount of rough ER. By bringing into balance the cell's metabolic demands and the lower levels of blood supply, nutrition, or trophic stimulation, a new equilibrium is achieved. *Early in the process, atrophic cells and tissues have diminished function, but cell death is minimal.* However, atrophy caused by gradually reduced blood supply may progress to the point at which cells are irreversibly injured and die, often by apoptosis. Cell death by apoptosis also contributes to the atrophy of endocrine organs after hormone withdrawal.

Mechanisms of Atrophy

Atrophy results from decreased protein synthesis and increased protein degradation. Protein synthesis decreases because of reduced trophic signals (e.g., those produced by growth receptors), which enhance uptake of nutrients and increase mRNA translation.

The degradation of cellular proteins occurs mainly by the ubiquitin-proteasome pathway. Nutrient deficiency and disuse may activate ubiquitin ligases, which attach the small peptide ubiquitin to cellular proteins and target these proteins for degradation in proteasomes. This pathway is also thought to be responsible for the accelerated proteolysis seen in a variety of catabolic conditions, including cancer-related cachexia. In many situations, atrophy is also accompanied by increased *autophagy*, marked by the appearance of increased numbers of autophagic vacuoles. Some of the cell debris within the autophagic vacuoles may resist digestion and persist in the cytoplasm as membrane-bound *residual bodies*. An example of residual bodies is *lipofuscin granules*, discussed later in the chapter. When present in sufficient amounts, they impart a brown discoloration to the tissue (*brown atrophy*). Autophagy is associated with various types of cell injury, as discussed earlier.

Metaplasia

Metaplasia is a reversible change in which one differentiated cell type (epithelial or mesenchymal) is replaced by another cell type. It often represents an adaptive response in which one cell type that is sensitive to a particular stress is replaced by another cell type that is better able to withstand the adverse environment.

The most common epithelial metaplasia is columnar to squamous (Fig. 2.28), as occurs in the respiratory tract in response to chronic irritation. In the habitual cigarette smoker, the normal ciliated columnar epithelial cells of the trachea and bronchi are often replaced by stratified squamous epithelial cells. Vitamin A (retinoic acid) deficiency can also induce squamous metaplasia in the respiratory epithelium and in the cornea, the latter with highly deleterious effects on vision (Chapter 9). Stones in the excretory ducts of the salivary glands, pancreas, or bile ducts, which are normally lined by secretory columnar epithelium, may produce irritation and pressure effects that lead to squamous metaplasia. In all these instances, rugged stratified squamous epithelium can survive under circumstances in which fragile specialized columnar epithelium might succumb. However, the change to metaplastic squamous cells comes with a price. In the respiratory tract, for example, although the epithelial lining becomes more durable, important mechanisms of protection against infection—mucus secretion and the ciliary action of the columnar epithelium—are lost. Thus, epithelial metaplasia, in most circumstances, represents an undesirable change. *Moreover, the influences that predispose to metaplasia, if persistent, can initiate malignant transformation in metaplastic*

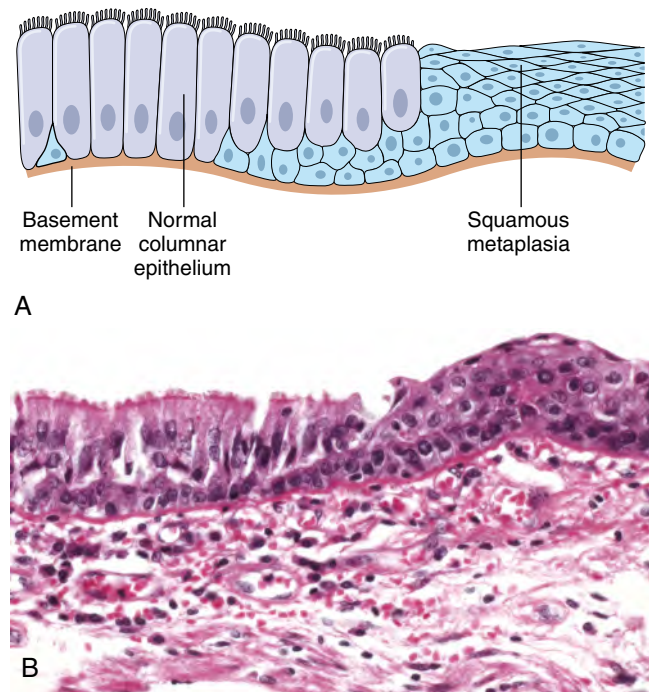


Fig. 2.28 Metaplasia of columnar to squamous epithelium. (A) Schematic diagram. (B) Metaplasia of columnar epithelium (*left*) to squamous epithelium (*right*) in a bronchus (as often occurs with smoking).

epithelium. The development of squamous cell carcinoma in areas of the lungs where the normal columnar epithelium has been replaced by squamous epithelium is one example.

Metaplasia from squamous to columnar type may also occur, as in *Barrett esophagus*, in which the esophageal squamous epithelium is replaced by intestinal-like columnar cells under the influence of refluxed gastric acid. As might be expected, the cancers that arise in these areas are typically gland-forming adenocarcinomas (Chapter 17).

Connective tissue metaplasia is the formation of cartilage, bone, or adipose cells (mesenchymal tissues) in tissues that normally do not contain these elements. For example, bone formation in muscle, designated *myositis ossificans*, occasionally occurs after intramuscular hemorrhage. This type of metaplasia is less clearly seen as an adaptive response and may be a result of cell or tissue injury. Unlike epithelial metaplasia, this type of metaplasia is not associated with increased cancer risk.

Mechanisms of Metaplasia

Metaplasia does not result from a change in the phenotype of an already differentiated cell type; rather, it results from either reprogramming of local tissue stem cells or, alternatively, colonization by differentiated cell populations from adjacent sites. In either case, the metaplastic change is stimulated by signals generated by cytokines, growth factors, and extracellular matrix components in the cells' environment. In the case of stem cell reprogramming, these external stimuli promote the expression of genes that drive cells toward a specific differentiation pathway. A direct link between transcription factor dysregulation and metaplasia is seen with vitamin A (retinoic acid) deficiency or excess, both of which may cause metaplasia. Retinoic acid regulates gene transcription directly through nuclear retinoid receptors (Chapter 9), which can influence the differentiation of progenitors derived from tissue stem cells.

KEY CONCEPTS

Cellular Adaptations to Stress

- **Hypertrophy:** increased cell and organ size, often in response to increased workload; induced by growth factors produced in response to mechanical stress or other stimuli; occurs in tissues with limited proliferative capacity. It may be physiologic (e.g., enlargement of the uterus in pregnancy) or pathologic (e.g., enlargement of the heart in hypertension or valvular disease).
- **Hyperplasia:** increased cell numbers in response to hormones and other growth factors; occurs in tissues whose cells can divide or contain abundant tissue stem cells. It may be physiologic in response to increased need (e.g., breast acini during lactation) or pathologic in response to inappropriate secretion of hormones (e.g., endometrial hyperplasia due to excessive estrogenic stimulation).
- **Atrophy:** decreased cell and organ size, because of decreased nutrient supply or disuse; associated with decreased synthesis of cellular building blocks and increased breakdown of proteins and cellular organelles via the ubiquitin-proteasome pathway and autophagy, respectively.
- **Metaplasia:** change in phenotype of differentiated cells, often in response to chronic irritation, that makes cells better able to withstand the stress; usually induced by altered differentiation pathway of tissue stem cells; may result in reduced function or increased propensity for malignant transformation.

INTRACELLULAR ACCUMULATIONS

One of the manifestations of metabolic derangements in cells is the intracellular accumulation of substances that may be harmless or cause further injury. These accumulations may be located in the cytoplasm, within organelles (typically lysosomes), or in the nucleus, and they may be composed of substances that are synthesized by the affected cells or are produced elsewhere.

There are four main mechanisms leading to abnormal intracellular accumulations (Fig. 2.29):

- **Inadequate removal** of a normal substance secondary to defects in production and transport, as in fatty change (steatosis) in the liver (Chapter 18).
- **Accumulation of an endogenous substance** because of genetic or acquired *defects in its folding, transport, or secretion*, as with mutated forms of α_1 -antitrypsin (Chapter 15).
- **Failure to degrade** a metabolite due to inherited enzyme deficiencies, typically lysosomal enzymes. The resulting disorders are called **lysosomal storage diseases** (Chapter 5).
- **Deposition and accumulation of an abnormal exogenous substance** when the cell has neither the enzymatic machinery to degrade the substance nor the ability to transport it to other sites. Accumulation of carbon or silica particles is an example of this type of alteration (Chapter 15).

In many cases, if the overload can be controlled or stopped, the accumulation is reversible. In inherited storage diseases, accumulation is progressive and may cause cellular injury, leading in some instances to death of the tissue and the patient.

Lipids

All major classes of lipids can accumulate in cells: triglycerides, cholesterol/cholesterol esters, and phospholipids. Phospholipids are components of the myelin figures found in necrotic cells. In addition, abnormal complexes of lipids and carbohydrates accumulate in the lysosomal storage diseases (Chapter 5). Triglyceride and cholesterol accumulations are discussed here.

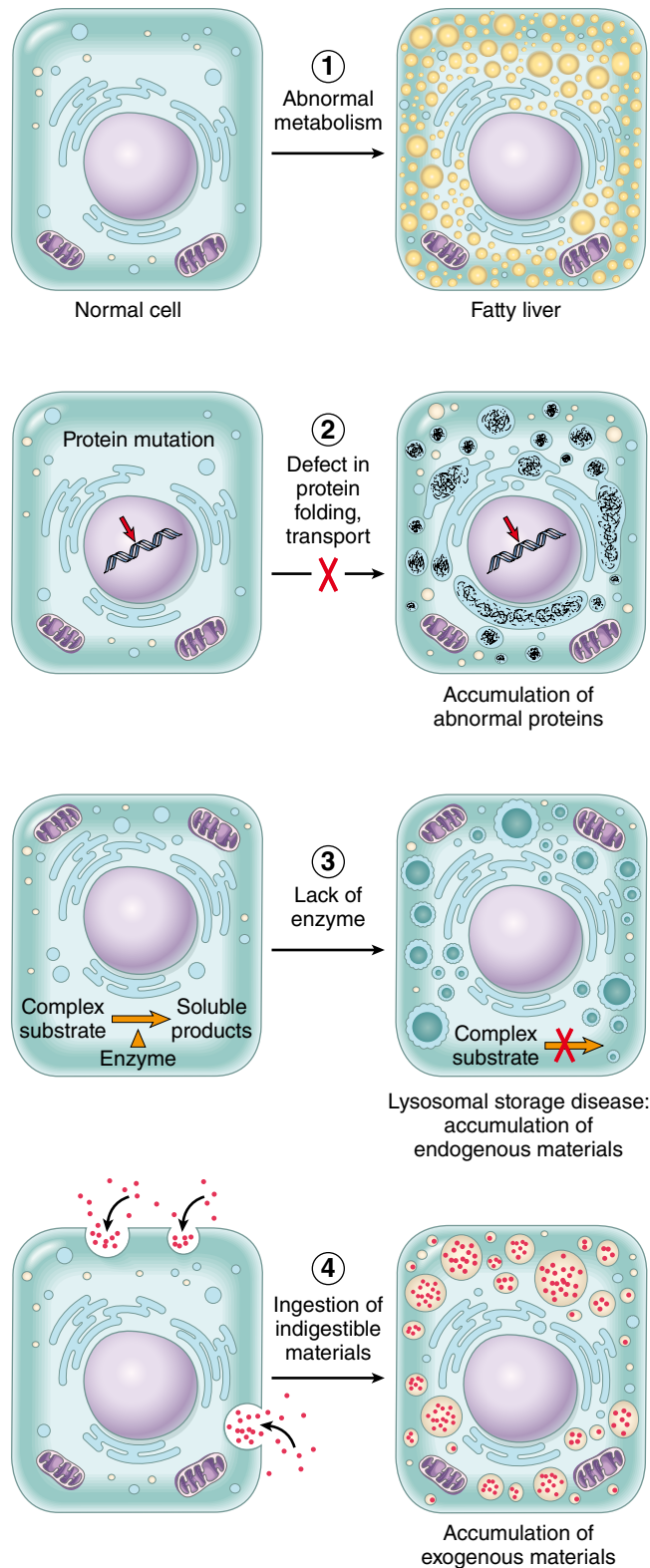


Fig. 2.29 Mechanisms of intracellular accumulations.

Steatosis (Fatty Change)

The terms **steatosis** and **fatty change** describe abnormal accumulations of triglycerides within parenchymal cells. Fatty change is often seen in the liver because it is the major organ involved in fat metabolism (Fig. 2.30), but it also occurs in the heart, muscle, and kidney. The causes

of steatosis include toxins, protein malnutrition, diabetes mellitus, obesity, and anoxia. In higher-income nations, the most common causes of significant fatty change in the liver (fatty liver) are excessive alcohol use and nonalcoholic fatty liver disease, which is often associated with diabetes and obesity. Fatty liver is discussed in more detail in Chapter 18.

Cholesterol and Cholesterol Esters

The cellular metabolism of cholesterol (Chapter 5) is tightly regulated such that most cells use cholesterol for the synthesis of cell membranes without intracellular accumulation of cholesterol or cholesterol esters. Accumulations manifested histologically by intracellular vacuoles are seen in several pathologic processes.

- **Atherosclerosis.** In atherosclerotic plaques, smooth muscle cells and macrophages within the intimal layer of the aorta and large arteries are filled with lipid vacuoles, most of which contain cholesterol and cholesterol esters. Such cells have a foamy appearance (foam cells), and aggregates of them in the intima produce the yellow cholesterol-laden atheromas characteristic of this serious disorder. Some of these fat-laden cells may rupture, releasing cholesterol and cholesterol esters into the extracellular space, where they may crystallize. Some crystals take the form of long needles that produce distinct clefts in tissue sections, while other small crystals are phagocytosed by macrophages and activate the inflammasome, contributing to local inflammation. The mechanisms of cholesterol accumulation and its pathogenic consequences in atherosclerosis are discussed in detail in Chapter 11.
- **Xanthomas.** Intracellular accumulation of cholesterol within macrophages is also characteristic of acquired and hereditary hyperlipidemic states. Clusters of foamy cells are found in the subepithelial connective tissue of the skin and in tendons, producing tumorous masses known as *xanthomas*.
- **Cholesterolosis.** This refers to the focal accumulations of cholesterol-laden macrophages in the lamina propria of the gallbladder (Fig. 2.31). The mechanism of accumulation is unknown.
- **Niemann-Pick disease, type C (NPC).** This lysosomal storage disease is caused by mutations affecting an enzyme involved in cholesterol trafficking, resulting in cholesterol accumulation in multiple organs (Chapter 5).

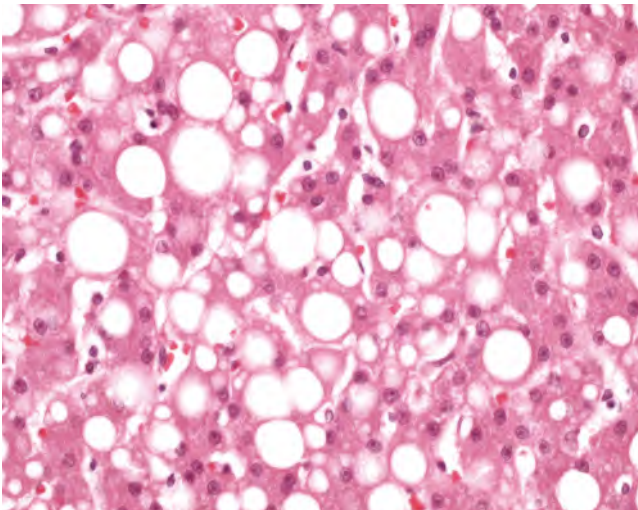


Fig. 2.30 Fatty liver. High-power detail of fatty change of the liver. In most cells, the well-preserved nucleus is squeezed into the displaced rim of cytoplasm about the fat vacuole. (Courtesy Dr. James Crawford, Department of Pathology, Hofstra Northwell School of Medicine, New York.)

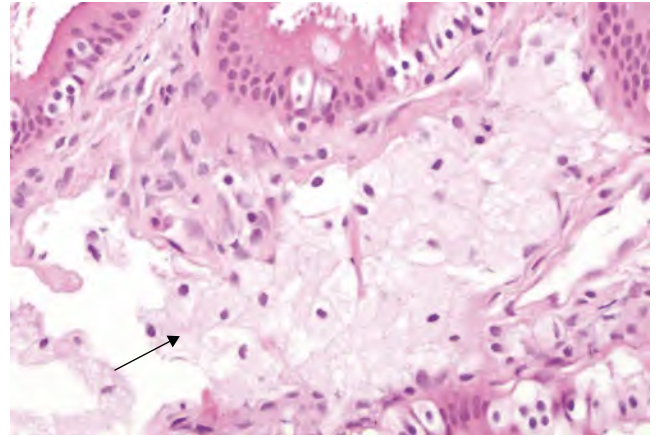


Fig. 2.31 Cholesterolosis. Cholesterol-laden macrophages (foam cells, arrow) in a focus of gallbladder cholesterolosis. (Courtesy Dr. Matthew Yeh, Department of Pathology, University of Washington, Seattle.)

Proteins

Intracellular accumulations of proteins usually appear as rounded, eosinophilic droplets, vacuoles, or aggregates in the cytoplasm. By electron microscopy, they can be amorphous, fibrillar, or crystalline in appearance. In some disorders, such as certain forms of amyloidosis, abnormal proteins deposit primarily in extracellular spaces (Chapter 6).

Excesses of proteins within cells sufficient to cause morphologically visible accumulation have diverse causes.

- **Reabsorption droplets in proximal renal tubules** are seen in renal diseases associated with protein loss in the urine (proteinuria). In the kidney, small amounts of protein filtered through the glomerulus are normally reabsorbed by pinocytosis in the proximal tubule. In disorders with heavy protein leakage across the glomerular filter, there is increased reabsorption of the protein into vesicles, and the protein appears as pink hyaline droplets within the cytoplasm of the tubular cell (Fig. 2.32). The process is reversible; if the proteinuria diminishes, the protein droplets are metabolized and disappear.
- **Increased production of normal proteins.** The proteins that accumulate may be normal secreted proteins that are produced in excessive amounts and accumulate within the ER, as occurs in certain plasma cells engaged in active synthesis of immunoglobulins. The ER becomes hugely distended, producing large, homogeneous eosinophilic inclusions called *Russell bodies*.
- **Defective intracellular transport and secretion of critical proteins.** In α_1 -antitrypsin deficiency, mutations in the protein significantly slow folding, resulting in the buildup of partially folded intermediates, which aggregate in the ER of the hepatocyte and are not secreted. The resultant deficiency of the circulating enzyme where it is needed in the lung causes emphysema (Chapter 15). In many of these protein-folding diseases, the pathology results not only from loss of protein function but also ER stress caused by the misfolded proteins, which initiates the unfolded protein response and culminates in cell death by apoptosis (discussed earlier).
- **Accumulation of cytoskeletal proteins.** There are several types of cytoskeletal proteins, including microtubules (20 to 25 nm in diameter), thin actin filaments (6 to 8 nm), thick myosin filaments (15 nm), and intermediate filaments (10 nm). Intermediate filaments, which provide a flexible intracellular scaffold that organizes the cytoplasm and resists forces applied to the cell, are divided into five classes: keratin filaments (characteristic of epithelial cells),

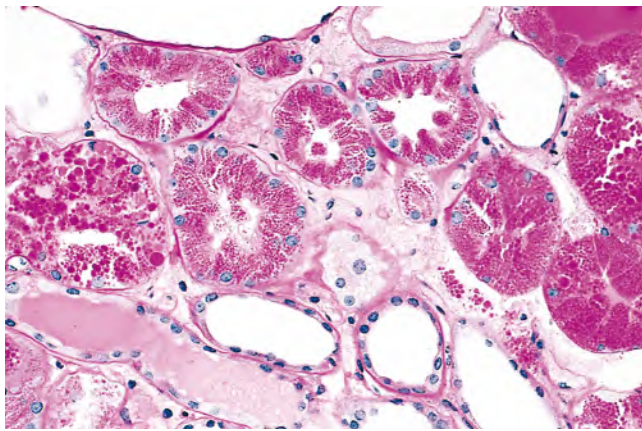


Fig. 2.32 Protein reabsorption droplets in the renal tubular epithelium. (Courtesy Dr. Helmut Rennke, Department of Pathology, Brigham and Women's Hospital, Boston.)

neurofilaments (neurons), desmin filaments (muscle cells), vimentin filaments (connective tissue cells), and glial filaments (astrocytes). Accumulations of keratin filaments and neurofilaments are associated with certain types of cell injury. *Alcoholic hyaline* is an eosinophilic cytoplasmic inclusion in liver cells that is characteristic of alcoholic liver disease and is composed predominantly of keratin intermediate filaments (Chapter 18). The *neurofibrillary tangle* found in the brain in Alzheimer disease contains neurofilaments and other proteins (Chapter 28).

- **Aggregation of abnormal proteins.** Abnormal or misfolded proteins may deposit in tissues and interfere with normal functions. The deposits can be intracellular, extracellular, or both, and the aggregates may either directly or indirectly cause pathologic changes. Certain forms of *amyloidosis* (Chapter 6) fall in this category of diseases. These disorders are sometimes called *proteinopathies* or *protein-aggregation diseases*. Finally, increasing evidence supports the idea that disease-related intracellular inclusion bodies (e.g., alcoholic hyaline, α_1 -antitrypsin aggregates) are dynamic, concentrated assemblies of proteins, called “membraneless organelles” or *biomolecular condensates*, which are formed via the process of liquid-liquid phase separation (Chapter 1); whether and how such inclusions contribute to disease progression remains an area of active investigation.

Hyaline Change

The term *hyaline* usually refers to an alteration within cells or in the extracellular space that gives a homogeneous, glassy, pink appearance in routine histologic sections stained with H&E. It is widely used as a descriptive histologic term rather than a specific marker for cell injury. This morphologic change is produced by a variety of alterations and does not represent a specific pattern of accumulation.

Intracellular hyaline accumulations of protein include reabsorption droplets, Russell bodies, and alcoholic hyaline (described earlier). *Extracellular hyaline* has been more difficult to analyze. Collagenous fibers in old scars may appear hyalinized, but the biochemical basis of this change is not clear. In long-standing hypertension and diabetes mellitus, the walls of arterioles, especially in the kidney, become hyalinized, resulting from extravasated plasma protein and deposition of basement membrane material.

Glycogen

Excessive intracellular deposits of glycogen are seen in patients with an abnormality in either glucose or glycogen metabolism. Glycogen

is a readily available source of glucose stored in the cytoplasm of healthy cells. Whatever the clinical setting, the glycogen masses appear as clear vacuoles within the cytoplasm because glycogen dissolves in aqueous fixatives; thus, it is most readily identified when tissues are fixed in absolute alcohol. Staining with Best carmine or the PAS reaction imparts a rose-to-violet color to glycogen, but can also stain protein-bound carbohydrates. Proof that staining is due to the presence of glycogen requires staining of a serial tissue section following treatment with diastase, an enzyme that hydrolyzes glycogen.

Diabetes mellitus is the prime example of a disorder of glucose metabolism. In this disease, glycogen is found in renal tubular epithelial cells, as well as within liver cells, β cells of the islets of Langerhans within the pancreas, and heart muscle cells.

Glycogen accumulates within select cells in a group of related genetic disorders that are collectively referred to as the *glycogen storage diseases*, or *glycogenoses* (Chapter 5). In these diseases, enzymatic defects in the synthesis or breakdown of glycogen result in massive accumulation, causing cell injury and cell death.

Pigments

Pigments are colored substances, some of which are normal constituents of cells (e.g., melanin), whereas others are abnormal and accumulate in cells under special circumstances. Pigments can be exogenous, coming from outside the body, or endogenous, synthesized within the body itself.

Exogenous Pigments

The most common exogenous pigment is carbon (coal dust), a ubiquitous air pollutant in urban areas. When inhaled, it is picked up by macrophages within the alveoli and then transported through lymphatic channels to lymph nodes in the tracheobronchial region. Accumulations of this pigment blacken the tissues of the lungs (*anthracosis*) and draining lymph nodes. In coal miners, the aggregates of carbon dust may induce a fibroblastic reaction or even emphysema, and thus cause a serious lung disease known as *coal worker's pneumoconiosis* (Chapter 15). *Tattooing* is an art form in which indigestible pigments are injected into the dermis, where they are phagocytosed by macrophages and remain for the lifetime of the embellished individual. The pigments do not usually evoke any inflammatory response.

Endogenous Pigments

Lipofuscin is an insoluble pigment, also known as lipochrome or wear-and-tear pigment. Lipofuscin is composed of polymers of lipids and phospholipids in complex with protein, suggesting that it is derived from lipid peroxidation of polyunsaturated lipids of intracellular membranes. Lipofuscin is not injurious to the cell or its functions. Its importance lies in its being a telltale sign of free radical injury and lipid peroxidation. The term is derived from the Latin (*fuscus*, brown), referring to brown lipid. In tissue sections, it appears in the cytoplasm a yellow-brown, finely granular, often perinuclear, pigment (Fig. 2.33). It is seen in cells undergoing slow, regressive changes and is particularly prominent in the liver and heart of aging patients or patients with severe malnutrition and cancer cachexia.

Melanin, derived from the Greek (*melas*, black), is an endogenous, brown-black, pigment formed when the enzyme tyrosinase catalyzes the oxidation of tyrosine to dihydroxyphenylalanine in melanocytes. It is discussed further in Chapter 25. For practical purposes, melanin is the *only endogenous brown-black pigment*. The only other that could be considered in this category is homogentisic acid, a black pigment only found in patients with *alkaptonuria*, a rare metabolic disease. Here the pigment is deposited in the skin, connective tissue, and cartilage, and the pigmentation is known as *ochronosis*.

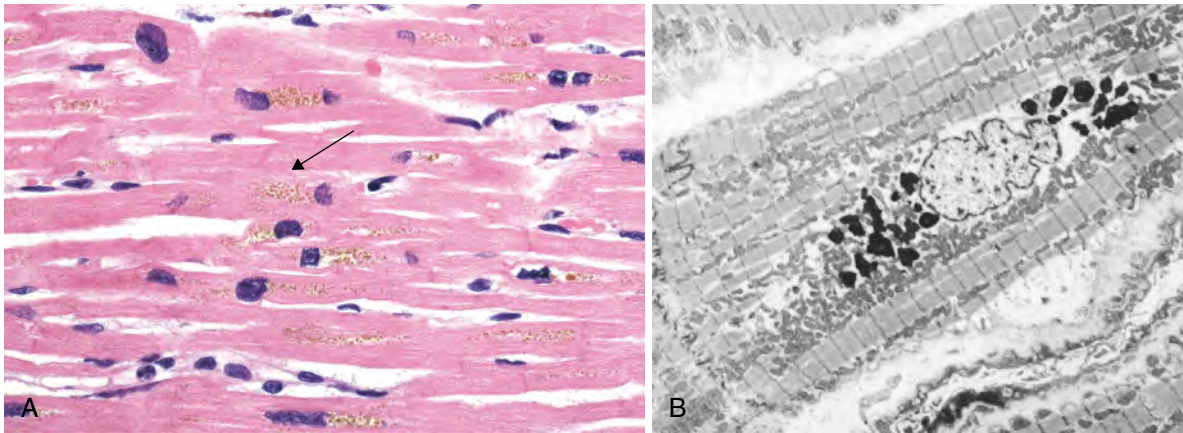


Fig. 2.33 Lipofuscin granules in cardiac myocytes. (A) Light microscopy (deposits indicated by arrow) and (B) electron microscopy (note the perinuclear, intralysosomal location).

Hemosiderin, a hemoglobin-derived, golden yellow-to-brown, granular or crystalline pigment is one of the major storage forms of iron. Iron metabolism and hemosiderin are considered in detail in Chapters 14 and 18. Iron is normally carried by a specific transport protein called *transferrin*. In cells, it is stored in association with a protein, apoferritin, as ferritin micelles. Ferritin is found in most cell types. When there is a local or systemic excess of iron, ferritin micelles aggregate to form *hemosiderin granules*, which are easily seen with the light microscope. Under normal conditions, small amounts of hemosiderin can be seen in the mononuclear phagocytes of the bone marrow, spleen, and liver, which are responsible for recycling of iron derived from hemoglobin during the breakdown of effete red blood cells.

Local or systemic excesses of iron cause hemosiderin to accumulate within cells. *Local excesses* result from hemorrhages in tissues. The best example of localized hemosiderosis is the common bruise. Extravasated red cells at the site of injury are phagocytosed over several days by macrophages, which break down the hemoglobin and recover the iron. After removal of iron, the heme moiety is converted first to biliverdin (“green bile”) and then to bilirubin (“red bile”). In parallel, the iron released from heme is incorporated into ferritin and eventually hemosiderin. These conversions account for the often dramatic play of colors seen in a healing bruise, which typically changes from red-blue to green-blue to golden-yellow before it is resolved.

When there is *systemic iron overload*, hemosiderin may be deposited in many organs and tissues, a condition called *hemosiderosis*. The main causes of hemosiderosis are (1) increased absorption of dietary iron due to an inborn error of metabolism called *hemochromatosis* (Chapter 18), (2) hemolytic anemias, in which excessive lysis of red cells leads to release of abnormal quantities of iron (Chapter 14), and (3) repeated blood transfusions, because transfused red cells constitute an exogenous iron load.

PATHOLOGIC CALCIFICATION

Pathologic calcification is the abnormal tissue deposition of calcium salts, together with smaller amounts of iron, magnesium, and other mineral salts. There are two forms of pathologic calcification. When the deposition occurs locally in dying tissues, it is known as *dystrophic calcification*; it occurs despite normal serum levels of calcium and in the absence of derangements in calcium metabolism. In contrast, the deposition of calcium salts in otherwise normal tissues is known as *metastatic calcification*, and it almost always results from hypercalcemia secondary to some disturbance in calcium metabolism.

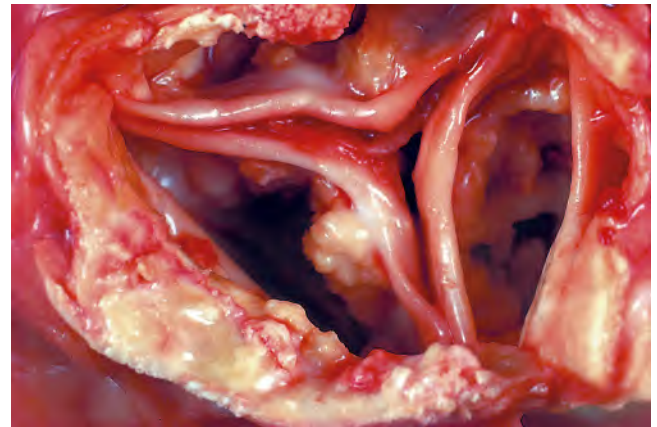


Fig. 2.34 Dystrophic calcification of the aortic valve. View looking down onto the unopened aortic valve in a heart with calcific aortic stenosis. The semilunar cusps are thickened and fibrotic, and behind each cusp are irregular masses of piled-up dystrophic calcification.

Dystrophic Calcification

Dystrophic calcification is encountered in areas of necrosis, whether they are of coagulative, caseous, or liquefactive type, and in foci of enzymatic necrosis of fat. Calcification is almost always present in the atheromas of advanced atherosclerosis. It also commonly develops in aging or damaged heart valves, further hampering their function (Fig. 2.34). Whatever the site of deposition, the calcium salts appear macroscopically as fine, white granules or clumps, often felt as gritty deposits. Sometimes a tuberculous lymph node is virtually converted to stone.

MORPHOLOGY

Histologically, with the usual H&E stain, calcium salts have a basophilic, amorphous granular, sometimes clumped appearance. They can be intracellular, extracellular, or in both locations. In the course of time, **heterotopic bone** may form in the focus of calcification. On occasion, single necrotic cells may constitute seed crystals that become encrusted by the mineral deposits. The progressive acquisition of outer layers may create lamellated configurations, called **psammoma bodies** because of their resemblance to grains of sand. Some types of papillary cancers (e.g., thyroid) are apt to develop psammoma bodies. In asbestosis, calcium and iron salts gather about long slender spicules of asbestos in the lung, creating exotic, beaded dumbbell forms known as **asbestos bodies** (Chapter 15).

Although dystrophic calcification may simply be a telltale sign of previous cell injury, it is often a cause of organ dysfunction. Such is the case in calcific valvular disease and atherosclerosis, as will become clear in further discussion of these diseases (Chapters 11 and 12). Serum calcium is normal in dystrophic calcification.

Metastatic Calcification

Metastatic calcification may occur in normal tissues whenever there is hypercalcemia. Hypercalcemia also accentuates dystrophic calcification. There are four principal causes of hypercalcemia: (1) *increased secretion of parathyroid hormone* (PTH) with subsequent bone resorption, as in hyperparathyroidism due to parathyroid tumors, and ectopic secretion of PTH-related protein by malignant tumors (Chapter 7); (2) *resorption of bone tissue*, secondary to primary tumors of bone marrow (e.g., multiple myeloma) or diffuse skeletal metastasis (e.g., breast cancer), accelerated bone turnover (e.g., Paget disease), or immobilization; (3) *vitamin D-related disorders*, including vitamin D intoxication, sarcoidosis (in which macrophages activate a vitamin D precursor), and idiopathic hypercalcemia of infancy (Williams syndrome), characterized by abnormal sensitivity to vitamin D; and (4) *renal failure*, which causes retention of phosphate, leading to secondary hyperparathyroidism. Less common causes include aluminum intoxication, which occurs in patients on chronic renal dialysis, and milk-alkali syndrome, which is due to excessive ingestion of calcium and absorbable antacids such as milk or calcium carbonate.

Metastatic calcification may occur widely throughout the body but principally affects the interstitial tissues of the gastric mucosa, kidneys, lungs, systemic arteries, and pulmonary veins. Although quite different in location, all of these tissues excrete acid and therefore have an internal alkaline compartment that predisposes them to metastatic calcification. In all these sites, the calcium salts morphologically resemble those described in dystrophic calcification. Thus, they may occur as noncrystalline amorphous deposits or, at other times, as hydroxyapatite crystals.

Usually, the mineral salts cause no clinical dysfunction, but on occasion, massive involvement of the lungs produces remarkable radiographs and respiratory compromise. Massive deposits in the kidney (nephrocalcinosis) may in time cause renal damage (Chapter 20).

KEY CONCEPTS

Abnormal Intracellular Depositions and Calcifications

Abnormal deposits of materials in cells and tissues are the result of excessive intake or defective transport or catabolism.

- **Deposition of lipids:**
 - **Fatty change:** Accumulation of free triglycerides in cells, resulting from excessive production or defective transport (often because of defects in synthesis of transport proteins); manifestation of reversible cell injury
 - **Cholesterol deposition:** Result of defective catabolism and excessive intake; in macrophages and smooth muscle cells of vessel walls in atherosclerosis
- **Deposition of proteins:** Reabsorbed proteins in kidney tubules; immunoglobulins in plasma cells
- **Deposition of glycogen:** In macrophages of patients with defects in lysosomal enzymes that break down glycogen (glycogen storage diseases) and in certain disorders of glycogen metabolism
- **Deposition of pigments:** Typically indigestible pigments, such as carbon, lipofuscin (breakdown product of lipid peroxidation), or iron (usually due to overload, as in hemosiderosis)
- **Pathologic calcifications:**
 - **Dystrophic calcification:** Deposition of calcium at sites of cell injury and necrosis
 - **Metastatic calcification:** Deposition of calcium in normal tissues, caused by hypercalcemia (usually a consequence of parathyroid hormone excess)

CELLULAR AGING

Mankind has pursued immortality from time immemorial. Toth and Hermes, Egyptian and Greek deities, are said to have discovered the elixir of youth and become immortal. Sadly, Toth and Hermes are nowhere to be found; hence the elixir remains a secret. Shakespeare probably characterized aging best in his elegant description of the seven ages of man. It begins at the moment of conception, involves the differentiation and maturation of the organism and its cells, at some variable point in time leads to the progressive loss of functional capacity characteristic of senescence, and ends in death.

Individuals age because their cells age. Although public attention on the aging process has traditionally focused on its cosmetic manifestations, aging has important health consequences, because age is one of the strongest independent risk factors for many chronic diseases, such as cancer, Alzheimer disease, and ischemic heart disease. Perhaps one of the most striking discoveries about cellular aging is that it is not simply a consequence of cells “running out of steam,” but in fact is regulated by genes that are evolutionarily conserved from yeast to worms to mammals.

Cellular aging is the result of a progressive decline in cellular function and viability caused by genetic abnormalities and the accumulation of cellular and molecular damage due to the effects of exposure to exogenous influences (Fig. 2.35). Studies in model systems have clearly established that aging is influenced by a limited number of genes, and genetic anomalies underlie syndromes resembling premature aging in humans as well. Such findings suggest that aging is associated with definable mechanistic alterations. Several mechanisms, some cell intrinsic and others environmentally induced, are believed to play a role in aging. These mechanisms have also been widely studied as potential avenues to promote longevity and healthy aging. However, the potential interventions described below are still largely experimental in nature, and their long-term safety and efficacy in humans is not fully understood.

DNA Damage

A variety of exogenous (physical, chemical, and biologic) agents and endogenous factors such as ROS threaten the integrity of nuclear and mitochondrial DNA. Although most DNA damage is repaired by DNA repair enzymes, some persists and accumulates as cells age. Several lines of evidence point to the importance of DNA repair in the aging process. Next-generation DNA sequencing studies have shown that the average hematopoietic stem cell suffers 14 new mutations per year, and it is likely that this accumulating damage explains why, like most cancers, the most common hematologic malignancies are diseases of the aged. Patients with *Werner syndrome* show premature aging, and the defective gene product is a DNA helicase, a protein involved in DNA replication and repair and other functions requiring DNA unwinding. A defect in this enzyme causes rapid accumulation of chromosomal damage that may mimic some aspects of the injury that normally accumulates during cellular aging. Genetic instability in somatic cells is also characteristic of other disorders in which patients display some of the manifestations of aging at an increased rate, such as *Bloom syndrome* and *ataxia-telangiectasia*, in which the mutated genes encode proteins involved in repairing double-strand breaks in DNA (Chapter 7).

Cellular Senescence

All normal cells have a limited capacity for replication, and after a fixed number of divisions cells become arrested in a terminally non-dividing state, known as *replicative senescence*. Aging is associated with progressive replicative senescence of cells. Cells from children

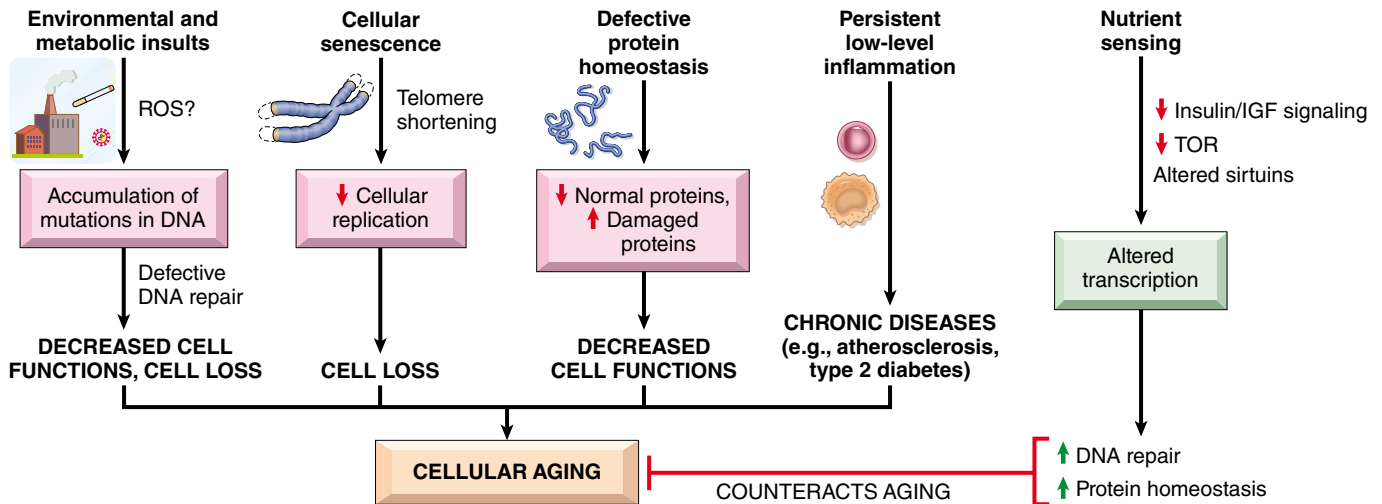


Fig. 2.35 Mechanisms that cause and counteract cellular aging. DNA damage, replicative senescence, decreased normal proteins, increased misfolded proteins, and persistent inflammation are among the best described mechanisms that promote cellular aging. Changes in nutrient sensing, exemplified by caloric restriction (not shown), counteract aging by modulating various signaling pathways. Among these pathways, the beneficial effects of reduced activation of the insulin/IGF and TOR signaling pathways on both lifespan and healthy aging are the most well understood. *IGF*, Insulin-like growth factor; *ROS*, reactive oxygen species; *TOR*, target of rapamycin.

have the capacity to undergo more rounds of replication than do cells from older people. Two mechanisms are believed to underlie cellular senescence:

- **Telomere attrition.** One mechanism of replicative senescence involves progressive shortening of telomeres, which ultimately results in cell cycle arrest. *Telomeres* are short repeated sequences of DNA present at the ends of linear chromosomes that are important for ensuring the complete replication of chromosome ends and for protecting the ends from fusion and degradation. When somatic cells replicate, a small section of the telomere is not duplicated, and telomeres become progressively shortened. As the telomeres become shorter, the ends of chromosomes cannot be protected and are seen as broken DNA, which signals cell cycle arrest. Telomere length is maintained by nucleotide addition mediated by an enzyme called *telomerase*. Telomerase is a specialized RNA-protein complex that uses its own RNA as a template for adding nucleotides to the ends of chromosomes. Telomerase is expressed in germ cells and is present at low levels in stem cells, but it is absent in most somatic tissues (Fig. 2.36). Therefore, as most somatic cells age, their telomeres become shorter and they exit the cell cycle, resulting in an inability to generate new cells to replace damaged ones. Conversely, in immortalized cancer cells, telomerase is usually reactivated, and telomere length is stabilized, allowing the cells to proliferate indefinitely. This is discussed more fully in Chapter 7. The causal links between telomere length and cellular senescence have been established in mouse models. Genetically engineered mice with shortened telomeres exhibit reduced life spans that can be restored to normal by telomerase activation. As discussed in other chapters, telomere shortening is also associated with premature development of diseases, such as pulmonary fibrosis (Chapter 15) and aplastic anemia (Chapter 14).
- **Activation of tumor suppressor genes.** In addition to telomere attrition, activation of certain tumor suppressor genes, particularly those encoded by the *CDKN2A* locus, also seems to be involved in controlling replicative senescence. The *CDKN2A* locus encodes two tumor suppressor proteins, expression of one of which, known as p16 or INK4a, is correlated with chronologic age in virtually all

human and mouse tissues examined. By controlling G1- to S-phase progression during the cell cycle (Chapter 1), p16 protects cells from uncontrolled mitogenic signals and pushes cells along the senescence pathway. This is discussed further in Chapter 7.

Defective Protein Homeostasis

Protein homeostasis involves two mechanisms: maintenance of proteins in their correctly folded conformations (mediated by chaperones) and degradation of misfolded, damaged, or unneeded proteins by the autophagy-lysosome and ubiquitin-proteasome systems. There is evidence that both normal folding and degradation of misfolded proteins are impaired with aging. Mutant mice deficient in chaperones of the heat shock protein family age rapidly, and conversely, those that over-express such chaperones are long lived. Similar data exist for the role of autophagy and proteasomal degradation of proteins. Of interest, administration of rapamycin, which inhibits the mTOR (mechanistic target of rapamycin) pathway, increases the life span of middle-aged mice. Rapamycin has multiple effects, including promotion of autophagy. Abnormal protein homeostasis can have many effects on cell survival, replication, and functions. In addition, it may lead to accumulation of misfolded proteins, which can trigger apoptosis.

Dysregulated Nutrient Sensing

Paradoxical though it may seem, eating less increases longevity. Caloric restriction increases life span in all eukaryotic species in which it has been tested, with encouraging results even in nonhuman primates and in a few unusually disciplined people who are the envy of others! Because of these observations, there has been much interest in deciphering the role of nutrient sensing in aging. Although incompletely understood, studies in model organisms have revealed two major neurohormonal circuits that regulate metabolism as important pathways to counteract aging, the *insulin and insulin-like growth factor 1 (IGF-1) signaling pathway* and *sirtuins*. Caloric restriction is proposed to increase longevity both by reducing the signaling intensity of the insulin/IGF-1 pathways and by increasing sirtuins.

- **IGF-1 signaling pathway.** IGF-1 is produced in many cell types in response to growth hormone secretion by the pituitary gland. IGF-1,

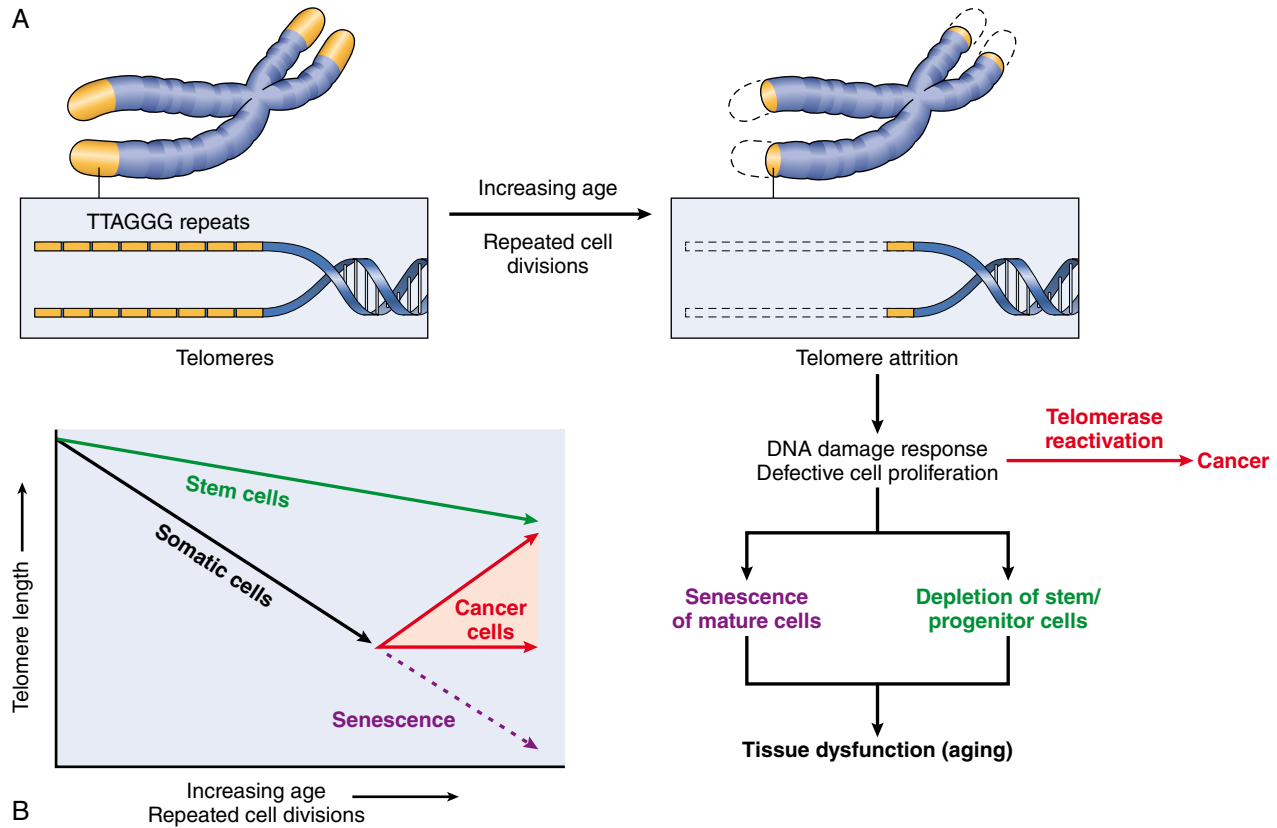


Fig. 2.36 Role of telomeres and telomerase in replicative senescence of cells. (A) Mechanisms and consequences of telomere attrition. Repeated cell division associated with aging leads to progressive shortening of telomeres, which triggers senescence and loss of stem and progenitor cell pools. (B) Telomere length is plotted against the number of cell divisions. Most somatic cells lack telomerase activity, and telomeres progressively shorten with increasing cell divisions until senescence or growth arrest occurs. Germ cells and stem cells both contain telomerase, but only germ cells have sufficient levels of the enzyme to stabilize telomere length completely. Cancer cells frequently reactivate telomerase and can reactivate telomeres. (Modified from Holt SE, Shay JW, Wright WE. Refining the telomere-telomerase hypothesis of aging and cancer. *Nat Biotechnol.* 1996;14:836-839.)

as indicated by its name, mimics intracellular signaling by insulin and thereby informs cells of the availability of glucose, promoting an anabolic state as well as cell growth and replication. Attenuation of IGF-1 signaling leads to lower rates of cell growth and metabolism and possibly reduced cellular damage. IGF-1 signaling has multiple downstream targets, among the most important is the downstream kinase *mechanistic target of rapamycin* (mTOR), as described above. Similar to reduced IGF-1 signaling, the inhibition of mTOR activity using rapamycin has been demonstrated to both extend lifespan and reduce age-related pathologic changes in diverse animal models.

- **Sirtuins.** Sirtuins are a family of NAD-dependent protein deacetylases. There are at least seven types of sirtuins in mammals that are distributed in different cellular compartments and have nonredundant functions designed to adapt bodily functions to various environmental stresses, including food deprivation and DNA damage. Sirtuins promote the expression of several genes whose products increase longevity and promote the health of organisms. These include proteins that inhibit metabolic activity, reduce apoptosis, stimulate protein folding, and counteract the harmful effects of oxygen free radicals. Sirtuins also increase insulin sensitivity and glucose metabolism, and may be targets for the treatment of diabetes.

An increase in sirtuins, particularly sirtuin-6, serves dual functions: the sirtuins (1) contribute to metabolic adaptations of caloric

restriction and (2) promote genomic integrity by activating DNA repair enzymes through deacylation. Although the antiaging effects of sirtuins have been widely publicized, much remains to be known before sirtuin-activating pills will be available to increase longevity. Nevertheless, optimistic wine lovers have been delighted to hear that a constituent of red wine may activate sirtuins and thus increase life span!

KEY CONCEPTS

Cellular Aging

- Cellular aging results from a combination of accumulating cellular damage (e.g., by free radicals), reduced capacity to divide (replicative senescence), reduced ability to repair damaged DNA, and defective protein homeostasis.
 - **Accumulation of DNA damage:** Defective DNA repair mechanisms; conversely, caloric restriction may activate DNA repair and slow aging in model organisms.
 - **Replicative senescence:** Reduced capacity of cells to divide secondary to progressive shortening of chromosomal ends (telomeres).
 - **Defective protein homeostasis:** Resulting from impaired chaperone and proteasome functions.
 - **Nutrient sensing system:** Caloric restriction increases longevity. Mediators include reduced IGF-1 signaling and increased sirtuins.

The various forms of cellular derangements and adaptations described in this chapter cover a wide spectrum, which includes adaptations in cell size, growth, and function; reversible and irreversible forms of acute cell injury; regulated cell death (e.g., in apoptosis); pathologic alterations in cell organelles; and less ominous forms of intracellular accumulations, including pigmentations. Reference is made to all of these alterations throughout this book, because all organ injury and ultimately all clinical disease arise from derangements in cell structure and function.

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